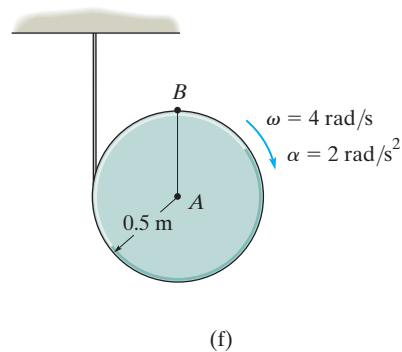
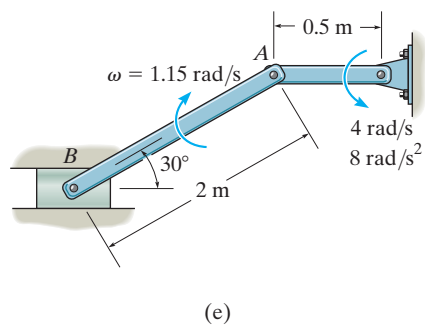
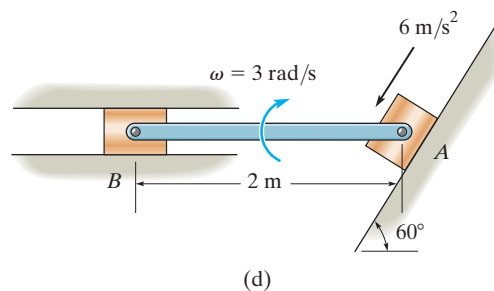
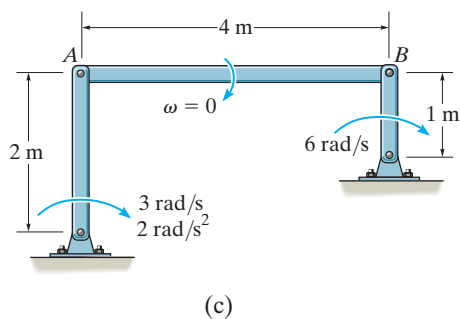
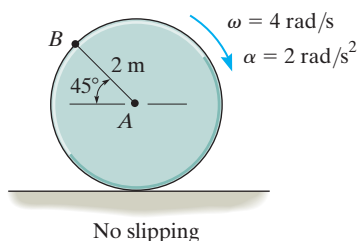
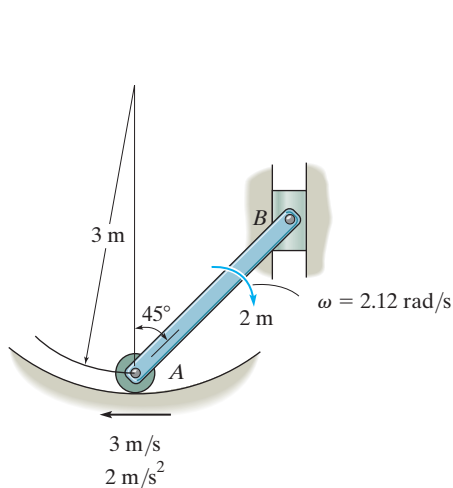


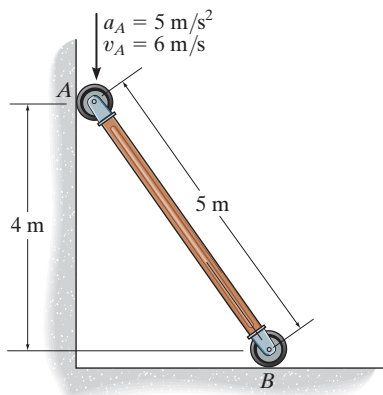
PRELIMINARY PROBLEM

P16-3. Set up the relative acceleration equation between points A and B . The angular velocity is given.



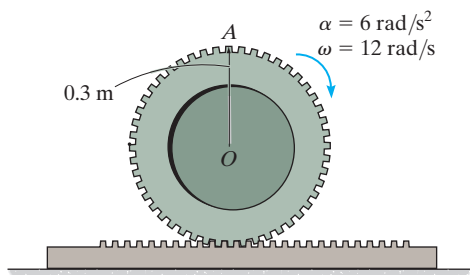
FUNDAMENTAL PROBLEMS

F16-19. At the instant shown, end A of the rod has the velocity and acceleration shown. Determine the angular acceleration of the rod and acceleration of end B of the rod.



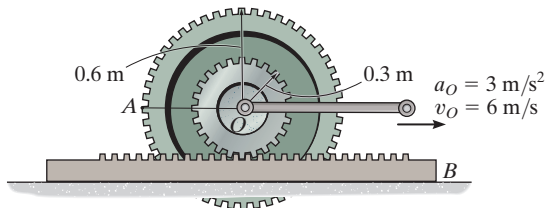
Prob. F16-19

F16-20. The gear rolls on the fixed rack with an angular velocity of $\omega = 12 \text{ rad/s}$ and angular acceleration of $\alpha = 6 \text{ rad/s}^2$. Determine the acceleration of point A.



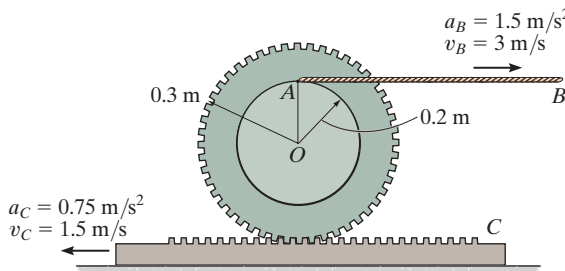
Prob. F16-20

F16-21. The gear rolls on the fixed rack B. At the instant shown, the center O of the gear moves with a velocity of $v_O = 6 \text{ m/s}$ and acceleration of $a_O = 3 \text{ m/s}^2$. Determine the angular acceleration of the gear and acceleration of point A at this instant.



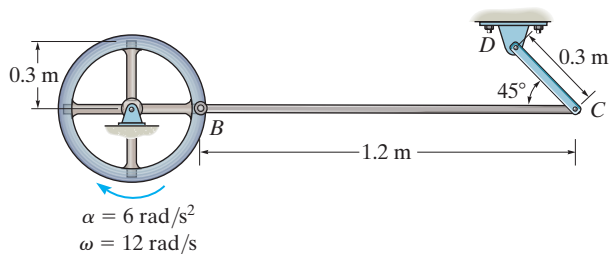
Prob. F16-21

F16-22. At the instant shown, cable AB has a velocity of 3 m/s and acceleration of 1.5 m/s^2 , while the gear rack has a velocity of 1.5 m/s and acceleration of 0.75 m/s^2 . Determine the angular acceleration of the gear at this instant.



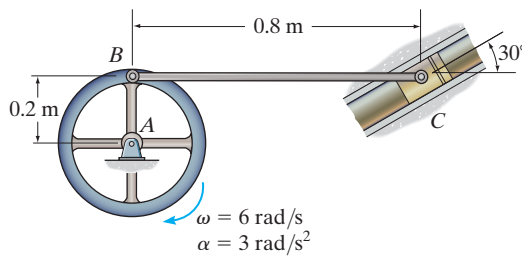
Prob. F16-22

F16-23. At the instant shown, the wheel rotates with an angular velocity of $\omega = 12 \text{ rad/s}$ and an angular acceleration of $\alpha = 6 \text{ rad/s}^2$. Determine the angular acceleration of link BC at the instant shown.



Prob. F16-23

F16-24. At the instant shown, wheel A rotates with an angular velocity of $\omega = 6 \text{ rad/s}$ and an angular acceleration of $\alpha = 3 \text{ rad/s}^2$. Determine the angular acceleration of link BC and the acceleration of piston C.

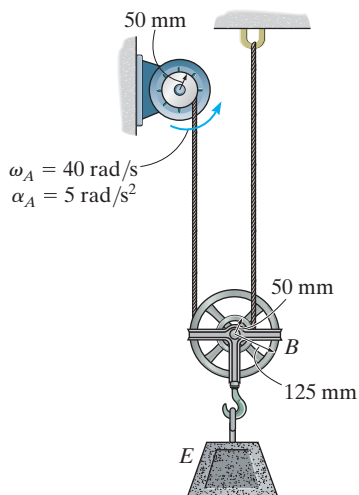


Prob. F16-24

PROBLEMS

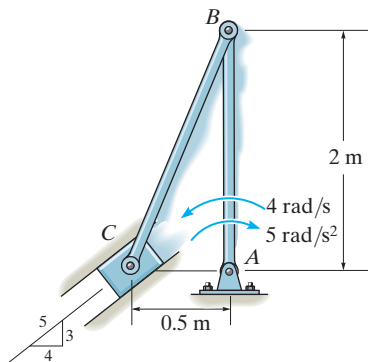
16–103. Pulley A rotates with the angular velocity and angular acceleration shown. Determine the angular acceleration of pulley B at the instant shown.

***16–104.** Pulley A rotates with the angular velocity and angular acceleration shown. Determine the acceleration of block E at the instant shown.



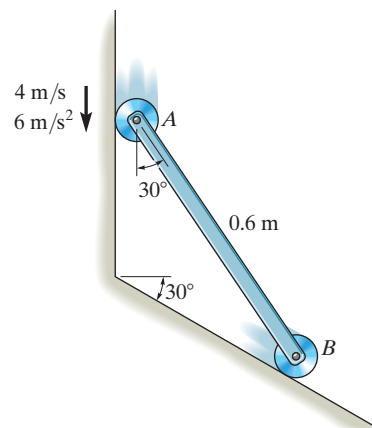
Probs. 16–103/104

16–105. Member AB has the angular motions shown. Determine the velocity and acceleration of the slider block C at this instant.



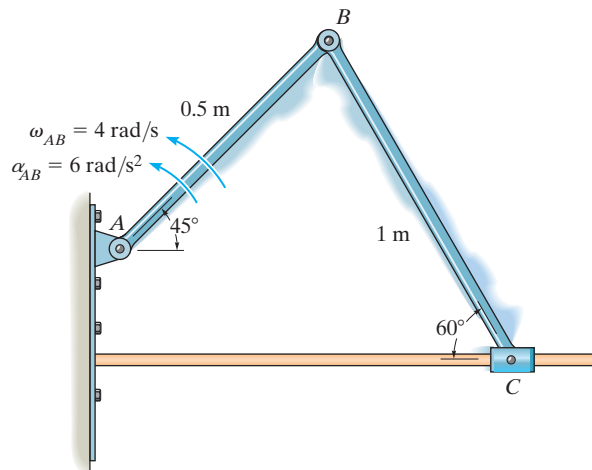
Prob. 16–105

16–106. At a given instant the roller A on the bar has the velocity and acceleration shown. Determine the velocity and acceleration of the roller B , and the bar's angular velocity and angular acceleration at this instant.



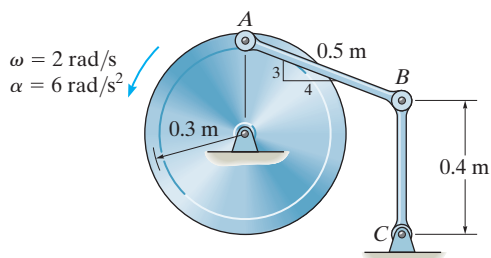
Prob. 16–106

16–107. Bar AB has the angular motions shown. Determine the velocity and acceleration of the slider block C at this instant.



Prob. 16–107

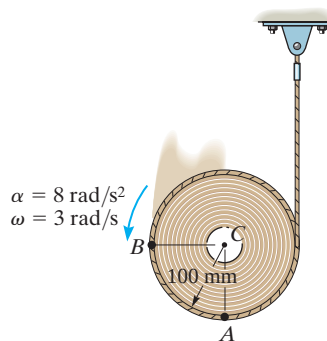
***16–108.** The flywheel rotates with angular velocity $\omega = 2 \text{ rad/s}$ and angular acceleration $\alpha = 6 \text{ rad/s}^2$. Determine the angular acceleration of links AB and BC at this instant.



Prob. 16–108

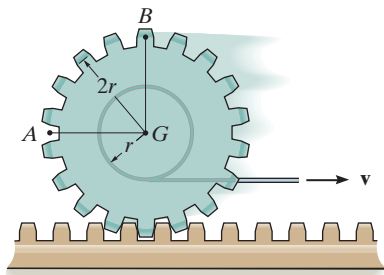
16–110. The reel of rope has the angular motion shown. Determine the velocity and acceleration of point A at the instant shown.

16–111. The reel of rope has the angular motion shown. Determine the velocity and acceleration of point B at the instant shown.



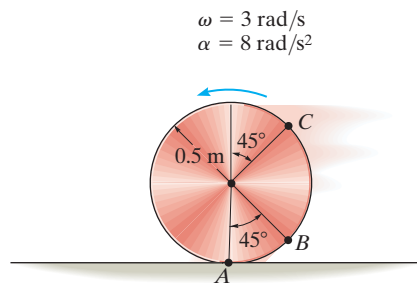
Probs. 16–110/111

16–109. A cord is wrapped around the inner spool of the gear. If it is pulled with a constant velocity $\mathbf{v}$, determine the velocities and accelerations of points A and B . The gear rolls on the fixed gear rack.



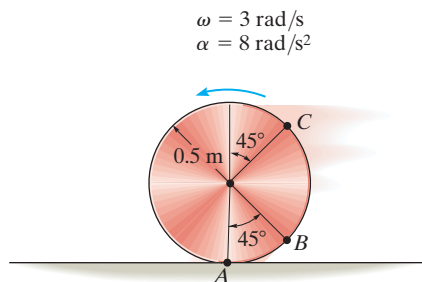
Prob. 16–109

***16–112.** The disk has an angular acceleration $\alpha = 8 \text{ rad/s}^2$ and angular velocity $\omega = 3 \text{ rad/s}$ at the instant shown. If it does not slip at A , determine the acceleration of point B .



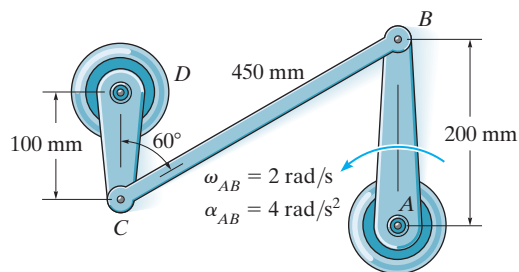
Prob. 16–112

16–113. The disk has an angular acceleration $\alpha = 8 \text{ rad/s}^2$ and angular velocity $\omega = 3 \text{ rad/s}$ at the instant shown. If it does not slip at A , determine the acceleration of point C .



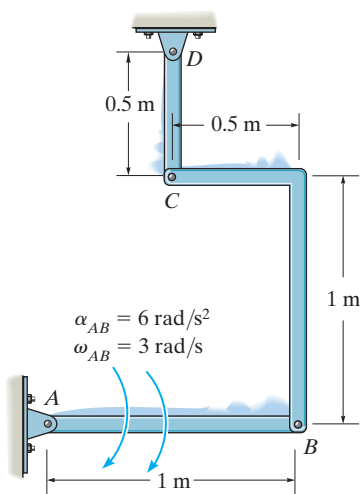
Prob. 16–113

16–114. Member AB has the angular motions shown. Determine the angular velocity and angular acceleration of members CB and DC .



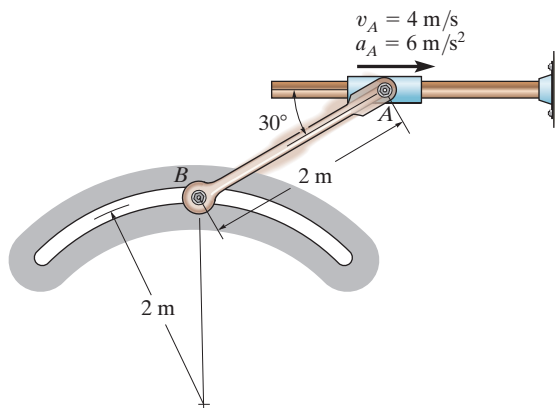
Prob. 16–114

16–115. Determine the angular acceleration of link CD if link AB has the angular velocity and angular acceleration shown.



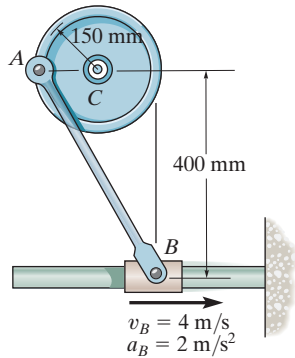
Prob. 16–115

***16–116.** At a given instant the slider block A is moving to the right with the motion shown. Determine the angular acceleration of link AB and the acceleration of point B at this instant.



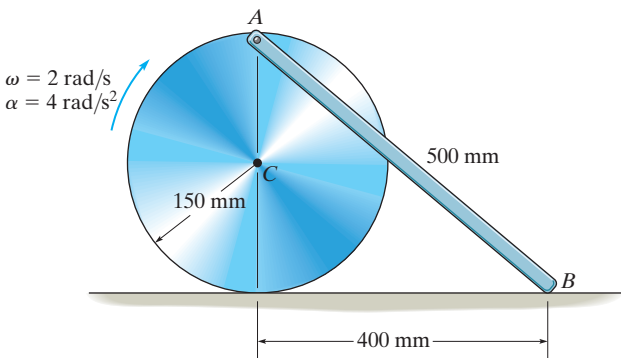
Prob. 16–116

16-117. The slider block has the motion shown. Determine the angular velocity and angular acceleration of the wheel at this instant.



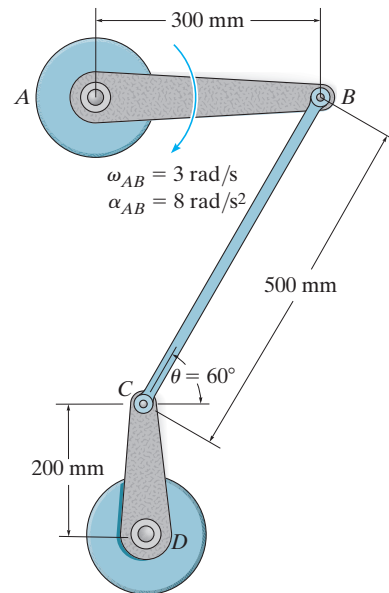
Prob. 16-117

16-118. The disk rolls without slipping such that it has an angular acceleration of $\alpha = 4 \text{ rad/s}^2$ and angular velocity of $\omega = 2 \text{ rad/s}$ at the instant shown. Determine the acceleration of points A and B on the link and the link's angular acceleration at this instant. Assume point A lies on the periphery of the disk, 150 mm from C.



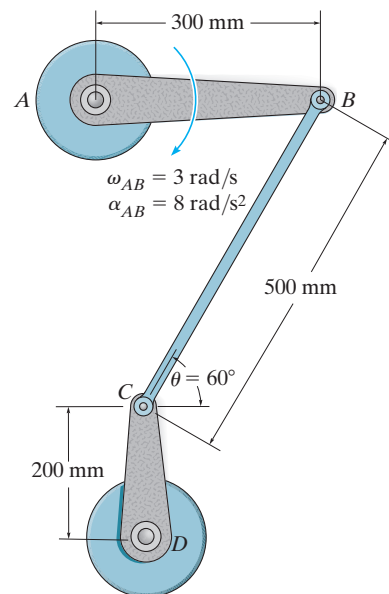
Prob. 16-118

16-119. If member AB has the angular motion shown, determine the angular velocity and angular acceleration of member CD at the instant shown.



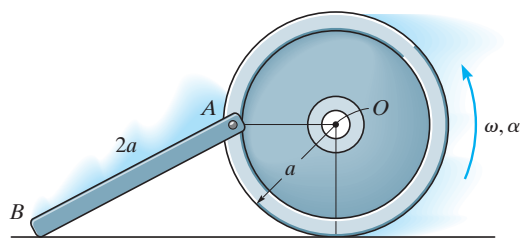
Prob. 16-119

***16-120.** If member AB has the angular motion shown, determine the velocity and acceleration of point C at the instant shown.



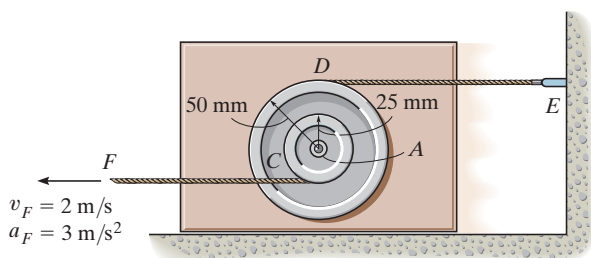
Prob. 16-120

16-121. The wheel rolls without slipping such that at the instant shown it has an angular velocity ω and angular acceleration α . Determine the velocity and acceleration of point B on the rod at this instant.



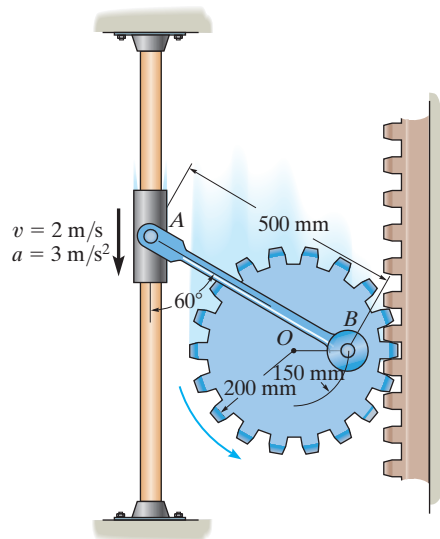
Prob. 16-121

16-122. A single pulley having both an inner and outer rim is pin connected to the block at A . As cord CF unwinds from the inner rim of the pulley with the motion shown, cord DE unwinds from the outer rim. Determine the angular acceleration of the pulley and the acceleration of the block at the instant shown.



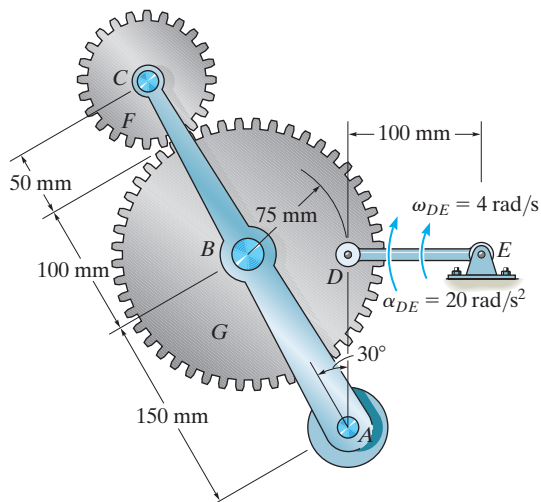
Prob. 16-122

16-123. The collar is moving downward with the motion shown. Determine the angular velocity and angular acceleration of the gear at the instant shown as it rolls along the fixed gear rack.



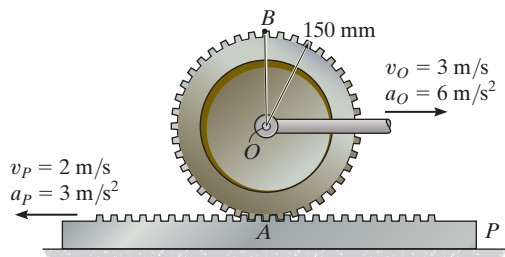
Prob. 16-123

***16-124.** The tied crank and gear mechanism gives rocking motion to crank AC , necessary for the operation of a printing press. If link DE has the angular motion shown, determine the respective angular velocities of gear F and crank AC at this instant, and the angular acceleration of crank AC .



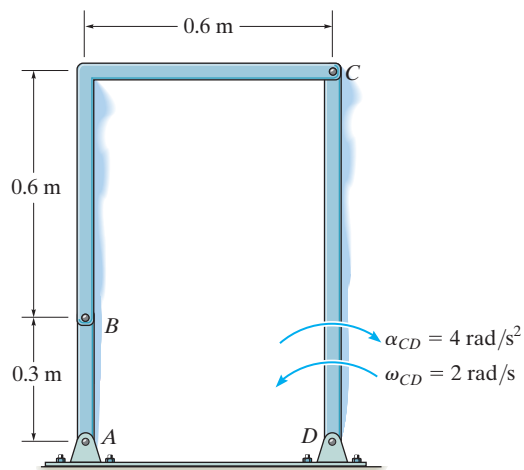
Prob. 16-124

16-125. The center O of the gear and the gear rack P move with the velocities and accelerations shown. Determine the angular acceleration of the gear and the acceleration of point B located at the rim of the gear at the instant shown.



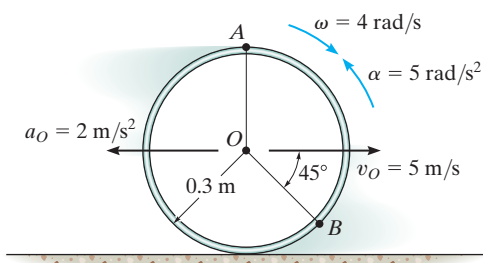
Prob. 16-125

16-127. Determine the angular acceleration of link AB if link CD has the angular velocity and angular deceleration shown.



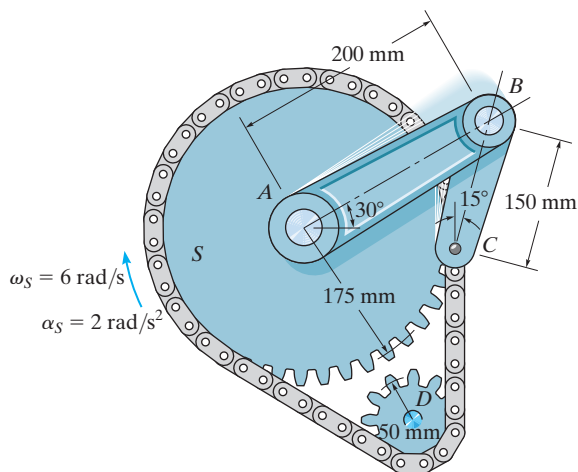
Prob. 16-127

16-126. The hoop is cast on the rough surface such that it has angular velocity $\omega = 4 \text{ rad/s}$ and an angular acceleration $\alpha = 5 \text{ rad/s}^2$. Also, its center has a velocity $v_O = 5 \text{ m/s}$ and a deceleration $a_O = 2 \text{ m/s}^2$. Determine the acceleration of point A at this instant.



Prob. 16-126

***16-128.** The mechanism produces intermittent motion of link AB . If the sprocket S is turning with an angular acceleration $\alpha_S = 2 \text{ rad/s}^2$ and has an angular velocity $\omega_S = 6 \text{ rad/s}$ at the instant shown, determine the angular velocity and angular acceleration of link AB at this instant. The sprocket S is mounted on a shaft which is *separate* from a collinear shaft attached to AB at A . The pin at C is attached to one of the chain links such that it moves vertically downward.



Prob. 16-128

16.8 Relative-Motion Analysis Using Rotating Axes

In the previous sections the relative-motion analysis for velocity and acceleration was described using a translating coordinate system. This type of analysis is useful for determining the motion of points on the *same* rigid body, or the motion of points located on several pin-connected bodies. In some problems, however, rigid bodies (mechanisms) are constructed such that *sliding* will occur at their connections. The kinematic analysis for such cases is best performed if the motion is analyzed using a coordinate system which both *translates* and *rotates*. Furthermore, this frame of reference is useful for analyzing the motions of two points on a mechanism which are *not* located in the *same* body and for specifying the kinematics of particle motion when the particle moves along a rotating path.

In the following analysis two equations will be developed which relate the velocity and acceleration of two points, one of which is the origin of a moving frame of reference subjected to both a translation and a rotation in the plane.*

Position. Consider the two points A and B shown in Fig. 16–32a. Their location is specified by the position vectors $\mathbf{r}_A$ and $\mathbf{r}_B$, which are measured with respect to the fixed X, Y, Z coordinate system. As shown in the figure, the “base point” A represents the origin of the x, y, z coordinate system, which is assumed to be both translating and rotating with respect to the X, Y, Z system. The position of B with respect to A is specified by the relative-position vector $\mathbf{r}_{B/A}$. The components of this vector may be expressed either in terms of unit vectors along the X, Y axes, i.e., $\mathbf{I}$ and $\mathbf{J}$, or by unit vectors along the x, y axes, i.e., $\mathbf{i}$ and $\mathbf{j}$. For the development which follows, $\mathbf{r}_{B/A}$ will be measured with respect to the moving x, y frame of reference. Thus, if B has coordinates (x_B, y_B) , Fig. 16–32a, then

$$\mathbf{r}_{B/A} = x_B \mathbf{i} + y_B \mathbf{j}$$

Using vector addition, the three position vectors in Fig. 16–32a are related by the equation

$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A} \quad (16-21)$$

At the instant considered, point A has a velocity $\mathbf{v}_A$ and an acceleration $\mathbf{a}_A$, while the angular velocity and angular acceleration of the x, y axes are Ω (omega) and $\dot{\Omega} = d\Omega/dt$, respectively.

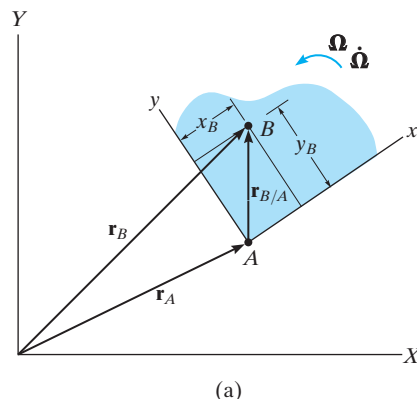


Fig. 16–32

*The more general, three-dimensional motion of the points is developed in Sec. 20.4.

Velocity. The velocity of point B is determined by taking the time derivative of Eq. 16–21, which yields

$$\mathbf{v}_B = \mathbf{v}_A + \frac{d\mathbf{r}_{B/A}}{dt} \quad (16-22)$$

The last term in this equation is evaluated as follows:

$$\begin{aligned} \frac{d\mathbf{r}_{B/A}}{dt} &= \frac{d}{dt}(x_B\mathbf{i} + y_B\mathbf{j}) \\ &= \frac{dx_B}{dt}\mathbf{i} + x_B\frac{d\mathbf{i}}{dt} + \frac{dy_B}{dt}\mathbf{j} + y_B\frac{d\mathbf{j}}{dt} \\ &= \left(\frac{dx_B}{dt}\mathbf{i} + \frac{dy_B}{dt}\mathbf{j}\right) + \left(x_B\frac{d\mathbf{i}}{dt} + y_B\frac{d\mathbf{j}}{dt}\right) \end{aligned} \quad (16-23)$$

The two terms in the first set of parentheses represent the components of velocity of point B as measured by an observer attached to the moving x, y, z coordinate system. These terms will be denoted by vector $(\mathbf{v}_{B/A})_{xyz}$. In the second set of parentheses the instantaneous time rate of change of the unit vectors $\mathbf{i}$ and $\mathbf{j}$ is measured by an observer located in the fixed X, Y, Z coordinate system. These changes, $d\mathbf{i}$ and $d\mathbf{j}$, are due *only* to the rotation $d\theta$ of the x, y, z axes, causing $\mathbf{i}$ to become $\mathbf{i}' = \mathbf{i} + d\mathbf{i}$ and $\mathbf{j}$ to become $\mathbf{j}' = \mathbf{j} + d\mathbf{j}$, Fig. 16–32*b*. As shown, the *magnitudes* of both $d\mathbf{i}$ and $d\mathbf{j}$ equal $1 d\theta$, since $i = i' = j = j' = 1$. The *direction* of $d\mathbf{i}$ is defined by $+\mathbf{j}$, since $d\mathbf{i}$ is tangent to the path described by the arrowhead of $\mathbf{i}$ in the limit as $\Delta t \rightarrow dt$. Likewise, $d\mathbf{j}$ acts in the $-\mathbf{i}$ direction, Fig. 16–32*b*. Hence,

$$\frac{d\mathbf{i}}{dt} = \frac{d\theta}{dt}(\mathbf{j}) = \Omega\mathbf{j} \quad \frac{d\mathbf{j}}{dt} = \frac{d\theta}{dt}(-\mathbf{i}) = -\Omega\mathbf{i}$$

Viewing the axes in three dimensions, Fig. 16–32*c*, and noting that $\Omega = \Omega\mathbf{k}$, we can express the above derivatives in terms of the cross product as

$$\frac{d\mathbf{i}}{dt} = \Omega \times \mathbf{i} \quad \frac{d\mathbf{j}}{dt} = \Omega \times \mathbf{j} \quad (16-24)$$

Substituting these results into Eq. 16–23 and using the distributive property of the vector cross product, we obtain

$$\frac{d\mathbf{r}_{B/A}}{dt} = (\mathbf{v}_{B/A})_{xyz} + \Omega \times (x_B\mathbf{i} + y_B\mathbf{j}) = (\mathbf{v}_{B/A})_{xyz} + \Omega \times \mathbf{r}_{B/A} \quad (16-25)$$

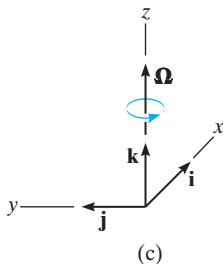
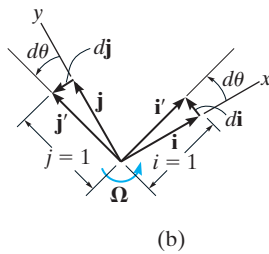


Fig. 16–32 (cont.)

Hence, Eq. 16–22 becomes

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\Omega} \times \mathbf{r}_{B/A} + (\mathbf{v}_{B/A})_{xyz} \quad (16-26)$$

where

$\mathbf{v}_B$ = velocity of B , measured from the X, Y, Z reference

$\mathbf{v}_A$ = velocity of the origin A of the x, y, z reference, measured from the X, Y, Z reference

$(\mathbf{v}_{B/A})_{xyz}$ = velocity of “ B with respect to A ,” as measured by an observer attached to the rotating x, y, z reference

$\boldsymbol{\Omega}$ = angular velocity of the x, y, z reference, measured from the X, Y, Z reference

$\mathbf{r}_{B/A}$ = position of B with respect to A

Comparing Eq. 16–26 with Eq. 16–16 ($\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\Omega} \times \mathbf{r}_{B/A}$), which is valid for a translating frame of reference, it can be seen that the only difference between these two equations is represented by the term $(\mathbf{v}_{B/A})_{xyz}$.

When applying Eq. 16–26 it is often useful to understand what each of the terms represents. In order of appearance, they are as follows:

$$\mathbf{v}_B \quad \left\{ \begin{array}{l} \text{absolute velocity of } B \\ \text{motion of } B \text{ observed} \\ \text{from the } X, Y, Z \text{ frame} \end{array} \right.$$

(equals)

$$\begin{array}{l} \mathbf{v}_A \quad \left\{ \begin{array}{l} \text{absolute velocity of the} \\ \text{origin of } x, y, z \text{ frame} \end{array} \right. \\ \quad \quad \quad \text{(plus)} \\ \boldsymbol{\Omega} \times \mathbf{r}_{B/A} \quad \left\{ \begin{array}{l} \text{angular velocity effect caused} \\ \text{by rotation of } x, y, z \text{ frame} \end{array} \right. \\ \quad \quad \quad \text{(plus)} \end{array} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{array}{l} \text{motion of } x, y, z \text{ frame} \\ \text{observed from the} \\ X, Y, Z \text{ frame} \end{array}$$

$$(\mathbf{v}_{B/A})_{xyz} \quad \left\{ \begin{array}{l} \text{velocity of } B \\ \text{with respect to } A \end{array} \right. \quad \left. \begin{array}{l} \\ \end{array} \right\} \begin{array}{l} \text{motion of } B \text{ observed} \\ \text{from the } x, y, z \text{ frame} \end{array}$$

Acceleration. The acceleration of B , observed from the X, Y, Z coordinate system, may be expressed in terms of its motion measured with respect to the rotating system of coordinates by taking the time derivative of Eq. 16–26.

$$\begin{aligned}\frac{d\mathbf{v}_B}{dt} &= \frac{d\mathbf{v}_A}{dt} + \frac{d\mathbf{\Omega}}{dt} \times \mathbf{r}_{B/A} + \mathbf{\Omega} \times \frac{d\mathbf{r}_{B/A}}{dt} + \frac{d(\mathbf{v}_{B/A})_{xyz}}{dt} \\ \mathbf{a}_B &= \mathbf{a}_A + \dot{\mathbf{\Omega}} \times \mathbf{r}_{B/A} + \mathbf{\Omega} \times \frac{d\mathbf{r}_{B/A}}{dt} + \frac{d(\mathbf{v}_{B/A})_{xyz}}{dt}\end{aligned}\quad (16-27)$$

Here $\dot{\mathbf{\Omega}} = d\mathbf{\Omega}/dt$ is the angular acceleration of the x, y, z coordinate system. Since $\mathbf{\Omega}$ is always perpendicular to the plane of motion, then $\dot{\mathbf{\Omega}}$ measures *only the change in magnitude* of $\mathbf{\Omega}$. The derivative $d\mathbf{r}_{B/A}/dt$ is defined by Eq. 16–25, so that

$$\mathbf{\Omega} \times \frac{d\mathbf{r}_{B/A}}{dt} = \mathbf{\Omega} \times (\mathbf{v}_{B/A})_{xyz} + \mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{r}_{B/A}) \quad (16-28)$$

Finding the time derivative of $(\mathbf{v}_{B/A})_{xyz} = (v_{B/A})_x \mathbf{i} + (v_{B/A})_y \mathbf{j}$,

$$\frac{d(\mathbf{v}_{B/A})_{xyz}}{dt} = \left[\frac{d(v_{B/A})_x}{dt} \mathbf{i} + \frac{d(v_{B/A})_y}{dt} \mathbf{j} \right] + \left[(v_{B/A})_x \frac{d\mathbf{i}}{dt} + (v_{B/A})_y \frac{d\mathbf{j}}{dt} \right]$$

The two terms in the first set of brackets represent the components of acceleration of point B as measured by an observer attached to the rotating coordinate system. These terms will be denoted by $(\mathbf{a}_{B/A})_{xyz}$. The terms in the second set of brackets can be simplified using Eqs. 16–24.

$$\frac{d(\mathbf{v}_{B/A})_{xyz}}{dt} = (\mathbf{a}_{B/A})_{xyz} + \mathbf{\Omega} \times (\mathbf{v}_{B/A})_{xyz}$$

Substituting this and Eq. 16–28 into Eq. 16–27 and rearranging terms,

$$\mathbf{a}_B = \mathbf{a}_A + \dot{\mathbf{\Omega}} \times \mathbf{r}_{B/A} + \mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{r}_{B/A}) + 2\mathbf{\Omega} \times (\mathbf{v}_{B/A})_{xyz} + (\mathbf{a}_{B/A})_{xyz} \quad (16-29)$$

where

$\mathbf{a}_B$ = acceleration of B , measured from the X, Y, Z reference

$\mathbf{a}_A$ = acceleration of the origin A of the x, y, z reference, measured from the X, Y, Z reference

$(\mathbf{a}_{B/A})_{xyz}, (\mathbf{v}_{B/A})_{xyz}$ = acceleration and velocity of B with respect to A , as measured by an observer attached to the rotating x, y, z reference

$\dot{\mathbf{\Omega}}, \mathbf{\Omega}$ = angular acceleration and angular velocity of the x, y, z reference, measured from the X, Y, Z reference

$\mathbf{r}_{B/A}$ = position of B with respect to A

If Eq. 16–29 is compared with Eq. 16–18, written in the form $\mathbf{a}_B = \mathbf{a}_A + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{B/A} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{B/A})$, which is valid for a translating frame of reference, it can be seen that the difference between these two equations is represented by the terms $2\boldsymbol{\Omega} \times (\mathbf{v}_{B/A})_{xyz}$ and $(\mathbf{a}_{B/A})_{xyz}$. In particular, $2\boldsymbol{\Omega} \times (\mathbf{v}_{B/A})_{xyz}$ is called the *Coriolis acceleration*, named after the French engineer G. C. Coriolis, who was the first to determine it. This term represents the difference in the acceleration of B as measured from nonrotating and rotating x, y, z axes. As indicated by the vector cross product, the Coriolis acceleration will *always* be perpendicular to both $\boldsymbol{\Omega}$ and $(\mathbf{v}_{B/A})_{xyz}$. It is an important component of the acceleration which must be considered whenever rotating reference frames are used. This often occurs, for example, when studying the accelerations and forces which act on rockets, long-range projectiles, or other bodies having motions whose measurements are significantly affected by the rotation of the earth.

The following interpretation of the terms in Eq. 16–29 may be useful when applying this equation to the solution of problems.

$\mathbf{a}_B$	$\left\{ \begin{array}{l} \text{absolute acceleration of } B \\ \text{(equals)} \end{array} \right.$	$\left. \begin{array}{l} \text{motion of } B \text{ observed} \\ \text{from the } X, Y, Z \text{ frame} \end{array} \right\}$
$\mathbf{a}_A$	$\left\{ \begin{array}{l} \text{absolute acceleration of the} \\ \text{origin of } x, y, z \text{ frame} \\ \text{(plus)} \end{array} \right.$	$\left. \begin{array}{l} \text{motion of} \\ x, y, z \text{ frame} \\ \text{observed from} \\ \text{the } X, Y, Z \text{ frame} \end{array} \right\}$
$\dot{\boldsymbol{\Omega}} \times \mathbf{r}_{B/A}$	$\left\{ \begin{array}{l} \text{angular acceleration effect} \\ \text{caused by rotation of } x, y, z \\ \text{frame} \\ \text{(plus)} \end{array} \right.$	
$\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{B/A})$	$\left\{ \begin{array}{l} \text{angular velocity effect caused} \\ \text{by rotation of } x, y, z \text{ frame} \\ \text{(plus)} \end{array} \right.$	
$2\boldsymbol{\Omega} \times (\mathbf{v}_{B/A})_{xyz}$	$\left\{ \begin{array}{l} \text{combined effect of } B \text{ moving} \\ \text{relative to } x, y, z \text{ coordinates} \\ \text{and rotation of } x, y, z \text{ frame} \\ \text{(plus)} \end{array} \right.$	
$(\mathbf{a}_{B/A})_{xyz}$	$\left\{ \begin{array}{l} \text{acceleration of } B \text{ with} \\ \text{respect to } A \end{array} \right.$	$\left. \begin{array}{l} \text{motion of } B \text{ observed} \\ \text{from the } x, y, z \text{ frame} \end{array} \right\}$

Procedure for Analysis

Equations 16–26 and 16–29 can be applied to the solution of problems involving the planar motion of particles or rigid bodies using the following procedure.

Coordinate Axes.

- Choose an appropriate location for the origin and proper orientation of the axes for both fixed X, Y, Z and moving x, y, z reference frames.
- Most often solutions are easily obtained if at the instant considered:
 1. the origins are coincident
 2. the corresponding axes are collinear
 3. the corresponding axes are parallel
- The moving frame should be selected fixed to the body or device along which the relative motion occurs.

Kinematic Equations.

- After defining the origin A of the moving reference and specifying the moving point B , Eqs. 16–26 and 16–29 should be written in symbolic form

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\Omega} \times \mathbf{r}_{B/A} + (\mathbf{v}_{B/A})_{xyz}$$

$$\mathbf{a}_B = \mathbf{a}_A + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{B/A} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{B/A}) + 2\boldsymbol{\Omega} \times (\mathbf{v}_{B/A})_{xyz} + (\mathbf{a}_{B/A})_{xyz}$$

- The Cartesian components of all these vectors may be expressed along either the X, Y, Z axes or the x, y, z axes. The choice is arbitrary provided a consistent set of unit vectors is used.
- Motion of the moving reference is expressed by $\mathbf{v}_A$, $\mathbf{a}_A$, $\boldsymbol{\Omega}$, and $\dot{\boldsymbol{\Omega}}$; and motion of B with respect to the moving reference is expressed by $\mathbf{r}_{B/A}$, $(\mathbf{v}_{B/A})_{xyz}$, and $(\mathbf{a}_{B/A})_{xyz}$.



The rotation of the dumping bin of the truck about point C is operated by the extension of the hydraulic cylinder AB . To determine the rotation of the bin due to this extension, we can use the equations of relative motion and fix the x, y axes to the cylinder so that the relative motion of the cylinder's extension occurs along the y axis.

EXAMPLE 16.18

At the instant $\theta = 60^\circ$, the rod in Fig. 16–33 has an angular velocity of 3 rad/s and an angular acceleration of 2 rad/s². At this same instant, collar C travels outward along the rod such that when $x = 0.2$ m the velocity is 2 m/s and the acceleration is 3 m/s², both measured relative to the rod. Determine the Coriolis acceleration and the velocity and acceleration of the collar at this instant.

SOLUTION

Coordinate Axes. The origin of both coordinate systems is located at point O , Fig. 16–33. Since motion of the collar is reported relative to the rod, the moving x, y, z frame of reference is *attached* to the rod.

Kinematic Equations.

$$\mathbf{v}_C = \mathbf{v}_O + \boldsymbol{\Omega} \times \mathbf{r}_{C/O} + (\mathbf{v}_{C/O})_{xyz} \quad (1)$$

$$\mathbf{a}_C = \mathbf{a}_O + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{C/O} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{C/O}) + 2\boldsymbol{\Omega} \times (\mathbf{v}_{C/O})_{xyz} + (\mathbf{a}_{C/O})_{xyz} \quad (2)$$

It will be simpler to express the data in terms of $\mathbf{i}, \mathbf{j}, \mathbf{k}$ component vectors rather than $\mathbf{I}, \mathbf{J}, \mathbf{K}$ components. Hence,

Motion of moving reference	Motion of C with respect to moving reference
$\mathbf{v}_O = \mathbf{0}$	$\mathbf{r}_{C/O} = \{0.2\mathbf{i}\} \text{ m}$
$\mathbf{a}_O = \mathbf{0}$	$(\mathbf{v}_{C/O})_{xyz} = \{2\mathbf{i}\} \text{ m/s}$
$\boldsymbol{\Omega} = \{-3\mathbf{k}\} \text{ rad/s}$	$(\mathbf{a}_{C/O})_{xyz} = \{3\mathbf{i}\} \text{ m/s}^2$
$\dot{\boldsymbol{\Omega}} = \{-2\mathbf{k}\} \text{ rad/s}^2$	

The Coriolis acceleration is defined as

$$\mathbf{a}_{\text{Cor}} = 2\boldsymbol{\Omega} \times (\mathbf{v}_{C/O})_{xyz} = 2(-3\mathbf{k}) \times (2\mathbf{i}) = \{-12\mathbf{j}\} \text{ m/s}^2 \quad \text{Ans.}$$

This vector is shown dashed in Fig. 16–33. If desired, it may be resolved into $\mathbf{I}, \mathbf{J}$ components acting along the X and Y axes, respectively.

The velocity and acceleration of the collar are determined by substituting the data into Eqs. 1 and 2 and evaluating the cross products, which yields

$$\begin{aligned} \mathbf{v}_C &= \mathbf{v}_O + \boldsymbol{\Omega} \times \mathbf{r}_{C/O} + (\mathbf{v}_{C/O})_{xyz} \\ &= \mathbf{0} + (-3\mathbf{k}) \times (0.2\mathbf{i}) + 2\mathbf{i} \\ &= \{2\mathbf{i} - 0.6\mathbf{j}\} \text{ m/s} \end{aligned}$$

Ans.

$$\begin{aligned} \mathbf{a}_C &= \mathbf{a}_O + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{C/O} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{C/O}) + 2\boldsymbol{\Omega} \times (\mathbf{v}_{C/O})_{xyz} + (\mathbf{a}_{C/O})_{xyz} \\ &= \mathbf{0} + (-2\mathbf{k}) \times (0.2\mathbf{i}) + (-3\mathbf{k}) \times [(-3\mathbf{k}) \times (0.2\mathbf{i})] + 2(-3\mathbf{k}) \times (2\mathbf{i}) + 3\mathbf{i} \\ &= \mathbf{0} - 0.4\mathbf{j} - 1.80\mathbf{i} - 12\mathbf{j} + 3\mathbf{i} \\ &= \{1.20\mathbf{i} - 12.4\mathbf{j}\} \text{ m/s}^2 \end{aligned}$$

Ans.

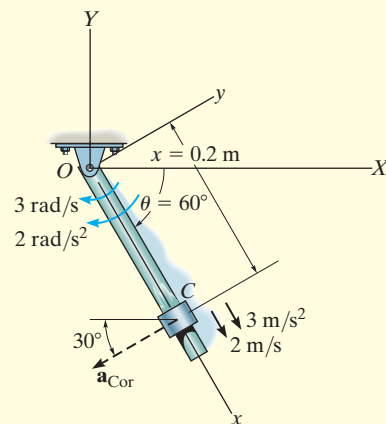


Fig. 16–33

EXAMPLE 16.19

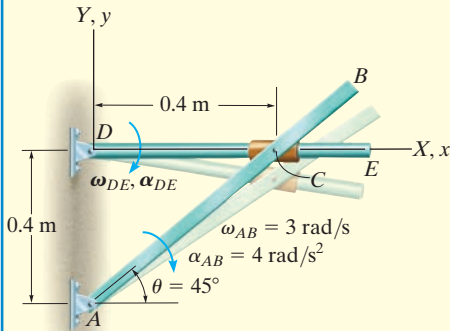


Fig. 16-34

Rod AB , shown in Fig. 16-34, rotates clockwise such that it has an angular velocity $\omega_{AB} = 3 \text{ rad/s}$ and angular acceleration $\alpha_{AB} = 4 \text{ rad/s}^2$ when $\theta = 45^\circ$. Determine the angular motion of rod DE at this instant. The collar at C is pin connected to AB and slides over rod DE .

SOLUTION

Coordinate Axes. The origin of both the fixed and moving frames of reference is located at D , Fig. 16-34. Furthermore, the x, y, z reference is attached to and rotates with rod DE so that the relative motion of the collar is easy to follow.

Kinematic Equations.

$$\mathbf{v}_C = \mathbf{v}_D + \boldsymbol{\Omega} \times \mathbf{r}_{C/D} + (\mathbf{v}_{C/D})_{xyz} \quad (1)$$

$$\mathbf{a}_C = \mathbf{a}_D + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{C/D} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{C/D}) + 2\boldsymbol{\Omega} \times (\mathbf{v}_{C/D})_{xyz} + (\mathbf{a}_{C/D})_{xyz} \quad (2)$$

All vectors will be expressed in terms of $\mathbf{i}, \mathbf{j}, \mathbf{k}$ components.

Motion of moving reference	Motion of C with respect to moving reference
$\mathbf{v}_D = \mathbf{0}$	$\mathbf{r}_{C/D} = \{0.4\mathbf{i}\} \text{ m}$
$\mathbf{a}_D = \mathbf{0}$	$(\mathbf{v}_{C/D})_{xyz} = (v_{C/D})_{xyz}\mathbf{i}$
$\boldsymbol{\Omega} = -\omega_{DE}\mathbf{k}$	$(\mathbf{a}_{C/D})_{xyz} = (a_{C/D})_{xyz}\mathbf{i}$
$\dot{\boldsymbol{\Omega}} = -\alpha_{DE}\mathbf{k}$	

Motion of C: Since the collar moves along a *circular path* of radius AC , its velocity and acceleration can be determined using Eqs. 16-9 and 16-14.

$$\mathbf{v}_C = \boldsymbol{\omega}_{AB} \times \mathbf{r}_{C/A} = (-3\mathbf{k}) \times (0.4\mathbf{i} + 0.4\mathbf{j}) = \{1.2\mathbf{i} - 1.2\mathbf{j}\} \text{ m/s}$$

$$\begin{aligned} \mathbf{a}_C &= \boldsymbol{\alpha}_{AB} \times \mathbf{r}_{C/A} - \omega_{AB}^2 \mathbf{r}_{C/A} \\ &= (-4\mathbf{k}) \times (0.4\mathbf{i} + 0.4\mathbf{j}) - (3)^2(0.4\mathbf{i} + 0.4\mathbf{j}) = \{-2\mathbf{i} - 5.2\mathbf{j}\} \text{ m/s}^2 \end{aligned}$$

Substituting the data into Eqs. 1 and 2, we have

$$\begin{aligned} \mathbf{v}_C &= \mathbf{v}_D + \boldsymbol{\Omega} \times \mathbf{r}_{C/D} + (\mathbf{v}_{C/D})_{xyz} \\ 1.2\mathbf{i} - 1.2\mathbf{j} &= \mathbf{0} + (-\omega_{DE}\mathbf{k}) \times (0.4\mathbf{i}) + (v_{C/D})_{xyz}\mathbf{i} \\ 1.2\mathbf{i} - 1.2\mathbf{j} &= \mathbf{0} - 0.4\omega_{DE}\mathbf{j} + (v_{C/D})_{xyz}\mathbf{i} \\ (v_{C/D})_{xyz} &= 1.2 \text{ m/s} \\ \omega_{DE} &= 3 \text{ rad/s} \quad \text{Ans.} \end{aligned}$$

$$\begin{aligned} \mathbf{a}_C &= \mathbf{a}_D + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{C/D} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{C/D}) + 2\boldsymbol{\Omega} \times (\mathbf{v}_{C/D})_{xyz} + (\mathbf{a}_{C/D})_{xyz} \\ -2\mathbf{i} - 5.2\mathbf{j} &= \mathbf{0} + (-\alpha_{DE}\mathbf{k}) \times (0.4\mathbf{i}) + (-3\mathbf{k}) \times [(-3\mathbf{k}) \times (0.4\mathbf{i})] \\ &\quad + 2(-3\mathbf{k}) \times (1.2\mathbf{i}) + (a_{C/D})_{xyz}\mathbf{i} \\ -2\mathbf{i} - 5.2\mathbf{j} &= -0.4\alpha_{DE}\mathbf{j} - 3.6\mathbf{i} - 7.2\mathbf{j} + (a_{C/D})_{xyz}\mathbf{i} \\ (a_{C/D})_{xyz} &= 1.6 \text{ m/s}^2 \\ \alpha_{DE} &= -5 \text{ rad/s}^2 = 5 \text{ rad/s}^2 \quad \text{Ans.} \end{aligned}$$

EXAMPLE 16.20

Planes *A* and *B* fly at the same elevation and have the motions shown in Fig. 16–35. Determine the velocity and acceleration of *A* as measured by the pilot of *B*.

SOLUTION

Coordinate Axes. Since the relative motion of *A* with respect to the pilot in *B* is being sought, the *x, y, z* axes are attached to plane *B*, Fig. 16–35. At the *instant* considered, the origin *B* coincides with the origin of the fixed *X, Y, Z* frame.

Kinematic Equations.

$$\mathbf{v}_A = \mathbf{v}_B + \boldsymbol{\Omega} \times \mathbf{r}_{A/B} + (\mathbf{v}_{A/B})_{xyz} \quad (1)$$

$$\mathbf{a}_A = \mathbf{a}_B + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{A/B} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{A/B}) + 2\boldsymbol{\Omega} \times (\mathbf{v}_{A/B})_{xyz} + (\mathbf{a}_{A/B})_{xyz} \quad (2)$$

Motion of Moving Reference:

$$\mathbf{v}_B = \{600\mathbf{j}\} \text{ km/h}$$

$$(a_B)_n = \frac{v_B^2}{\rho} = \frac{(600)^2}{400} = 900 \text{ km/h}^2$$

$$\mathbf{a}_B = (\mathbf{a}_B)_n + (\mathbf{a}_B)_t = \{900\mathbf{i} - 100\mathbf{j}\} \text{ km/h}^2$$

$$\boldsymbol{\Omega} = \frac{v_B}{\rho} = \frac{600 \text{ km/h}}{400 \text{ km}} = 1.5 \text{ rad/h} \curvearrowright \quad \boldsymbol{\Omega} = \{-1.5\mathbf{k}\} \text{ rad/h}$$

$$\dot{\boldsymbol{\Omega}} = \frac{(a_B)_t}{\rho} = \frac{100 \text{ km/h}^2}{400 \text{ km}} = 0.25 \text{ rad/h}^2 \curvearrowright \quad \dot{\boldsymbol{\Omega}} = \{0.25\mathbf{k}\} \text{ rad/h}^2$$

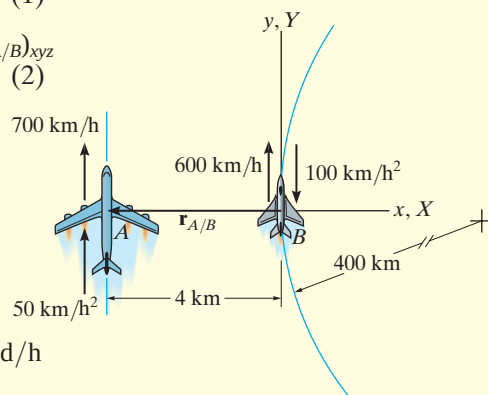


Fig. 16–35

Motion of A with Respect to Moving Reference:

$$\mathbf{r}_{A/B} = \{-4\mathbf{i}\} \text{ km} \quad (\mathbf{v}_{A/B})_{xyz} = ? \quad (\mathbf{a}_{A/B})_{xyz} = ?$$

Substituting the data into Eqs. 1 and 2, realizing that $\mathbf{v}_A = \{700\mathbf{j}\} \text{ km/h}$ and $\mathbf{a}_A = \{50\mathbf{j}\} \text{ km/h}^2$, we have

$$\mathbf{v}_A = \mathbf{v}_B + \boldsymbol{\Omega} \times \mathbf{r}_{A/B} + (\mathbf{v}_{A/B})_{xyz}$$

$$700\mathbf{j} = 600\mathbf{j} + (-1.5\mathbf{k}) \times (-4\mathbf{i}) + (\mathbf{v}_{A/B})_{xyz}$$

$$(\mathbf{v}_{A/B})_{xyz} = \{94\mathbf{j}\} \text{ km/h} \quad \text{Ans.}$$

$$\mathbf{a}_A = \mathbf{a}_B + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{A/B} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{A/B}) + 2\boldsymbol{\Omega} \times (\mathbf{v}_{A/B})_{xyz} + (\mathbf{a}_{A/B})_{xyz}$$

$$50\mathbf{j} = (900\mathbf{i} - 100\mathbf{j}) + (0.25\mathbf{k}) \times (-4\mathbf{i})$$

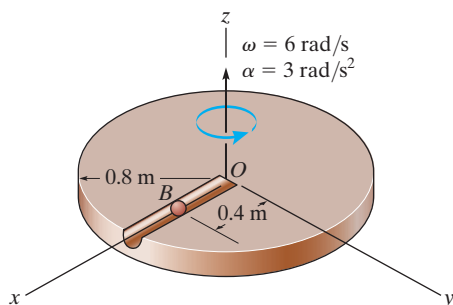
$$+ (-1.5\mathbf{k}) \times [(-1.5\mathbf{k}) \times (-4\mathbf{i})] + 2(-1.5\mathbf{k}) \times (94\mathbf{j}) + (\mathbf{a}_{A/B})_{xyz}$$

$$(\mathbf{a}_{A/B})_{xyz} = \{-119\mathbf{i} + 151\mathbf{j}\} \text{ km/h}^2 \quad \text{Ans.}$$

NOTE: The solution of this problem should be compared with that of Example 12.26, where it is seen that $(v_{B/A})_{xyz} \neq (v_{A/B})_{xyz}$ and $(a_{B/A})_{xyz} \neq (a_{A/B})_{xyz}$.

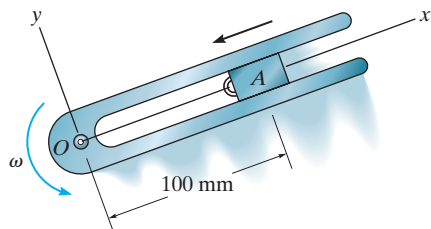
PROBLEMS

16-129. At the instant shown, ball B is rolling along the slot in the disk with a velocity of 600 mm/s and an acceleration of 150 mm/s², both measured relative to the disk and directed away from O . If at the same instant the disk has the angular velocity and angular acceleration shown, determine the velocity and acceleration of the ball at this instant.



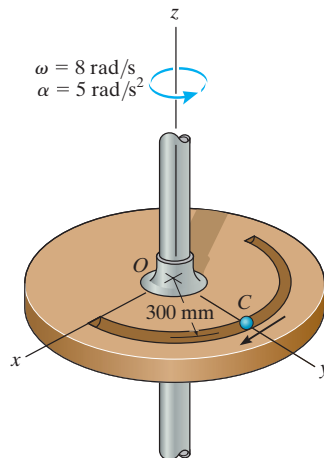
Prob. 16-129

16-130. Block A , which is attached to a cord, moves along the slot of a horizontal forked rod. At the instant shown, the cord is pulled down through the hole at O with an acceleration of 4 m/s² and its velocity is 2 m/s. Determine the acceleration of the block at this instant. The rod rotates about O with a constant angular velocity $\omega = 4$ rad/s.



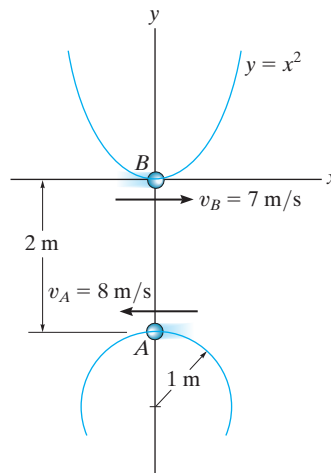
Prob. 16-130

16-131. Ball C moves with a speed of 3 m/s, which is increasing at a constant rate of 1.5 m/s², both measured relative to the circular plate and directed as shown. At the same instant the plate rotates with the angular velocity and angular acceleration shown. Determine the velocity and acceleration of the ball at this instant.



Prob. 16-131

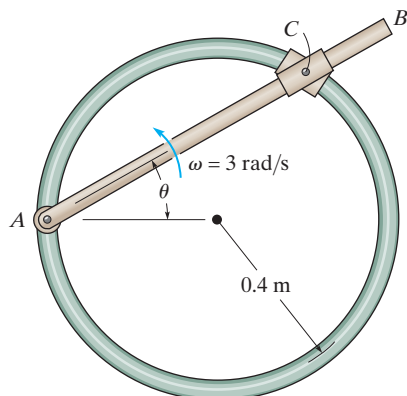
***16-132.** Particles B and A move along the parabolic and circular paths, respectively. If B has a velocity of 7 m/s in the direction shown and its speed is increasing at 4 m/s², while A has a velocity of 8 m/s in the direction shown and its speed is decreasing at 6 m/s², determine the relative velocity and relative acceleration of B with respect to A .



Prob. 16-132

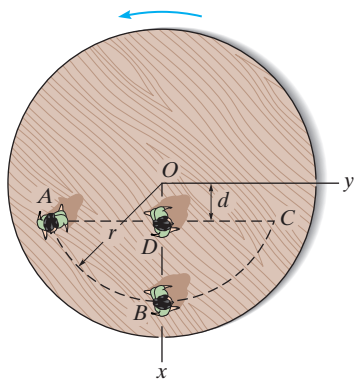
16–133. Rod AB rotates counterclockwise with a constant angular velocity $\omega = 3 \text{ rad/s}$. Determine the velocity of point C located on the double collar when $\theta = 30^\circ$. The collar consists of two pin-connected slider blocks which are constrained to move along the circular path and the rod AB .

16–134. Rod AB rotates counterclockwise with a constant angular velocity $\omega = 3 \text{ rad/s}$. Determine the velocity and acceleration of point C located on the double collar when $\theta = 45^\circ$. The collar consists of two pin-connected slider blocks which are constrained to move along the circular path and the rod AB .



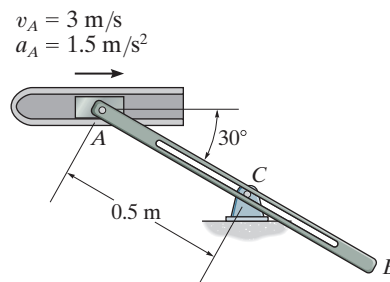
Probs. 16–133/134

16–135. A girl stands at A on a platform which is rotating with an angular acceleration $\alpha = 0.2 \text{ rad/s}^2$ and at the instant shown has an angular velocity $\omega = 0.5 \text{ rad/s}$. If she walks at a constant speed $v = 0.75 \text{ m/s}$ measured relative to the platform, determine her acceleration (a) when she reaches point D in going along the path ADC , $d = 1 \text{ m}$; and (b) when she reaches point B if she follows the path ABC , $r = 3 \text{ m}$.



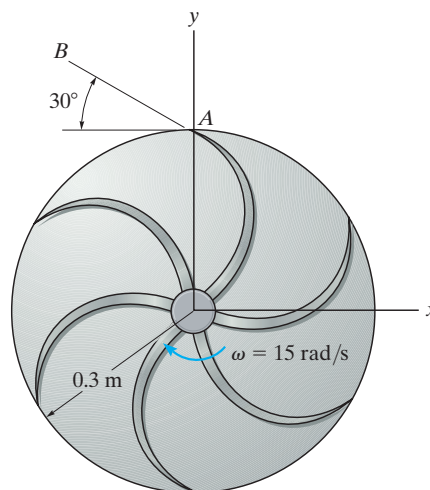
Prob. 16–135

***16–136.** If the piston is moving with a velocity $v_A = 3 \text{ m/s}$ and acceleration of $a_A = 1.5 \text{ m/s}^2$, determine the angular velocity and angular acceleration of the slotted link at the instant shown. Link AB slides freely along its slot on the fixed peg C .



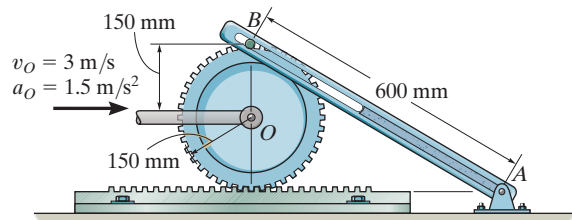
Prob. 16–136

16–137. Water leaves the impeller of the centrifugal pump with a velocity of 25 m/s and acceleration of 30 m/s^2 , both measured relative to the impeller along the blade line AB . Determine the velocity and acceleration of a water particle at A as it leaves the impeller at the instant shown. The impeller rotates with a constant angular velocity of $\omega = 15 \text{ rad/s}$.



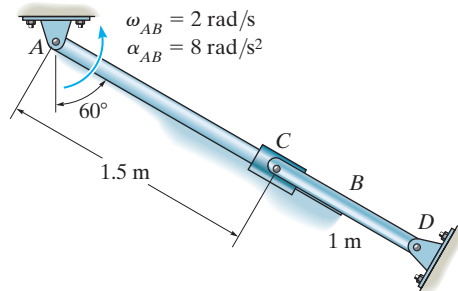
Prob. 16–137

16-138. Peg B on the gear slides freely along the slot in link AB . If the gear's center O moves with the velocity and acceleration shown, determine the angular velocity and angular acceleration of the link at this instant.



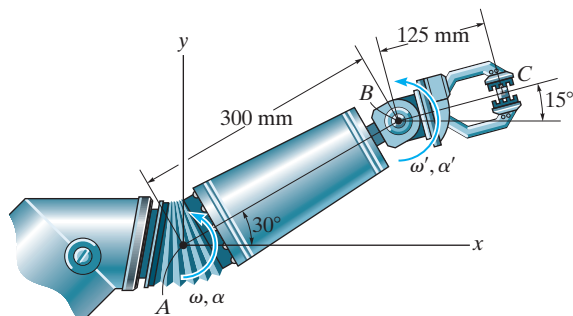
Prob. 16-138

16-139. The collar C is pinned to rod CD while it slides on rod AB . If rod AB has an angular velocity of 2 rad/s and an angular acceleration of 8 rad/s^2 , both acting counterclockwise, determine the angular velocity and the angular acceleration of rod CD at the instant shown.



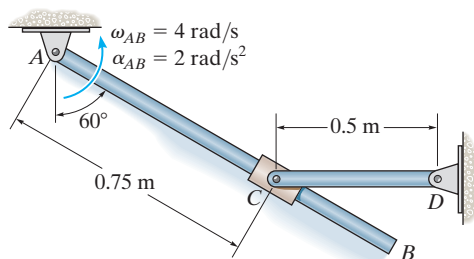
Prob. 16-139

***16-140.** At the instant shown, the robotic arm AB is rotating counterclockwise at $\omega = 5 \text{ rad/s}$ and has an angular acceleration $\alpha = 2 \text{ rad/s}^2$. Simultaneously, the grip BC is rotating counterclockwise at $\omega' = 6 \text{ rad/s}$ and $\alpha' = 2 \text{ rad/s}^2$, both measured relative to a fixed reference. Determine the velocity and acceleration of the object held at the grip C .



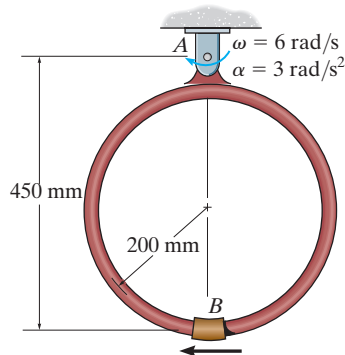
Prob. 16-140

16-141. At the instant shown rod AB has an angular velocity $\omega_{AB} = 4 \text{ rad/s}$ and an angular acceleration $\alpha_{AB} = 2 \text{ rad/s}^2$. Determine the angular velocity and angular acceleration of rod CD at this instant. The collar at C is pin connected to CD and slides freely along AB .



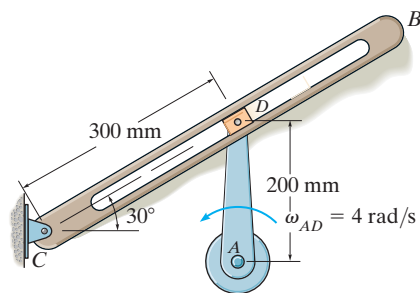
Prob. 16-141

16-142. Collar B moves to the left with a speed of 5 m/s , which is increasing at a constant rate of 1.5 m/s^2 , relative to the hoop, while the hoop rotates with the angular velocity and angular acceleration shown. Determine the magnitudes of the velocity and acceleration of the collar at this instant.



Prob. 16-142

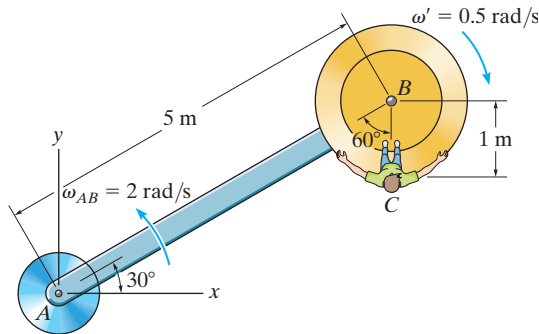
16-143. Block D of the mechanism is confined to move within the slot of member CB . If link AD is rotating at a constant rate of $\omega_{AD} = 4 \text{ rad/s}$, determine the angular velocity and angular acceleration of member CB at the instant shown.



Prob. 16-143

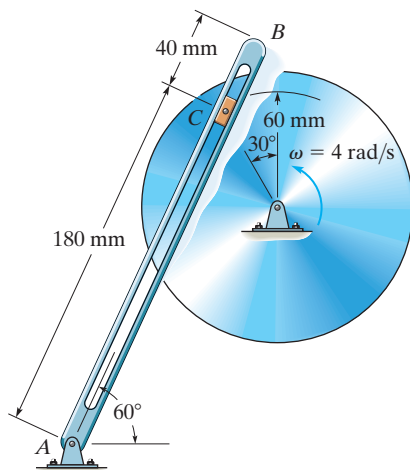
***16–144.** A ride in an amusement park consists of a rotating arm AB having a constant angular velocity $\omega_{AB} = 2 \text{ rad/s}$ about point A and a car mounted at the end of the arm which has a constant angular velocity $\omega' = \{-0.5\mathbf{k}\} \text{ rad/s}$, measured relative to the arm. At the instant shown, determine the velocity and acceleration of the passenger at C .

16–145. A ride in an amusement park consists of a rotating arm AB that has an angular acceleration of $\alpha_{AB} = 1 \text{ rad/s}^2$ when $\omega_{AB} = 2 \text{ rad/s}$ at the instant shown. Also at this instant the car mounted at the end of the arm has a relative angular acceleration of $\alpha' = \{-0.6\mathbf{k}\} \text{ rad/s}^2$ when $\omega' = \{-0.5\mathbf{k}\} \text{ rad/s}$. Determine the velocity and acceleration of the passenger C at this instant.



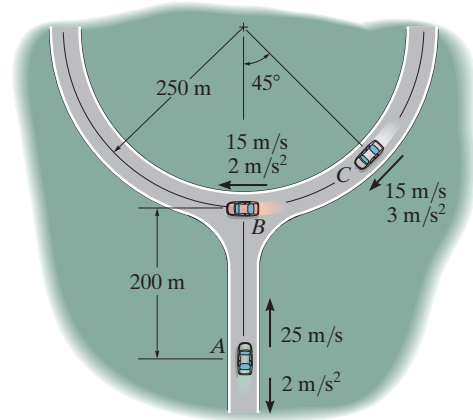
Probs. 16–144/145

16–146. If the slider block C is fixed to the disk that has a constant counterclockwise angular velocity of 4 rad/s , determine the angular velocity and angular acceleration of the slotted arm AB at the instant shown.



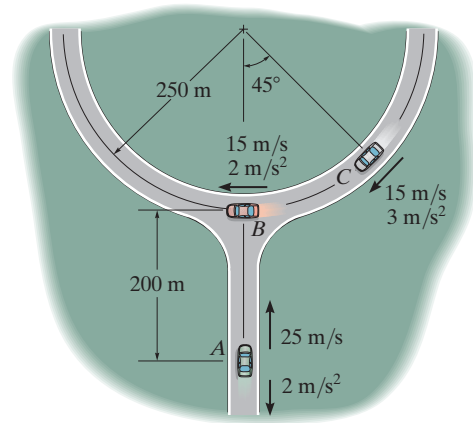
Prob. 16–146

16–147. At the instant shown, car A travels with a speed of 25 m/s , which is decreasing at a constant rate of 2 m/s^2 , while car C travels with a speed of 15 m/s , which is increasing at a constant rate of 3 m/s^2 . Determine the velocity and acceleration of car A with respect to car C .



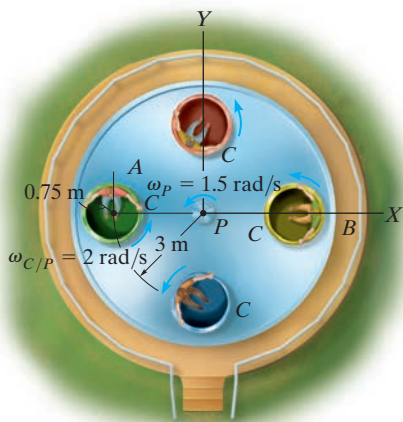
Prob. 16–147

***16–148.** At the instant shown, car B travels with a speed of 15 m/s , which is increasing at a constant rate of 2 m/s^2 , while car C travels with a speed of 15 m/s , which is increasing at a constant rate of 3 m/s^2 . Determine the velocity and acceleration of car B with respect to car C .



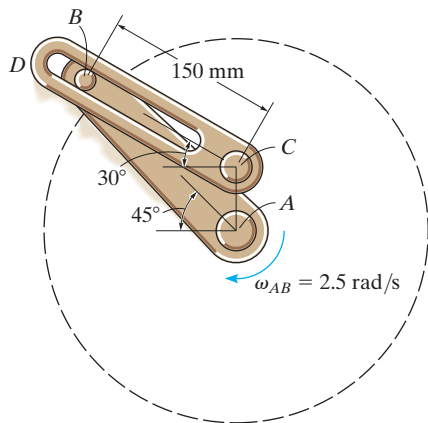
Prob. 16–148

16-149. A ride in an amusement park consists of a rotating platform P , having constant angular velocity $\omega_P = 1.5 \text{ rad/s}$, and four cars, C , mounted on the platform, which have constant angular velocities $\omega_{C/P} = 2 \text{ rad/s}$ measured relative to the platform. Determine the velocity and acceleration of the passenger at B at the instant shown.



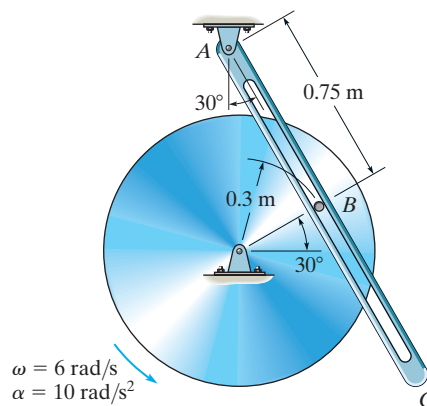
Prob. 16-149

16-150. The two-link mechanism serves to amplify angular motion. Link AB has a pin at B which is confined to move within the slot of link CD . If at the instant shown, AB (input) has an angular velocity of $\omega_{AB} = 2.5 \text{ rad/s}$, determine the angular velocity of CD (output) at this instant.



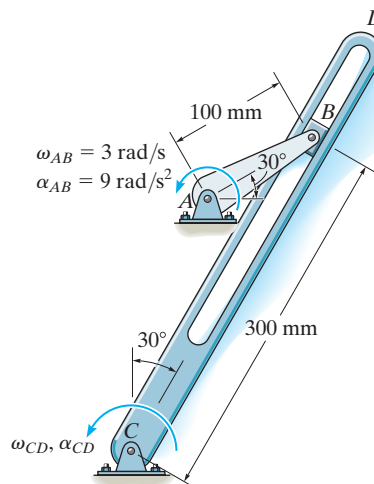
Prob. 16-150

16-151. The disk rotates with the angular motion shown. Determine the angular velocity and angular acceleration of the slotted link AC at this instant. The peg at B is fixed to the disk.



Prob. 16-151

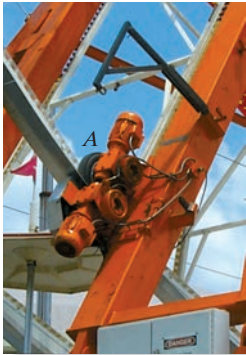
***16-152.** The “quick-return” mechanism consists of a crank AB , slider block B , and slotted link CD . If the crank has the angular motion shown, determine the angular motion of the slotted link at this instant.



Prob. 16-152

CONCEPTUAL PROBLEMS

C16-1. An electric motor turns the tire at A at a constant angular velocity, and friction then causes the tire to roll without slipping on the inside rim of the Ferris wheel. Using appropriate numerical values, determine the magnitude of the velocity and acceleration of passengers in one of the baskets. Do passengers in the other baskets experience this same motion? Explain.



Prob. C16-1

C16-2. The crank AB turns counterclockwise at a constant rate ω causing the connecting arm CD and rocking beam DE to move. Draw a sketch showing the location of the IC for the connecting arm when $\theta = 0^\circ, 90^\circ, 180^\circ$, and 270° . Also, how was the curvature of the head at E determined, and why is it curved in this way?



Prob. C16-2

C16-3. The bi-fold hangar door is opened by cables that move upward at a constant speed of 0.5 m/s. Determine the angular velocity of BC and the angular velocity of AB when $\theta = 45^\circ$. Panel BC is pinned at C and has a height which is the same as the height of BA . Use appropriate numerical values to explain your result.



Prob. C16-3

C16-4. If the tires do not slip on the pavement, determine the points on the tire that have a maximum and minimum speed and the points that have a maximum and minimum acceleration. Use appropriate numerical values for the car's speed and tire size to explain your result.



Prob. C16-4

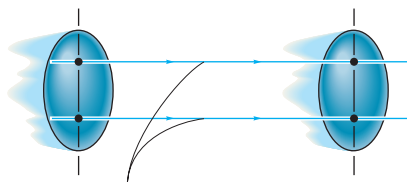
CHAPTER REVIEW

Rigid-Body Planar Motion

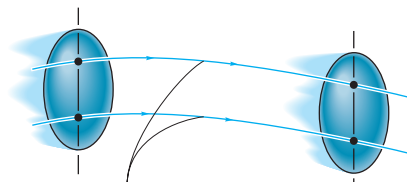
A rigid body undergoes three types of planar motion: translation, rotation about a fixed axis, and general plane motion.

Translation

When a body has rectilinear translation, all the particles of the body travel along parallel straight-line paths. If the paths have the same radius of curvature, then curvilinear translation occurs. Provided we know the motion of one of the particles, then the motion of all of the others is also known.



Path of rectilinear translation



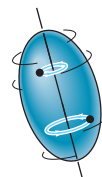
Path of curvilinear translation

Rotation about a Fixed Axis

For this type of motion, all of the particles move along circular paths. Here, all line segments in the body undergo the same angular displacement, angular velocity, and angular acceleration.

Once the angular motion of the body is known, then the velocity of any particle a distance r from the axis can be obtained.

The acceleration of any particle has two components. The tangential component accounts for the change in the magnitude of the velocity, and the normal component accounts for the change in the velocity's direction.



Rotation about a fixed axis

$$\omega = d\theta/dt$$

$$\omega = \omega_0 + \alpha_c t$$

$$\alpha = d\omega/dt \quad \text{or}$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha_c t^2$$

$$\alpha d\theta = \omega d\omega$$

$$\omega^2 = \omega_0^2 + 2\alpha_c(\theta - \theta_0)$$

Constant α_c

$$v = \omega r$$

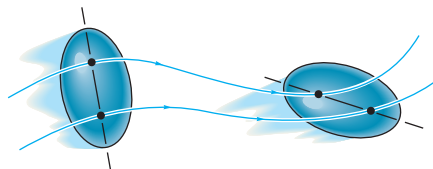
$$a_t = \alpha r, \quad a_n = \omega^2 r$$

General Plane Motion

When a body undergoes general plane motion, it simultaneously translates and rotates. There are several methods for analyzing this motion.

Absolute Motion Analysis

If the motion of a point on a body or the angular motion of a line is known, then it may be possible to relate this motion to that of another point or line using an absolute motion analysis. To do so, linear position coordinates s or angular position coordinates θ are established (measured from a fixed point or line). These position coordinates are then related using the geometry of the body. The time derivative of this equation gives the relationship between the velocities and/or the angular velocities. A second time derivative relates the accelerations and/or the angular accelerations.



General plane motion

Relative-Motion using Translating Axes

General plane motion can also be analyzed using a relative-motion analysis between two points A and B located on the body. This method considers the motion in parts: first a translation of the selected base point A , then a relative “rotation” of the body about point A , which is measured from a translating axis. Since the relative motion is viewed as circular motion about the base point, point B will have a velocity $\mathbf{v}_{B/A}$ that is tangent to the circle. It also has two components of acceleration, $(\mathbf{a}_{B/A})_t$ and $(\mathbf{a}_{B/A})_n$. It is also important to realize that $\mathbf{a}_A$ and $\mathbf{a}_B$ will have tangential and normal components if these points move along curved paths.

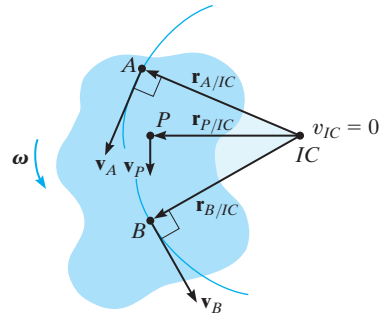
$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$$

$$\mathbf{a}_B = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{B/A} - \omega^2 \mathbf{r}_{B/A}$$

Instantaneous Center of Zero Velocity

If the base point A is selected as having zero velocity, then the relative velocity equation becomes $\mathbf{v}_B = \boldsymbol{\omega} \times \mathbf{r}_{B/A}$. In this case, motion appears as if the body rotates about an instantaneous axis passing through A .

The instantaneous center of rotation (IC) can be established provided the directions of the velocities of any two points on the body are known, or the velocity of a point and the angular velocity are known. Since a radial line r will always be perpendicular to each velocity, then the IC is at the point of intersection of these two radial lines. Its measured location is determined from the geometry of the body. Once it is established, then the velocity of any point P on the body can be determined from $v = \omega r$, where r extends from the IC to point P .

*Relative Motion using Rotating Axes*

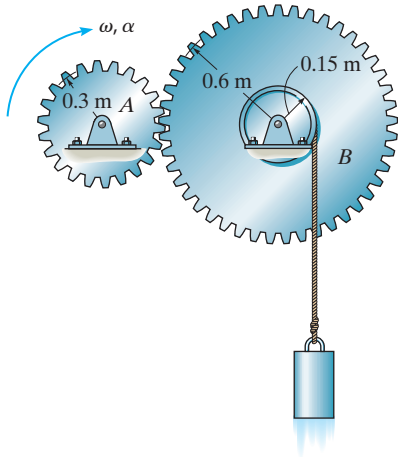
Problems that involve connected members that slide relative to one another or points not located on the same body can be analyzed using a relative-motion analysis referenced from a rotating frame. This gives rise to the term $2\boldsymbol{\Omega} \times (\mathbf{v}_{B/A})_{xyz}$ that is called the Coriolis acceleration.

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\Omega} \times \mathbf{r}_{B/A} + (\mathbf{v}_{B/A})_{xyz}$$

$$\mathbf{a}_B = \mathbf{a}_A + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{B/A} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{B/A}) + 2\boldsymbol{\Omega} \times (\mathbf{v}_{B/A})_{xyz} + (\mathbf{a}_{B/A})_{xyz}$$

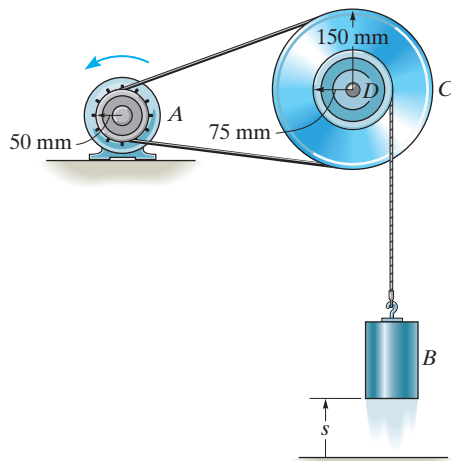
REVIEW PROBLEMS

R16-1. The hoisting gear A has an initial angular velocity of 60 rad/s and a constant deceleration of 1 rad/s^2 . Determine the velocity and deceleration of the block which is being hoisted by the hub on gear B when $t = 3 \text{ s}$.



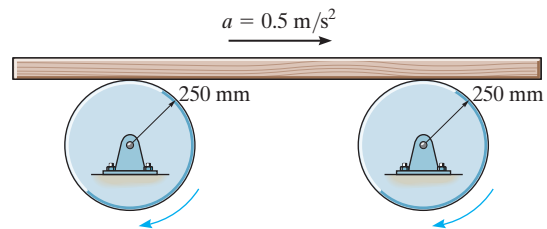
Prob. R16-1

R16-2. Starting at $(\omega_A)_0 = 3 \text{ rad/s}$, when $\theta = 0$, $s = 0$, pulley A is given an angular acceleration $\alpha = (0.6\theta) \text{ rad/s}^2$, where θ is in radians. Determine the speed of block B when it has risen $s = 0.5 \text{ m}$. The pulley has an inner hub D which is fixed to C and turns with it.



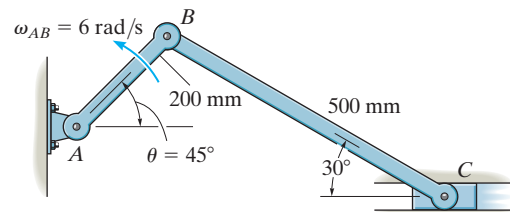
Prob. R16-2

R16-3. The board rests on the surface of two drums. At the instant shown, it has an acceleration of 0.5 m/s^2 to the right, while at the same instant, points on the outer rim of each drum have an acceleration with a magnitude of 3 m/s^2 . If the board does not slip on the drums, determine its speed due to the motion.



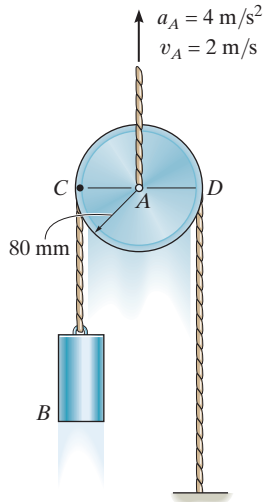
Prob. R16-3

R16-4. If bar AB has an angular velocity $\omega_{AB} = 6 \text{ rad/s}$, determine the velocity of the slider block C at the instant shown.



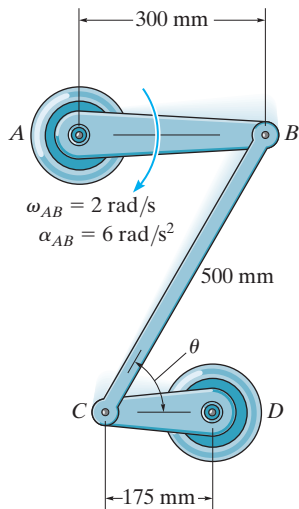
Prob. R16-4

R16-5. The center of the pulley is being lifted vertically with an acceleration of 4 m/s^2 at the instant it has a velocity of 2 m/s . If the cable does not slip on the pulley's surface, determine the accelerations of the cylinder B and point C on the pulley.



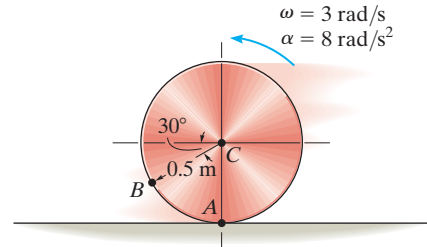
Prob. R16-5

R16-6. At the instant shown, link AB has an angular velocity $\omega_{AB} = 2 \text{ rad/s}$ and an angular acceleration $\alpha_{AB} = 6 \text{ rad/s}^2$. Determine the acceleration of the pin at C and the angular acceleration of link CB at this instant, when $\theta = 60^\circ$.



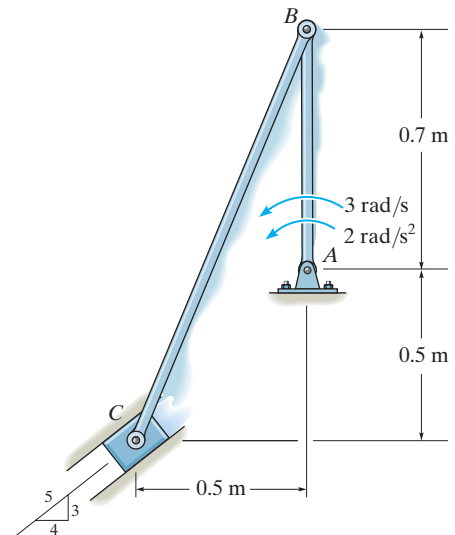
Prob. R16-6

R16-7. The disk is moving to the left such that it has an angular acceleration $\alpha = 8 \text{ rad/s}^2$ and angular velocity $\omega = 3 \text{ rad/s}$ at the instant shown. If it does not slip at A , determine the acceleration of point B .



Prob. R16-7

R16-8. At the given instant member AB has the angular motions shown. Determine the velocity and acceleration of the slider block C at this instant.



Prob. R16-8

Chapter 17



(© Surasaki/Fotolia)

Tractors and other heavy equipment can be subjected to severe loadings due to dynamic loadings as they accelerate. In this chapter we will show how to determine these loadings for planar motion.

Planar Kinetics of a Rigid Body: Force and Acceleration

CHAPTER OBJECTIVES

- To introduce the methods used to determine the mass moment of inertia of a body.
- To develop the planar kinetic equations of motion for a symmetric rigid body.
- To discuss applications of these equations to bodies undergoing translation, rotation about a fixed axis, and general plane motion.



Video Solutions are available for selected questions in this chapter.

17.1 Mass Moment of Inertia

Since a body has a definite size and shape, an applied nonconcurrent force system can cause the body to both translate and rotate. The translational aspects of the motion were studied in Chapter 13 and are governed by the equation $\mathbf{F} = m\mathbf{a}$. It will be shown in the next section that the rotational aspects, caused by a moment $\mathbf{M}$, are governed by an equation of the form $\mathbf{M} = I\boldsymbol{\alpha}$. The symbol I in this equation is termed the mass moment of inertia. By comparison, the *moment of inertia* is a measure of the resistance of a body to *angular acceleration* ($\mathbf{M} = I\boldsymbol{\alpha}$) in the same way that *mass* is a measure of the body's resistance to *acceleration* ($\mathbf{F} = m\mathbf{a}$).

The flywheel on the engine of this tractor has a large moment of inertia about its axis of rotation. Once it is set into motion, it will be difficult to stop, and this in turn will prevent the engine from stalling and instead will allow it to maintain a constant power.



We define the *moment of inertia* as the integral of the “second moment” about an axis of all the elements of mass dm which compose the body.* For example, the body’s moment of inertia about the z axis in Fig. 17–1 is

$$I = \int_m r^2 dm \quad (17-1)$$

Here the “moment arm” r is the perpendicular distance from the z axis to the arbitrary element dm . Since the formulation involves r , the value of I is different for each axis about which it is computed. In the study of planar kinetics, the axis chosen for analysis generally passes through the body’s mass center G and is always perpendicular to the plane of motion. The moment of inertia about this axis will be denoted as I_G . Since r is squared in Eq. 17–1, the mass moment of inertia is always a *positive* quantity. Common units used for its measurement are $\text{kg} \cdot \text{m}^2$.

If the body consists of material having a variable density, $\rho = \rho(x, y, z)$, the elemental mass dm of the body can be expressed in terms of its density and volume as $dm = \rho dV$. Substituting dm into Eq. 17–1, the body’s moment of inertia is then computed using *volume elements* for integration; i.e.,

$$I = \int_V r^2 \rho dV \quad (17-2)$$

*Another property of the body, which measures the symmetry of the body’s mass with respect to a coordinate system, is the product of inertia. This property applies to the three-dimensional motion of a body and will be discussed in Chapter 21.

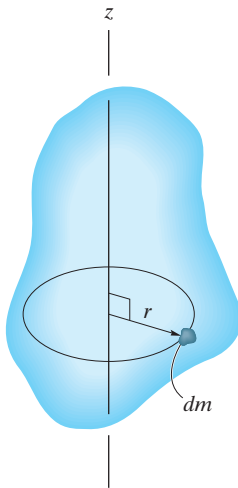


Fig. 17–1

In the special case of ρ being a *constant*, this term may be factored out of the integral, and the integration is then purely a function of geometry,

$$I = \rho \int_V r^2 dV \quad (17-3)$$

When the volume element chosen for integration has infinitesimal dimensions in all three directions, Fig. 17-2a, the moment of inertia of the body must be determined using “triple integration.” The integration process can, however, be simplified to a *single integration* provided the chosen volume element has a differential size or thickness in only *one direction*. Shell or disk elements are often used for this purpose.

Procedure for Analysis

To obtain the moment of inertia by integration, we will consider only symmetric bodies having volumes which are generated by revolving a curve about an axis. An example of such a body is shown in Fig. 17-2a. Two types of differential elements can be chosen.

Shell Element.

- If a *shell element* having a height z , radius $r = y$, and thickness dy is chosen for integration, Fig. 17-2b, then the volume is $dV = (2\pi y)(z)dy$.
- This element may be used in Eq. 17-2 or 17-3 for determining the moment of inertia I_z of the body about the z axis, since the *entire element*, due to its “thinness,” lies at the *same* perpendicular distance $r = y$ from the z axis (see Example 17.1).

Disk Element.

- If a *disk element* having a radius y and a thickness dz is chosen for integration, Fig. 17-2c, then the volume is $dV = (\pi y^2)dz$.
- This element is *finite* in the radial direction, and consequently its parts *do not* all lie at the *same radial distance* r from the z axis. As a result, Eq. 17-2 or 17-3 *cannot* be used to determine I_z directly. Instead, to perform the integration it is first necessary to determine the moment of inertia *of the element* about the z axis and then integrate this result (see Example 17.2).

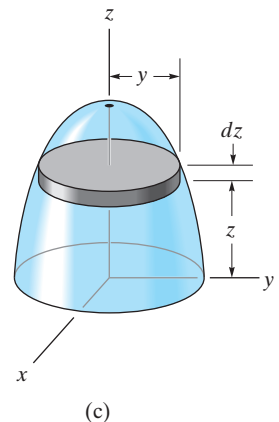
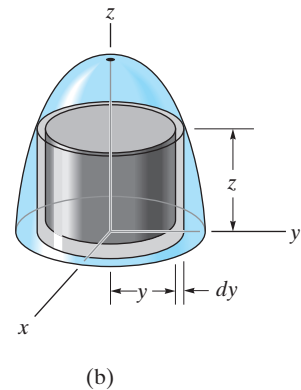
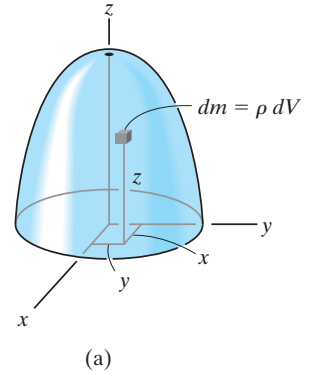
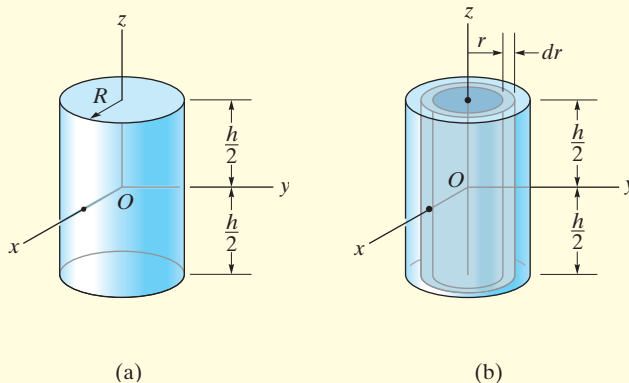


Fig. 17-2

EXAMPLE 17.1

Determine the moment of inertia of the cylinder shown in Fig. 17-3a about the z axis. The density of the material, ρ , is constant.

**Fig. 17-3****SOLUTION**

Shell Element. This problem can be solved using the *shell element* in Fig. 17-3b and a single integration. The volume of the element is $dV = (2\pi r)(h) dr$, so that its mass is $dm = \rho dV = \rho(2\pi hr dr)$. Since the *entire element* lies at the same distance r from the z axis, the moment of inertia of the element is

$$dI_z = r^2 dm = \rho 2\pi h r^3 dr$$

Integrating over the entire region of the cylinder yields

$$I_z = \int_m r^2 dm = \rho 2\pi h \int_0^R r^3 dr = \frac{\rho \pi}{2} R^4 h$$

The mass of the cylinder is

$$m = \int_m dm = \rho 2\pi h \int_0^R r dr = \rho \pi h R^2$$

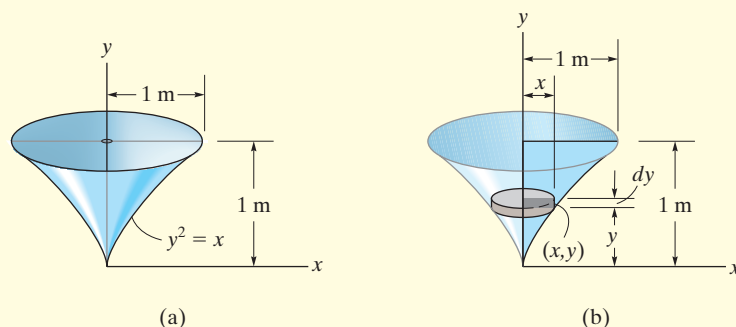
so that

$$I_z = \frac{1}{2} m R^2$$

Ans.

EXAMPLE 17.2

If the density of the material is 3000 kg/m^3 , determine the moment of inertia of the solid in Fig. 17-4a about the y axis.

**Fig. 17-4****SOLUTION**

Disk Element. The moment of inertia will be found using a *disk element*, as shown in Fig. 17-4b. Here the element intersects the curve at the arbitrary point (x, y) and has a mass

$$dm = \rho dV = \rho(\pi x^2) dy$$

Although all portions of the element are *not* located at the same distance from the y axis, it is still possible to determine the moment of inertia dI_y of the element about the y axis. In the preceding example it was shown that the moment of inertia of a cylinder about its longitudinal axis is $I = \frac{1}{2}mR^2$, where m and R are the mass and radius of the cylinder. Since the height is not involved in this formula, the disk itself can be thought of as a cylinder. Thus, for the disk element in Fig. 17-4b, we have

$$dI_y = \frac{1}{2}(dm)x^2 = \frac{1}{2}[\rho(\pi x^2) dy]x^2$$

Substituting $x = y^2$, $\rho = 3000 \text{ kg/m}^3$, and integrating with respect to y , from $y = 0$ to $y = 1 \text{ m}$, yields the moment of inertia for the entire solid.

$$I_y = \frac{\pi(3000 \text{ kg/m}^3)}{2} \int_0^1 x^4 dy = \frac{\pi(3000)}{2} \int_0^1 y^8 dy = 524 \text{ kg} \cdot \text{m}^2 \quad \text{Ans.}$$

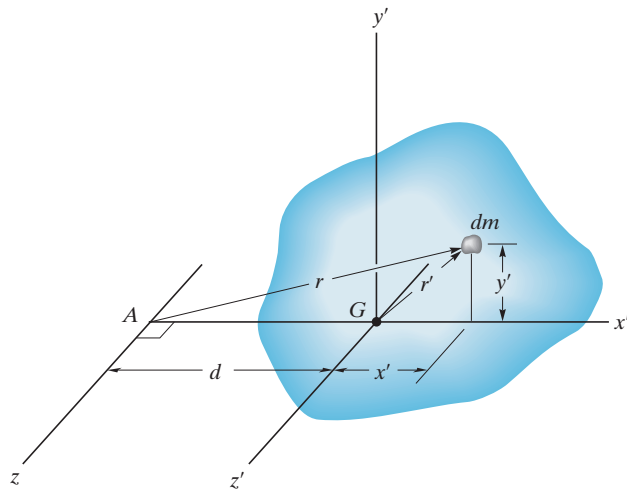


Fig. 17-5

Parallel-Axis Theorem. If the moment of inertia of the body about an axis passing through the body's mass center is known, then the moment of inertia about any other *parallel axis* can be determined by using the *parallel-axis theorem*. This theorem can be derived by considering the body shown in Fig. 17-5. Here the z' axis passes through the mass center G , whereas the corresponding *parallel z axis* lies at a constant distance d away. Selecting the differential element of mass dm , which is located at point (x', y') , and using the Pythagorean theorem, $r^2 = (d + x')^2 + y'^2$, we can express the moment of inertia of the body about the z axis as

$$\begin{aligned} I &= \int_m r^2 dm = \int_m [(d + x')^2 + y'^2] dm \\ &= \int_m (x'^2 + y'^2) dm + 2d \int_m x' dm + d^2 \int_m dm \end{aligned}$$

Since $r'^2 = x'^2 + y'^2$, the first integral represents I_G . The second integral equals *zero*, since the z' axis passes through the body's mass center, i.e., $\int x' dm = \bar{x}'m = 0$ since $\bar{x}' = 0$. Finally, the third integral

represents the total mass m of the body. Hence, the moment of inertia about the z axis can be written as

$$I = I_G + md^2 \quad (17-4)$$

where

I_G = moment of inertia about the z' axis passing through the mass center G

m = mass of the body

d = perpendicular distance between the parallel z and z' axes

Radius of Gyration. Occasionally, the moment of inertia of a body about a specified axis is reported in handbooks using the *radius of gyration*, k . This is a geometrical property which has units of length. When it and the body's mass m are known, the body's moment of inertia is determined from the equation

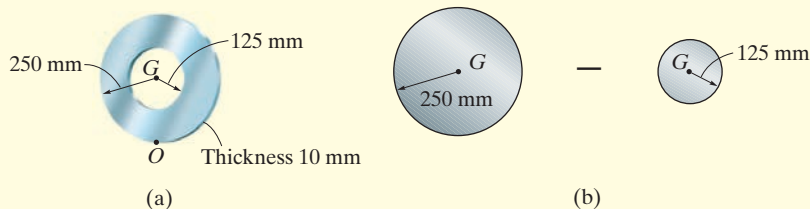
$$I = mk^2 \quad \text{or} \quad k = \sqrt{\frac{I}{m}} \quad (17-5)$$

Note the *similarity* between the definition of k in this formula and r in the equation $dI = r^2 dm$, which defines the moment of inertia of an elemental mass dm of the body about an axis.

Composite Bodies. If a body consists of a number of simple shapes such as disks, spheres, and rods, the moment of inertia of the body about any axis can be determined by adding algebraically the moments of inertia of all the composite shapes computed about the axis. Algebraic addition is necessary since a composite part must be considered as a negative quantity if it has already been counted as a piece of another part—for example, a “hole” subtracted from a solid plate. The parallel-axis theorem is needed for the calculations if the center of mass of each composite part does not lie on the axis. For the calculation, then, $I = \Sigma(I_G + md^2)$. Here I_G for each of the composite parts is determined by integration, or for simple shapes, such as rods and disks, it can be found from a table, such as the one given on the inside back cover of this book.

EXAMPLE 17.3

If the plate shown in Fig. 17–6a has a density of 8000 kg/m^3 and a thickness of 10 mm, determine its moment of inertia about an axis directed perpendicular to the page and passing through point O .

**Fig. 17–6****SOLUTION**

The plate consists of two composite parts, the 250-mm-radius disk *minus* a 125-mm-radius disk, Fig. 17–6b. The moment of inertia about O can be determined by computing the moment of inertia of each of these parts about O and then adding the results *algebraically*. The calculations are performed by using the parallel-axis theorem in conjunction with the data listed in the table on the inside back cover.

Disk. The moment of inertia of a disk about the centroidal axis perpendicular to the plane of the disk is $I_G = \frac{1}{2}mr^2$. The mass center of the disk is located at a distance of 0.25 m from point O . Thus,

$$\begin{aligned} m_d &= \rho_d V_d = 8000 \text{ kg/m}^3 [\pi(0.25 \text{ m})^2(0.01 \text{ m})] = 15.71 \text{ kg} \\ (I_d)_O &= \frac{1}{2}m_d r_d^2 + m_d d^2 \\ &= \frac{1}{2}(15.71 \text{ kg})(0.25 \text{ m})^2 + (15.71 \text{ kg})(0.25 \text{ m})^2 \\ &= 1.473 \text{ kg} \cdot \text{m}^2 \end{aligned}$$

Hole. For the 125-mm-radius disk (hole), we have

$$\begin{aligned} m_h &= \rho_h V_h = 8000 \text{ kg/m}^3 [\pi(0.125 \text{ m})^2(0.01 \text{ m})] = 3.927 \text{ kg} \\ (I_h)_O &= \frac{1}{2}m_h r_h^2 + m_h d^2 \\ &= \frac{1}{2}(3.927 \text{ kg})(0.125 \text{ m})^2 + (3.927 \text{ kg})(0.25 \text{ m})^2 \\ &= 0.276 \text{ kg} \cdot \text{m}^2 \end{aligned}$$

The moment of inertia of the plate about point O is therefore

$$\begin{aligned} I_O &= (I_d)_O - (I_h)_O \\ &= 1.473 \text{ kg} \cdot \text{m}^2 - 0.276 \text{ kg} \cdot \text{m}^2 \\ &= 1.20 \text{ kg} \cdot \text{m}^2 \end{aligned}$$

Ans.

EXAMPLE 17.4

The pendulum in Fig. 17–7 is suspended from the pin at O and consists of two thin rods. Rod OA has a mass of 12 kg, and BC has a mass of 9 kg. Determine the moment of inertia of the pendulum about an axis passing through (a) point O , and (b) the mass center G of the pendulum.

SOLUTION

Part (a). Using the table on the inside back cover, the moment of inertia of rod OA about an axis perpendicular to the page and passing through point O of the rod is $I_O = \frac{1}{3}ml^2$. Hence,

$$(I_{OA})_O = \frac{1}{3}ml^2 = \frac{1}{3}(12 \text{ kg})(2 \text{ m})^2 = 16 \text{ kg} \cdot \text{m}^2$$

This same value can be obtained using $I_G = \frac{1}{12}ml^2$ and the parallel-axis theorem.

$$\begin{aligned}(I_{OA})_O &= \frac{1}{12}ml^2 + md^2 = \frac{1}{12}(12 \text{ kg})(2 \text{ m})^2 + (12 \text{ kg})(1 \text{ m})^2 \\ &= 16 \text{ kg} \cdot \text{m}^2\end{aligned}$$

For rod BC we have

$$\begin{aligned}(I_{BC})_O &= \frac{1}{12}ml^2 + md^2 = \frac{1}{12}(9 \text{ kg})(1.5 \text{ m})^2 + (9 \text{ kg})(2 \text{ m})^2 \\ &= 37.6875 \text{ kg} \cdot \text{m}^2\end{aligned}$$

The moment of inertia of the pendulum about O is therefore

$$I_O = 16 + 37.6875 = 53.6875 \text{ kg} \cdot \text{m}^2 = 53.7 \text{ kg} \cdot \text{m}^2 \quad \text{Ans.}$$

Part (b). The mass center G will be located relative to point O . Assuming this distance to be $\bar{y}$, Fig. 17–7, and using the formula for determining the mass center, we have

$$\bar{y} = \frac{\sum \tilde{y}m}{\sum m} = \frac{1(12) + 2(9)}{(12) + (9)} = 1.4286 \text{ m}$$

The moment of inertia I_G may be found in the same manner as I_O , which requires successive applications of the parallel-axis theorem to transfer the moments of inertia of rods OA and BC to G . A more direct solution, however, involves using the result for I_O , i.e.,

$$I_O = I_G + md^2; \quad 53.6875 \text{ kg} \cdot \text{m}^2 = I_G + (21 \text{ kg})(1.4286 \text{ m})^2$$

$$I_G = 10.8 \text{ kg} \cdot \text{m}^2 \quad \text{Ans.}$$

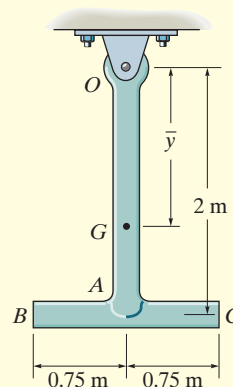
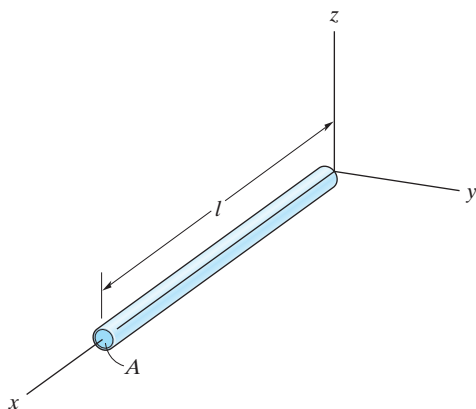


Fig. 17–7

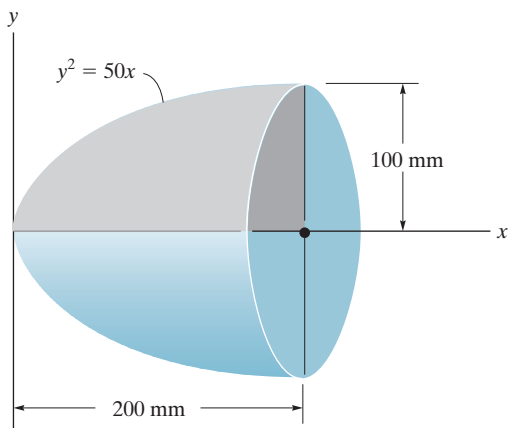
PROBLEMS

17-1. Determine the moment of inertia I_y for the slender rod. The rod's density ρ and cross-sectional area A are constant. Express the result in terms of the rod's total mass m .



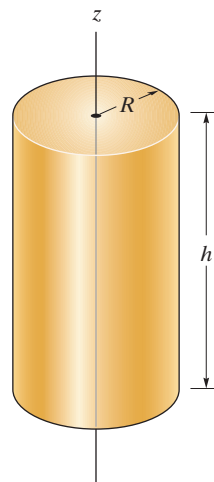
Prob. 17-1

17-2. The paraboloid is formed by revolving the shaded area around the x axis. Determine the radius of gyration k_x . The density of the material is $\rho = 5 \text{ Mg/m}^3$.



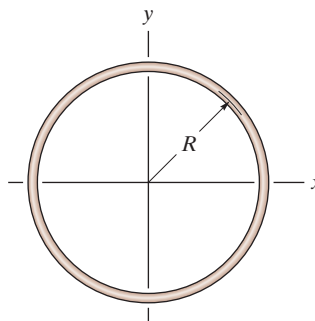
Prob. 17-2

17-3. The solid cylinder has an outer radius R , height h , and is made from a material having a density that varies from its center as $\rho = k + ar^2$, where k and a are constants. Determine the mass of the cylinder and its moment of inertia about the z axis.



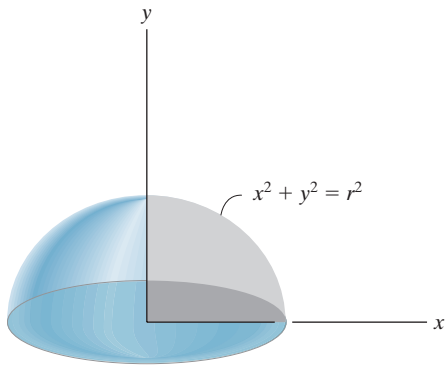
Prob. 17-3

***17-4.** Determine the moment of inertia of the thin ring about the z axis. The ring has a mass m .



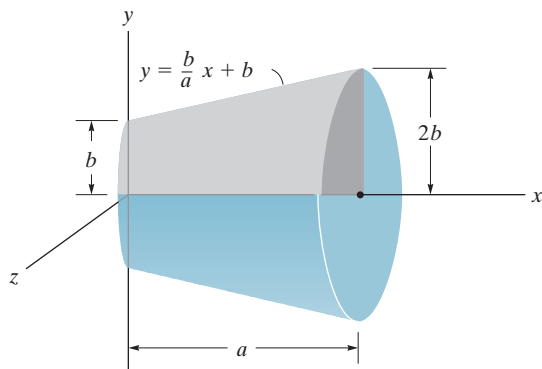
Prob. 17-4

17-5. The hemisphere is formed by rotating the shaded area around the y axis. Determine the moment of inertia I_y and express the result in terms of the total mass m of the hemisphere. The material has a constant density ρ .



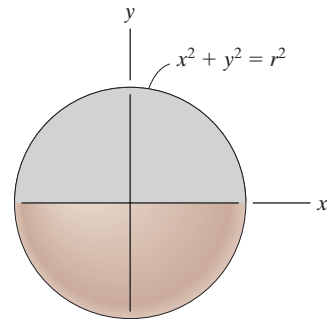
Prob. 17-5

17-6. The frustum is formed by rotating the shaded area around the x axis. Determine the moment of inertia I_x and express the result in terms of the total mass m of the frustum. The frustum has a constant density ρ .



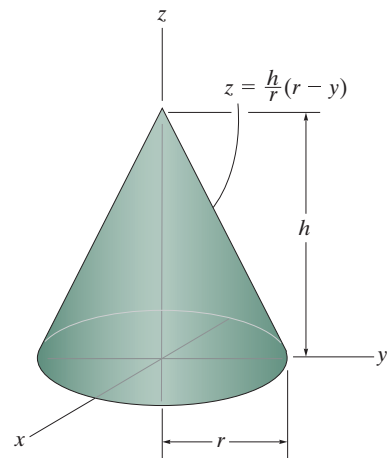
Prob. 17-6

17-7. The sphere is formed by revolving the shaded area around the x axis. Determine the moment of inertia I_x and express the result in terms of the total mass m of the sphere. The material has a constant density ρ .



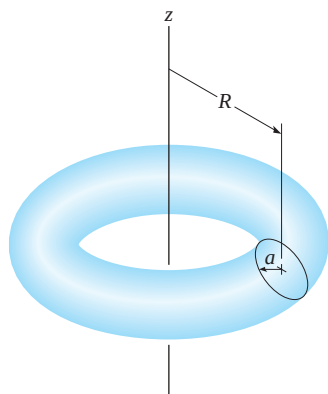
Prob. 17-7

***17-8.** Determine the mass moment of inertia I_z of the cone formed by revolving the shaded area around the z axis. The density of the material is ρ . Express the result in terms of the mass m of the cone.



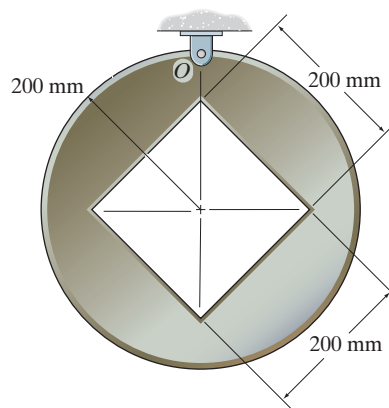
Prob. 17-8

17-9. Determine the moment of inertia I_z of the torus. The mass of the torus is m and the density ρ is constant.
Suggestion: Use a shell element.



Prob. 17-9

***17-12.** Determine the mass moment of inertia of the thin plate about an axis perpendicular to the page and passing through point O . The material has a mass per unit area of 20 kg/m^2 .

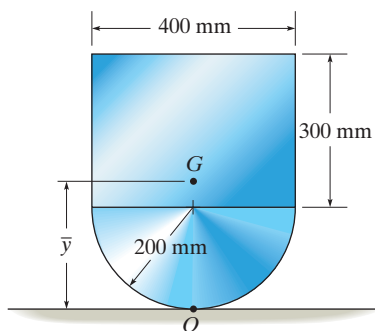


Prob. 17-12

17

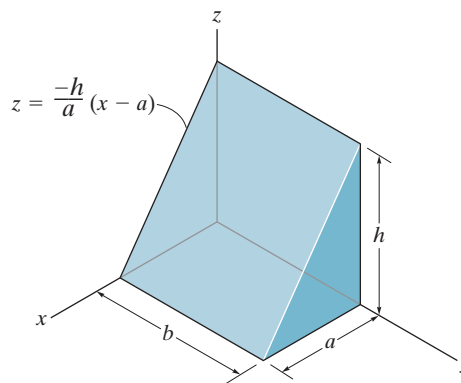
17-10. Determine the location $\bar{y}$ of the center of mass G of the assembly and then calculate the moment of inertia about an axis perpendicular to the page and passing through G . The block has a mass of 3 kg and the semicylinder has a mass of 5 kg.

17-11. Determine the moment of inertia of the assembly about an axis perpendicular to the page and passing through point O . The block has a mass of 3 kg, and the semicylinder has a mass of 5 kg.



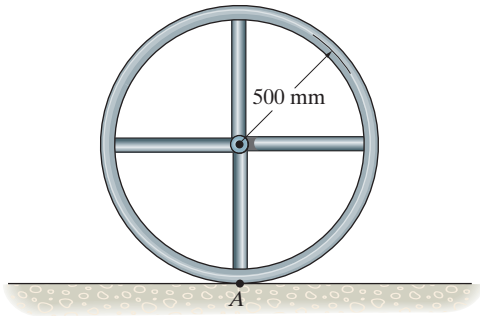
Probs. 17-10/11

17-13. Determine the moment of inertia of the homogeneous triangular prism with respect to the y axis. Express the result in terms of the mass m of the prism. *Hint:* For integration, use thin plate elements parallel to the x - y plane and having a thickness dz .



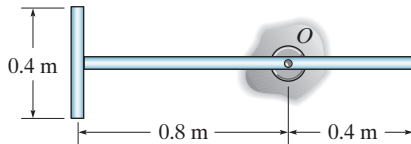
Prob. 17-13

17–14. The wheel consists of a thin ring having a mass of 10 kg and four spokes made from slender rods and each having a mass of 2 kg. Determine the wheel's moment of inertia about an axis perpendicular to the page and passing through point A .



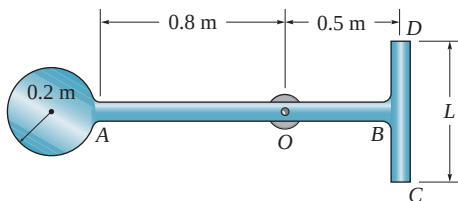
Prob. 17–14

17–15. The assembly is made of the slender rods that have a mass per unit length of 3 kg/m. Determine the mass moment of inertia of the assembly about an axis perpendicular to the page and passing through point O .



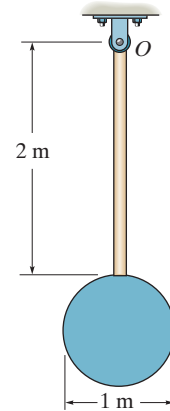
Prob. 17–15

***17–16.** The assembly consists of a disk having a mass of 6 kg and slender rods AB and DC which have a mass per unit length of 2 kg/m. Determine the length L of DC so that the center of mass is at the bearing O . What is the moment of inertia of the assembly about an axis perpendicular to the page and passing through O ?



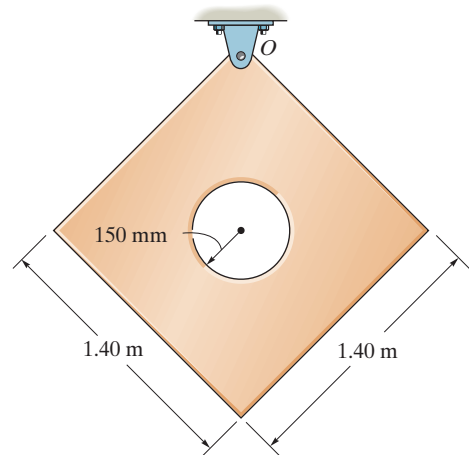
Prob. 17–16

17–17. The pendulum consists of a 4-kg circular plate and a 2-kg slender rod. Determine the radius of gyration of the pendulum about an axis perpendicular to the page and passing through point O .



Prob. 17–17

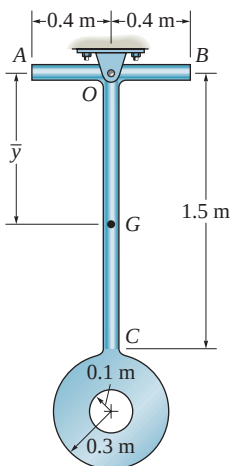
17–18. Determine the moment of inertia about an axis perpendicular to the page and passing through the pin at O . The thin plate has a hole in its center. Its thickness is 50 mm, and the material has a density $\rho = 50 \text{ kg/m}^3$.



Prob. 17–18

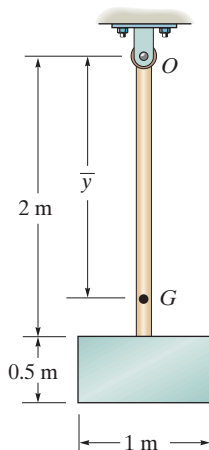
17–19. The pendulum consists of two slender rods AB and OC which have a mass per unit length of 3 kg/m . The thin circular plate has a mass per unit area of 12 kg/m^2 . Determine the location $\bar{y}$ of the center of mass G of the pendulum, then calculate the moment of inertia of the pendulum about an axis perpendicular to the page and passing through G .

***17–20.** The pendulum consists of two slender rods AB and OC which have a mass per unit length of 3 kg/m . The thin circular plate has a mass per unit area of 12 kg/m^2 . Determine the moment of inertia of the pendulum about an axis perpendicular to the page and passing through the pin at O .



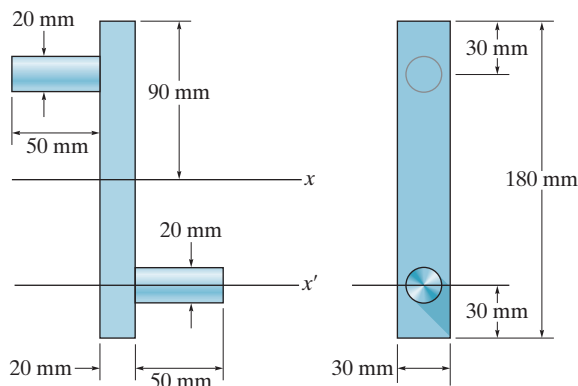
Probs. 17–19/20

17–21. The pendulum consists of the 3-kg slender rod and the 5-kg thin plate. Determine the location $\bar{y}$ of the center of mass G of the pendulum; then calculate the moment of inertia of the pendulum about an axis perpendicular to the page and passing through G .



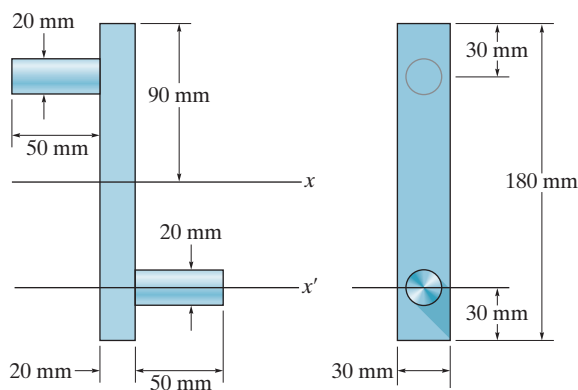
Prob. 17–21

17–22. Determine the moment of inertia of the overhung crank about the x axis. The material is steel having a density of $\rho = 7.85 \text{ Mg/m}^3$.



Prob. 17–22

17–23. Determine the moment of inertia of the overhung crank about the x' axis. The material is steel having a density of $\rho = 7.85 \text{ Mg/m}^3$.



Prob. 17–23

17.2 Planar Kinetic Equations of Motion

In the following analysis we will limit our study of planar kinetics to rigid bodies which, along with their loadings, are considered to be *symmetrical* with respect to a fixed reference plane.* Since the motion of the body can be viewed within the reference plane, all the forces (and couple moments) acting on the body can then be projected onto the plane. An example of an arbitrary body of this type is shown in Fig. 17–8a. Here the *inertial frame of reference* x, y, z has its origin *coincident* with the arbitrary point P in the body. By definition, *these axes do not rotate and are either fixed or translate with constant velocity.*

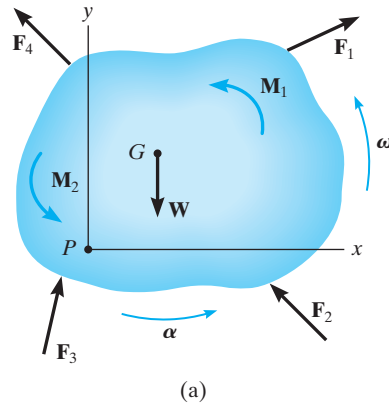


Fig. 17–8

Equation of Translational Motion. The external forces acting on the body in Fig. 17–8a represent the effect of gravitational, electrical, magnetic, or contact forces between adjacent bodies. Since this force system has been considered previously in Sec. 13.3 for the analysis of a system of particles, the resulting Eq. 13–6 can be used here, in which case

$$\Sigma \mathbf{F} = m\mathbf{a}_G$$

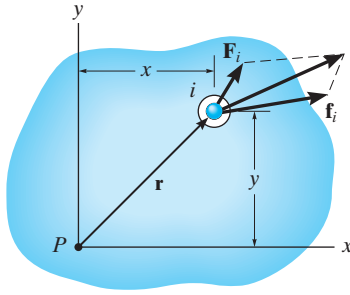
This equation is referred to as the *translational equation of motion* for the mass center of a rigid body. It states that *the sum of all the external forces acting on the body is equal to the body's mass times the acceleration of its mass center G .*

For motion of the body in the x – y plane, the translational equation of motion may be written in the form of two independent scalar equations, namely,

$$\Sigma F_x = m(a_G)_x$$

$$\Sigma F_y = m(a_G)_y$$

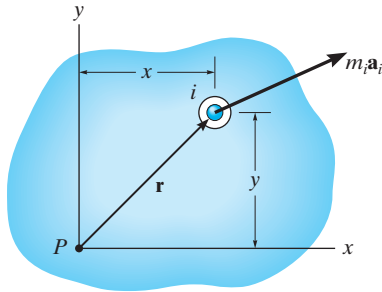
*By doing this, the rotational equation of motion reduces to a rather simplified form. The more general case of body shape and loading is considered in Chapter 21.



Particle free-body diagram

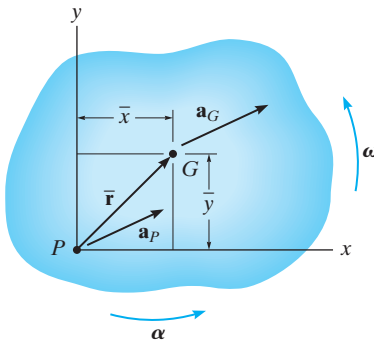
(b)

||



Particle kinetic diagram

(c)



(d)

Fig. 17-8 (cont.)

Equation of Rotational Motion. We will now determine the effects caused by the moments of the external force system computed about an axis perpendicular to the plane of motion (the z axis) and passing through point P . As shown on the free-body diagram of the i th particle, Fig. 17-8b, $\mathbf{F}_i$ represents the *resultant external force* acting on the particle, and $\mathbf{f}_i$ is the *resultant of the internal forces* caused by interactions with adjacent particles. If the particle has a mass m_i and its acceleration is $\mathbf{a}_i$, then its kinetic diagram is shown in Fig. 17-8c. Summing moments about point P , we require

$$\mathbf{r} \times \mathbf{F}_i + \mathbf{r} \times \mathbf{f}_i = \mathbf{r} \times m_i \mathbf{a}_i$$

or

$$(\mathbf{M}_P)_i = \mathbf{r} \times m_i \mathbf{a}_i$$

The moments about P can also be expressed in terms of the acceleration of point P , Fig. 17-8d. If the body has an angular acceleration α and angular velocity ω , then using Eq. 16-18 we have

$$\begin{aligned} (\mathbf{M}_P)_i &= m_i \mathbf{r} \times (\mathbf{a}_P + \alpha \times \mathbf{r} - \omega^2 \mathbf{r}) \\ &= m_i [\mathbf{r} \times \mathbf{a}_P + \mathbf{r} \times (\alpha \times \mathbf{r}) - \omega^2 (\mathbf{r} \times \mathbf{r})] \end{aligned}$$

The last term is zero, since $\mathbf{r} \times \mathbf{r} = \mathbf{0}$. Expressing the vectors with Cartesian components and carrying out the cross-product operations yields

$$\begin{aligned} (M_P)_i \mathbf{k} &= m_i \{ (\bar{x}\mathbf{i} + \bar{y}\mathbf{j}) \times [(a_P)_x \mathbf{i} + (a_P)_y \mathbf{j}] \\ &\quad + (\bar{x}\mathbf{i} + \bar{y}\mathbf{j}) \times [\alpha \mathbf{k} \times (\bar{x}\mathbf{i} + \bar{y}\mathbf{j})] \} \\ (M_P)_i \mathbf{k} &= m_i [-\bar{y}(a_P)_x + \bar{x}(a_P)_y + \alpha \bar{x}^2 + \alpha \bar{y}^2] \mathbf{k} \\ \zeta (M_P)_i &= m_i [-\bar{y}(a_P)_x + \bar{x}(a_P)_y + \alpha \bar{r}^2] \end{aligned}$$

Letting $m_i \rightarrow dm$ and integrating with respect to the entire mass m of the body, we obtain the resultant moment equation

$$\zeta \Sigma M_P = - \left(\int_m \bar{y} dm \right) (a_P)_x + \left(\int_m \bar{x} dm \right) (a_P)_y + \left(\int_m \bar{r}^2 dm \right) \alpha$$

Here ΣM_P represents only the moment of the *external forces* acting on the body about point P . The resultant moment of the internal forces is zero, since for the entire body these forces occur in equal and opposite collinear pairs and thus the moment of each pair of forces about P cancels. The integrals in the first and second terms on the right are used to locate the body's center of mass G with respect to P , since $\bar{y}m = \int \bar{y} dm$ and $\bar{x}m = \int \bar{x} dm$, Fig. 17-8d. Also, the last integral represents the body's moment of inertia about the z axis, i.e., $I_P = \int \bar{r}^2 dm$. Thus,

$$\zeta \Sigma M_P = -\bar{y}m(a_P)_x + \bar{x}m(a_P)_y + I_P \alpha \quad (17-6)$$

It is possible to reduce this equation to a simpler form if point P coincides with the mass center G for the body. If this is the case, then $\bar{x} = \bar{y} = 0$, and therefore*

$$\Sigma M_G = I_G \alpha \quad (17-7)$$

This rotational equation of motion states that the sum of the moments of all the external forces about the body's mass center G is equal to the product of the moment of inertia of the body about an axis passing through G and the body's angular acceleration.

Equation 17-6 can also be rewritten in terms of the x and y components of $\mathbf{a}_G$ and the body's moment of inertia I_G . If point G is located at $(\bar{x}, \bar{y})$, Fig. 17-8d, then by the parallel-axis theorem, $I_P = I_G + m(\bar{x}^2 + \bar{y}^2)$. Substituting into Eq. 17-6 and rearranging terms, we get

$$\zeta \Sigma M_P = \bar{y}m[-(a_P)_x + \bar{y}\alpha] + \bar{x}m[(a_P)_y + \bar{x}\alpha] + I_G \alpha \quad (17-8)$$

From the kinematic diagram of Fig. 17-8d, $\mathbf{a}_P$ can be expressed in terms of $\mathbf{a}_G$ as

$$\mathbf{a}_G = \mathbf{a}_P + \boldsymbol{\alpha} \times \bar{\mathbf{r}} - \omega^2 \bar{\mathbf{r}}$$

$$(a_G)_x \mathbf{i} + (a_G)_y \mathbf{j} = (a_P)_x \mathbf{i} + (a_P)_y \mathbf{j} + \alpha \mathbf{k} \times (\bar{x} \mathbf{i} + \bar{y} \mathbf{j}) - \omega^2 (\bar{x} \mathbf{i} + \bar{y} \mathbf{j})$$

Carrying out the cross product and equating the respective $\mathbf{i}$ and $\mathbf{j}$ components yields the two scalar equations

$$(a_G)_x = (a_P)_x - \bar{y}\alpha - \bar{x}\omega^2$$

$$(a_G)_y = (a_P)_y + \bar{x}\alpha - \bar{y}\omega^2$$

From these equations, $[-(a_P)_x + \bar{y}\alpha] = [-(a_G)_x - \bar{x}\omega^2]$ and $[(a_P)_y + \bar{x}\alpha] = [(a_G)_y + \bar{y}\omega^2]$. Substituting these results into Eq. 17-8 and simplifying gives

$$\zeta \Sigma M_P = -\bar{y}m(a_G)_x + \bar{x}m(a_G)_y + I_G \alpha \quad (17-9)$$

This important result indicates that when moments of the external forces shown on the free-body diagram are summed about point P , Fig. 17-8e, they are equivalent to the sum of the “kinetic moments” of the components of $m\mathbf{a}_G$ about P plus the “kinetic moment” of $I_G \alpha$, Fig. 17-8f. In other words, when the “kinetic moments,” $\Sigma(\mathcal{M}_k)_P$, are computed, Fig. 17-8f, the vectors $m(\mathbf{a}_G)_x$ and $m(\mathbf{a}_G)_y$ are treated as sliding vectors; that is, they can act at any point along their line of action. In a similar manner, $I_G \alpha$ can be treated as a free vector and can therefore act at any point. It is important to keep in mind, however, that $m\mathbf{a}_G$ and $I_G \alpha$ are not the same as a force or a couple moment. Instead, they are caused by the external effects of forces and couple moments acting on the body. With this in mind we can therefore write Eq. 17-9 in a more general form as

$$\Sigma M_P = \Sigma(\mathcal{M}_k)_P \quad (17-10)$$

*It also reduces to this same simple form $\Sigma M_P = I_P \alpha$ if point P is a *fixed point* (see Eq. 17-16) or the acceleration of point P is directed along the line PG .

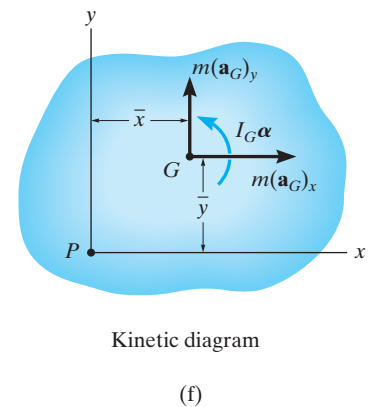
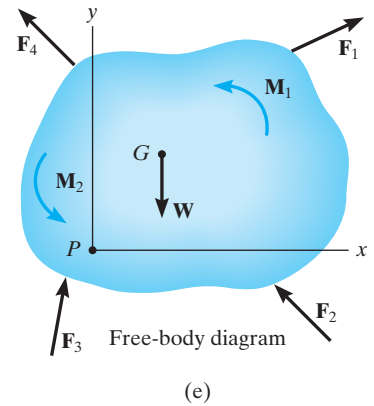
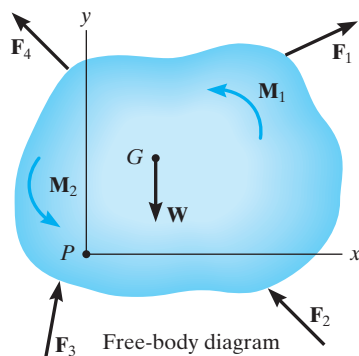
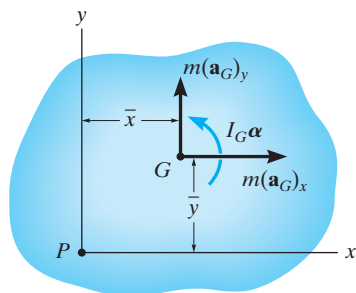


Fig. 17-8 (cont.)



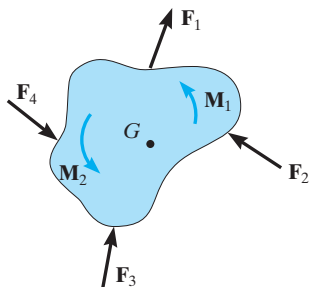
(e)



Kinetic diagram

(f)

Fig. 17-8 (cont.)



(a)

Fig. 17-9

General Application of the Equations of Motion. To summarize this analysis, *three* independent scalar equations can be written to describe the general plane motion of a symmetrical rigid body.

$$\Sigma F_x = m(a_G)_x$$

$$\Sigma F_y = m(a_G)_y$$

$$\Sigma M_G = I_G \alpha$$

or

$$\Sigma M_P = \Sigma (\mathcal{M}_k)_P \quad (17-11)$$

When applying these equations, one should *always* draw a free-body diagram, Fig. 17-8e, in order to account for the terms involved in ΣF_x , ΣF_y , ΣM_G , or ΣM_P . In some problems it may also be helpful to draw the *kinetic diagram* for the body, Fig. 17-8f. This diagram graphically accounts for the terms $m(a_G)_x$, $m(a_G)_y$, and $I_G \alpha$. It is especially convenient when used to determine the components of $m\mathbf{a}_G$ and the moment of these components in $\Sigma (\mathcal{M}_k)_P$.*

17.3 Equations of Motion: Translation

When the rigid body in Fig. 17-9a undergoes a *translation*, all the particles of the body have the *same acceleration*. Furthermore, $\alpha = 0$, in which case the rotational equation of motion applied at point G reduces to a simplified form, namely, $\Sigma M_G = 0$. Application of this and the force equations of motion will now be discussed for each of the two types of translation.

Rectilinear Translation. When a body is subjected to *rectilinear translation*, all the particles of the body (slab) travel along parallel straight-line paths. The free-body and kinetic diagrams are shown in Fig. 17-9b. Since $I_G \alpha = 0$, only $m\mathbf{a}_G$ is shown on the kinetic diagram. Hence, the equations of motion which apply in this case become

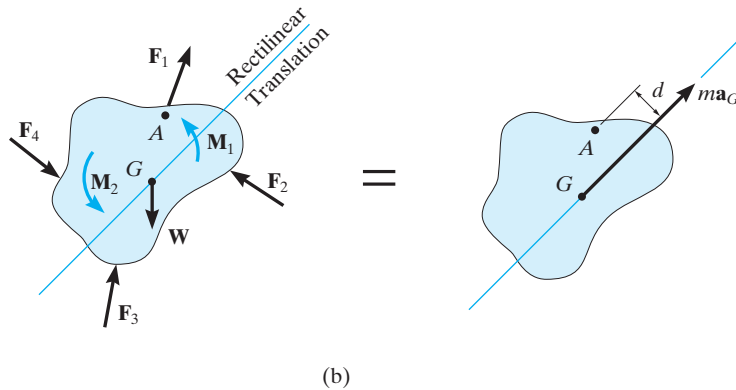
$$\Sigma F_x = m(a_G)_x$$

$$\Sigma F_y = m(a_G)_y$$

$$\Sigma M_G = 0$$

$$(17-12)$$

*For this reason, the kinetic diagram will be used in the solution of an example problem whenever $\Sigma M_P = \Sigma (\mathcal{M}_k)_P$ is applied.



It is also possible to sum moments about other points on or off the body, in which case the moment of $m\mathbf{a}_G$ must be taken into account. For example, if point A is chosen, which lies at a perpendicular distance d from the line of action of $m\mathbf{a}_G$, the following moment equation applies:

$$\zeta + \Sigma M_A = \Sigma (\mathcal{M}_k)_A; \quad \Sigma M_A = (ma_G)d$$

Here the sum of moments of the external forces and couple moments about A (ΣM_A , free-body diagram) equals the moment of $m\mathbf{a}_G$ about A ($\Sigma (\mathcal{M}_k)_A$, kinetic diagram).

Curvilinear Translation. When a rigid body is subjected to *curvilinear translation*, all the particles of the body have the same accelerations as they travel along *curved paths* as noted in Sec.16.1. For analysis, it is often convenient to use an inertial coordinate system having an origin which coincides with the body's mass center at the instant considered, and axes which are oriented in the normal and tangential directions to the path of motion, Fig. 17-9c. The three scalar equations of motion are then

$$\begin{aligned} \Sigma F_n &= m(a_G)_n \\ \Sigma F_t &= m(a_G)_t \\ \Sigma M_G &= 0 \end{aligned} \quad (17-13)$$

If moments are summed about the arbitrary point B , Fig. 17-9c, then it is necessary to account for the moments, $\Sigma (\mathcal{M}_k)_B$, of the two components $m(\mathbf{a}_G)_n$ and $m(\mathbf{a}_G)_t$ about this point. From the kinetic diagram, h and e represent the perpendicular distances (or “moment arms”) from B to the lines of action of the components. The required moment equation therefore becomes

$$\zeta + \Sigma M_B = \Sigma (\mathcal{M}_k)_B; \quad \Sigma M_B = e[m(a_G)_t] - h[m(a_G)_n]$$

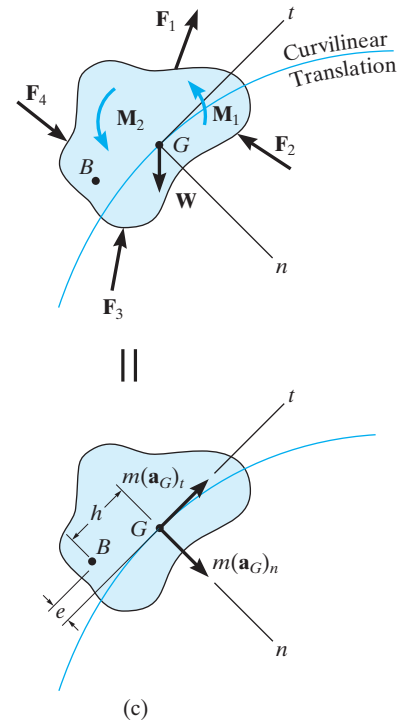
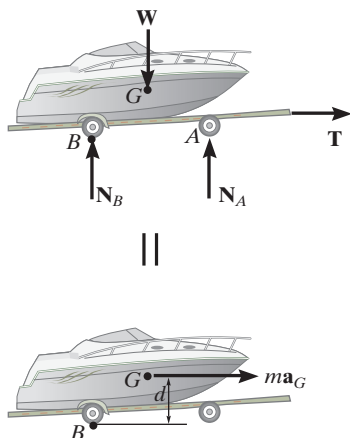


Fig. 17-9



The free-body and kinetic diagrams for this boat and trailer are drawn first in order to apply the equations of motion. Here the forces on the free-body diagram cause the effect shown on the kinetic diagram. If moments are summed about the mass center, G , then $\sum M_G = 0$. However, if moments are summed about point B then $\zeta + \sum M_B = ma_G(d)$.



17

Procedure for Analysis

Kinetic problems involving rigid-body *translation* can be solved using the following procedure.

Free-Body Diagram.

- Establish the x, y or n, t inertial coordinate system and draw the free-body diagram in order to account for all the external forces and couple moments that act on the body.
- The direction and sense of the acceleration of the body's mass center $\mathbf{a}_G$ should be established.
- Identify the unknowns in the problem.
- If it is decided that the rotational equation of motion $\sum M_P = \sum (\mathcal{M}_k)_P$ is to be used in the solution, then consider drawing the kinetic diagram, since it graphically accounts for the components $m(\mathbf{a}_G)_x$, $m(\mathbf{a}_G)_y$ or $m(\mathbf{a}_G)_t$, $m(\mathbf{a}_G)_n$ and is therefore convenient for “visualizing” the terms needed in the moment sum $\sum (\mathcal{M}_k)_P$.

Equations of Motion.

- Apply the three equations of motion in accordance with the established sign convention.
- To simplify the analysis, the moment equation $\sum M_G = 0$ can be replaced by the more general equation $\sum M_P = \sum (\mathcal{M}_k)_P$, where point P is usually located at the intersection of the lines of action of as many unknown forces as possible.
- If the body is in contact with a *rough surface* and slipping occurs, use the friction equation $F = \mu_k N$. Remember, $\mathbf{F}$ always acts on the body so as to oppose the motion of the body relative to the surface it contacts.

Kinematics.

- Use kinematics to determine the velocity and position of the body.
- For rectilinear translation with *variable acceleration*

$$a_G = dv_G/dt \quad a_G ds_G = v_G dv_G$$
- For rectilinear translation with *constant acceleration*

$$v_G = (v_G)_0 + a_G t \quad v_G^2 = (v_G)_0^2 + 2a_G[s_G - (s_G)_0]$$

$$s_G = (s_G)_0 + (v_G)_0 t + \frac{1}{2} a_G t^2$$
- For curvilinear translation
$$(a_G)_n = v_G^2/\rho$$

$$(a_G)_t = dv_G/dt \quad (a_G)_t ds_G = v_G dv_G$$

EXAMPLE 17.5

The car shown in Fig. 17–10a has a mass of 2 Mg and a center of mass at G . Determine the acceleration if the rear “driving” wheels are always slipping, whereas the front wheels are free to rotate. Neglect the mass of the wheels. The coefficient of kinetic friction between the wheels and the road is $\mu_k = 0.25$.

SOLUTION I

Free-Body Diagram. As shown in Fig. 17–10b, the rear-wheel frictional force F_B pushes the car forward, and since *slipping occurs*, $F_B = 0.25N_B$. The frictional forces acting on the *front wheels* are *zero*, since these wheels have negligible mass.* There are three unknowns in the problem, N_A , N_B , and a_G . Here we will sum moments about the mass center. The car (point G) accelerates to the left, i.e., in the negative x direction, Fig. 17–10b.

Equations of Motion.

$$\pm \Sigma F_x = m(a_G)_x; \quad -0.25N_B = -(2000 \text{ kg})a_G \quad (1)$$

$$+\uparrow \Sigma F_y = m(a_G)_y; \quad N_A + N_B - 2000(9.81) \text{ N} = 0 \quad (2)$$

$$\zeta + \Sigma M_G = 0; \quad -N_A(1.25 \text{ m}) - 0.25N_B(0.3 \text{ m}) + N_B(0.75 \text{ m}) = 0 \quad (3)$$

Solving,

$$a_G = 1.59 \text{ m/s}^2 \leftarrow$$

$$N_A = 6.88 \text{ kN}$$

$$N_B = 12.7 \text{ kN}$$

Ans.

SOLUTION II

Free-Body and Kinetic Diagrams. If the “moment” equation is applied about point A , then the unknown N_A will be eliminated from the equation. To “visualize” the moment of ma_G about A , we will include the kinetic diagram as part of the analysis, Fig. 17–10c.

Equation of Motion.

$$\zeta + \Sigma M_A = \Sigma (\mathcal{M}_k)_A; \quad N_B(2 \text{ m}) - [2000(9.81) \text{ N}](1.25 \text{ m}) = (2000 \text{ kg})a_G(0.3 \text{ m})$$

Solving this and Eq. 1 for a_G leads to a simpler solution than that obtained from Eqs. 1 to 3.

*With negligible wheel mass, $I\alpha = 0$ and the frictional force at A required to turn the wheel is zero. If the wheels’ mass were included, then the solution would be more involved, since a general-plane-motion analysis of the wheels would have to be considered (see Sec. 17.5).

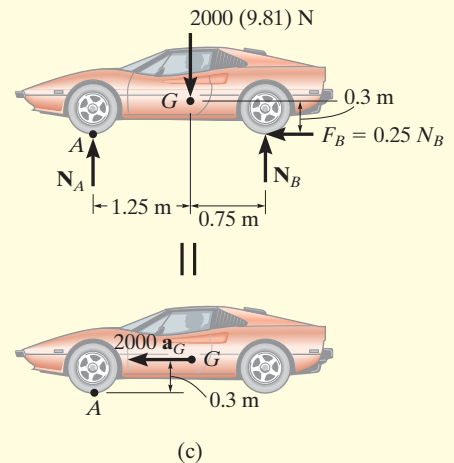
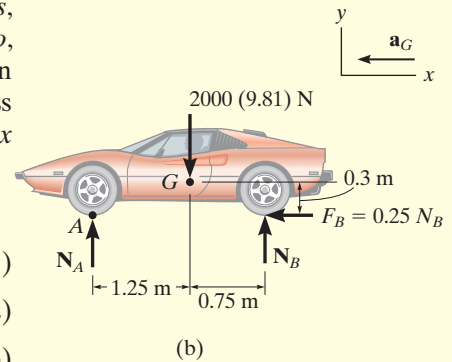
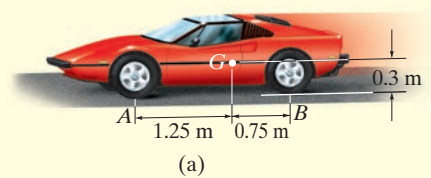
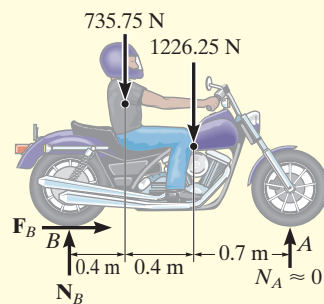


Fig. 17–10

EXAMPLE 17.6



The motorcycle shown in Fig. 17–11a has a mass of 125 kg and a center of mass at G_1 , while the rider has a mass of 75 kg and a center of mass at G_2 . Determine the minimum coefficient of static friction between the wheels and the pavement in order for the rider to do a “wheelie,” i.e., lift the front wheel off the ground as shown in the photo. What acceleration is necessary to do this? Neglect the mass of the wheels and assume that the front wheel is free to roll.



||

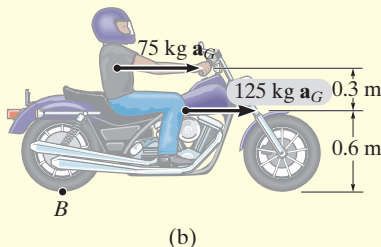
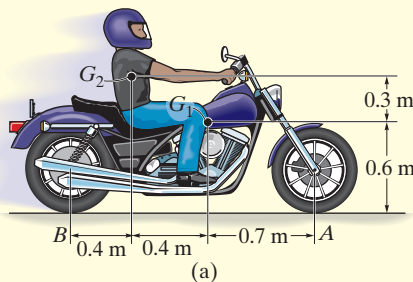


Fig. 17–11



SOLUTION

Free-Body and Kinetic Diagrams. In this problem we will consider both the motorcycle and the rider as a single *system*. It is possible first to determine the location of the center of mass for this “system” by using the equations $\bar{x} = \Sigma \tilde{x}m / \Sigma m$ and $\bar{y} = \Sigma \tilde{y}m / \Sigma m$. Here, however, we will consider the weight and mass of the motorcycle and rider separately as shown on the free-body and kinetic diagrams, Fig. 17–11b. Both of these parts move with the *same* acceleration. We have assumed that the front wheel is *about* to leave the ground, so that the normal reaction $N_A \approx 0$. The three unknowns in the problem are N_B , F_B , and a_G .

Equations of Motion.

$$\pm \Sigma F_x = m(a_G)_x; \quad F_B = (75 \text{ kg} + 125 \text{ kg})a_G \quad (1)$$

$$+ \uparrow \Sigma F_y = m(a_G)_y; \quad N_B - 735.75 \text{ N} - 1226.25 \text{ N} = 0$$

$$\zeta + \Sigma M_B = \Sigma (\mathcal{M}_k)_B; \quad -(735.75 \text{ N})(0.4 \text{ m}) - (1226.25 \text{ N})(0.8 \text{ m}) =$$

$$-(75 \text{ kg } a_G)(0.9 \text{ m}) - (125 \text{ kg } a_G)(0.6 \text{ m}) \quad (2)$$

Solving,

$$a_G = 8.95 \text{ m/s}^2 \rightarrow \quad \text{Ans.}$$

$$N_B = 1962 \text{ N}$$

$$F_B = 1790 \text{ N}$$

Thus the minimum coefficient of static friction is

$$(\mu_s)_{\min} = \frac{F_B}{N_B} = \frac{1790 \text{ N}}{1962 \text{ N}} = 0.912 \quad \text{Ans.}$$

EXAMPLE 17.7

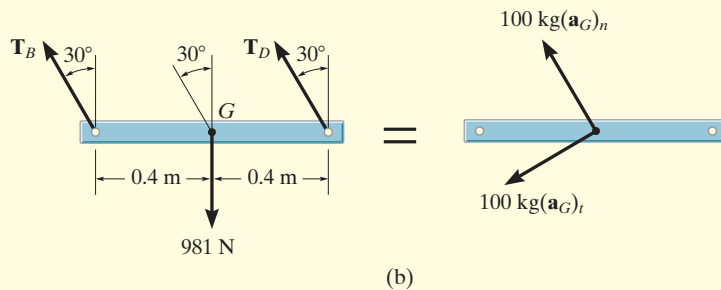
The 100-kg beam BD shown in Fig. 17–12a is supported by two rods having negligible mass. Determine the force developed in each rod if at the instant $\theta = 30^\circ$, $\omega = 6 \text{ rad/s}$.

SOLUTION

Free-Body and Kinetic Diagrams. The beam moves with *curvilinear translation* since all points on the beam move along circular paths, each path having the same radius of 0.5 m, but different centers of curvature. Using normal and tangential coordinates, the free-body and kinetic diagrams for the beam are shown in Fig. 17–12b. Because of the *translation*, G has the *same* motion as the pin at B , which is connected to both the rod and the beam. Note that the tangential component of acceleration acts downward to the left due to the clockwise direction of α , Fig. 17–12c. Furthermore, the normal component of acceleration is *always* directed toward the center of curvature (toward point A for rod AB). Since the angular velocity of AB is 6 rad/s when $\theta = 30^\circ$, then

$$(a_G)_n = \omega^2 r = (6 \text{ rad/s})^2 (0.5 \text{ m}) = 18 \text{ m/s}^2$$

The three unknowns are T_B , T_D , and $(a_G)_t$.

**Fig. 17–12****Equations of Motion.**

$$+\nearrow \Sigma F_n = m(a_G)_n; T_B + T_D - 981 \cos 30^\circ \text{ N} = 100 \text{ kg}(18 \text{ m/s}^2) \quad (1)$$

$$+\searrow \Sigma F_t = m(a_G)_t; 981 \sin 30^\circ = 100 \text{ kg}(a_G)_t \quad (2)$$

$$\zeta + \Sigma M_G = 0; -(T_B \cos 30^\circ)(0.4 \text{ m}) + (T_D \cos 30^\circ)(0.4 \text{ m}) = 0 \quad (3)$$

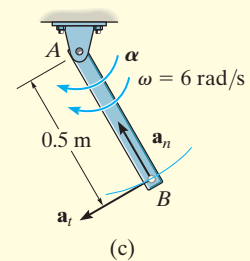
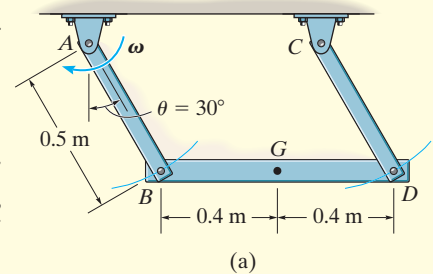
Simultaneous solution of these three equations gives

$$T_B = T_D = 1.32 \text{ kN}$$

Ans.

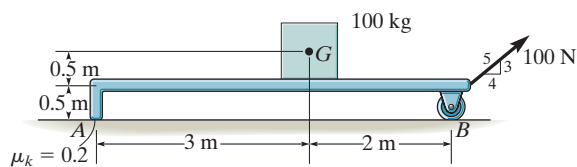
$$(a_G)_t = 4.905 \text{ m/s}^2$$

NOTE: It is also possible to apply the equations of motion along horizontal and vertical x, y axes, but the solution becomes more involved.

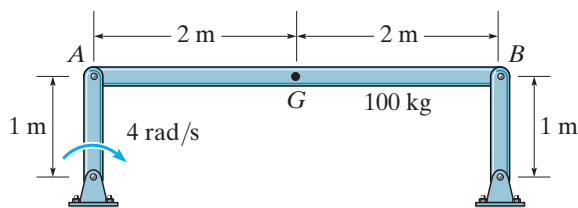


PRELIMINARY PROBLEMS

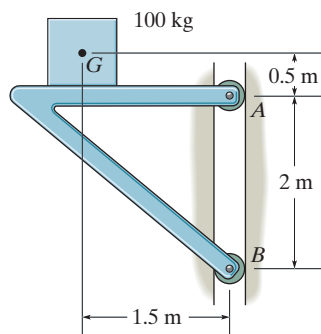
P17-1. Draw the free-body and kinetic diagrams of the object AB .



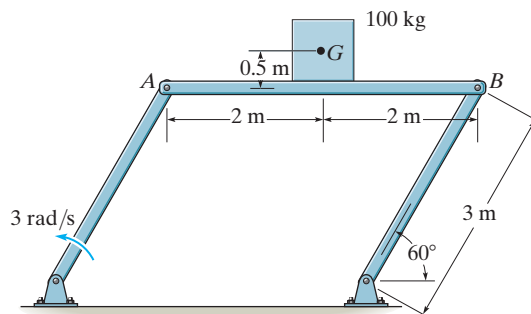
(a)



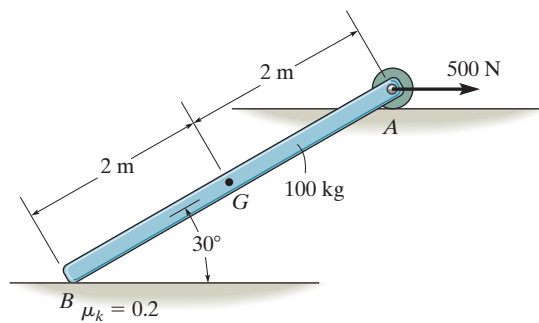
(d)



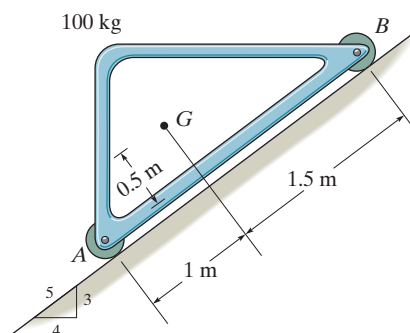
(b)



(e)



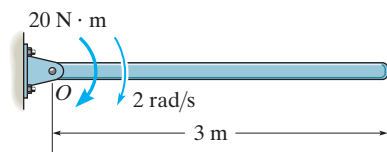
(c)



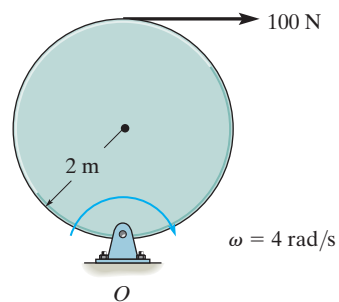
(f)

Prob. P17-1

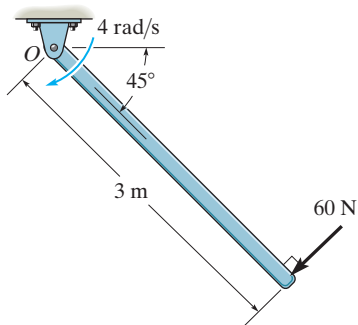
P17-2. Draw the free-body and kinetic diagrams of the 100-kg object.



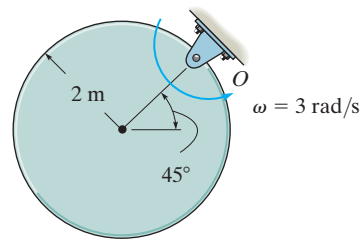
(a)



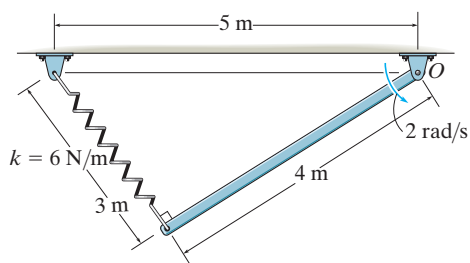
(d)



(b)

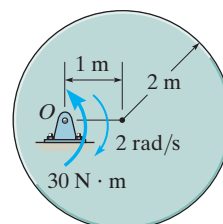


(e)



Unstretched length of spring is 1 m.

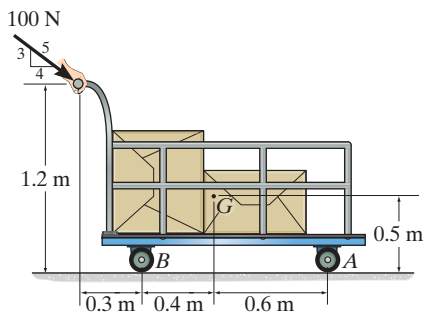
(c)



(f)

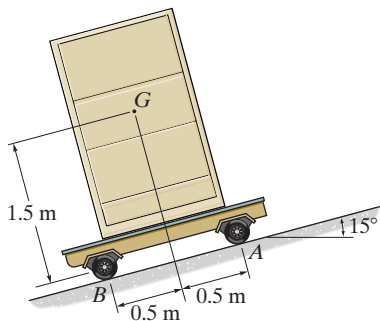
FUNDAMENTAL PROBLEMS

F17-1. The cart and its load have a total mass of 100 kg. Determine the acceleration of the cart and the normal reactions on the pair of wheels at A and B. Neglect the mass of the wheels.



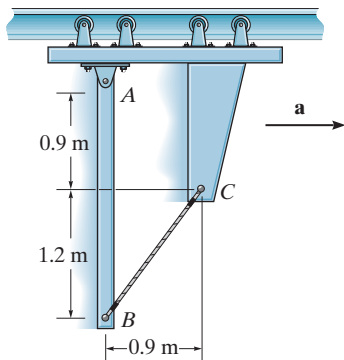
Prob. F17-1

F17-2. If the 80-kg cabinet is allowed to roll down the inclined plane, determine the acceleration of the cabinet and the normal reactions on the pair of rollers at A and B that have negligible mass.



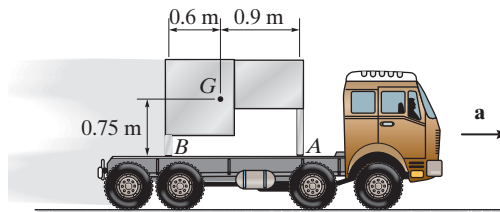
Prob. F17-2

F17-3. The 10-kg link AB is pinned to a moving frame at A and held in a vertical position by means of a string BC which can support a maximum tension of 50 N. Determine the maximum acceleration of the frame without breaking the string. What are the corresponding components of reaction at the pin A?



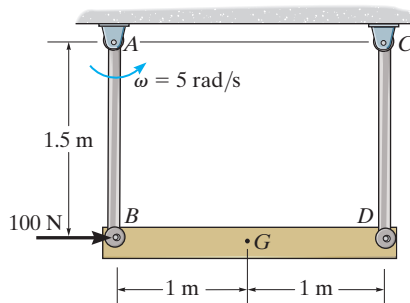
Prob. F17-3

F17-4. Determine the maximum acceleration of the truck without causing the assembly to move relative to the truck. Also what is the corresponding normal reaction on legs A and B? The 100-kg table has a mass center at G and the coefficient of static friction between the legs of the table and the bed of the truck is $\mu_s = 0.2$.



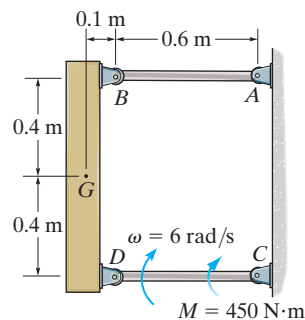
Prob. F17-4

F17-5. At the instant shown both rods of negligible mass swing with a counterclockwise angular velocity of $\omega = 5 \text{ rad/s}$, while the 50-kg bar is subjected to the 100-N horizontal force. Determine the tension developed in the rods and the angular acceleration of the rods at this instant.



Prob. F17-5

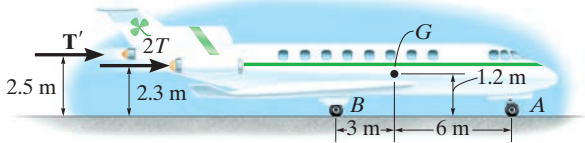
F17-6. At the instant shown, link CD rotates with an angular velocity of $\omega = 6 \text{ rad/s}$. If it is subjected to a couple moment $M = 450 \text{ N}\cdot\text{m}$, determine the force developed in link AB, the horizontal and vertical component of reaction on pin D, and the angular acceleration of link CD at this instant. The block has a mass of 50 kg and center of mass at G. Neglect the mass of links AB and CD.



Prob. F17-6

PROBLEMS

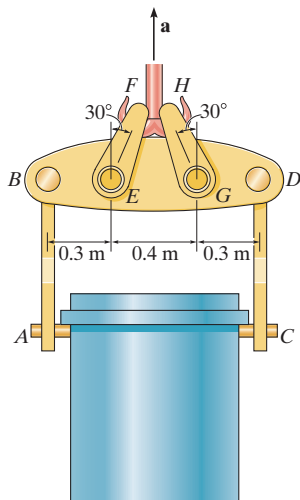
***17–24.** The jet aircraft has a total mass of 22 Mg and a center of mass at G . Initially at take-off the engines provide a thrust $2T = 4$ kN and $T' = 1.5$ kN. Determine the acceleration of the plane and the normal reactions on the nose wheel at A and each of the *two* wing wheels located at B . Neglect the mass of the wheels and, due to low velocity, neglect any lift caused by the wings.



Prob. 17–24

17–25. The 4-Mg uniform canister contains nuclear waste material encased in concrete. If the mass of the spreader beam BD is 50 kg, determine the force in each of the links AB , CD , EF , and GH when the system is lifted with an acceleration of $a = 2$ m/s² for a short period of time.

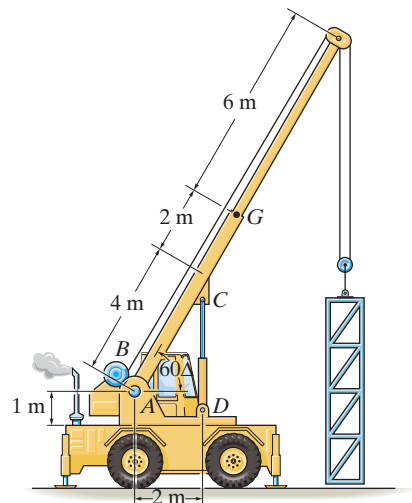
17–26. The 4-Mg uniform canister contains nuclear waste material encased in concrete. If the mass of the spreader beam BD is 50 kg, determine the largest vertical acceleration a of the system so that each of the links AB and CD are not subjected to a force greater than 30 kN and links EF and GH are not subjected to a force greater than 34 kN.



Probs. 17–25/26

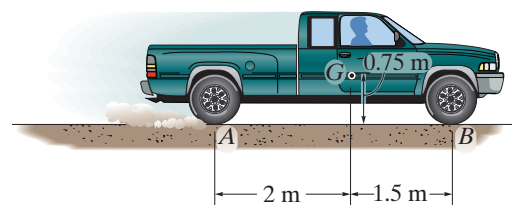
17–27. The assembly has a mass of 8 Mg and is hoisted using the boom and pulley system. If the winch at B draws in the cable with an acceleration of 2 m/s², determine the compressive force in the hydraulic cylinder needed to support the boom. The boom has a mass of 2 Mg and mass center at G .

***17–28.** The assembly has a mass of 4 Mg and is hoisted using the winch at B . Determine the greatest acceleration of the assembly so that the compressive force in the hydraulic cylinder supporting the boom does not exceed 180 kN. What is the tension in the supporting cable? The boom has a mass of 2 Mg and mass center at G .



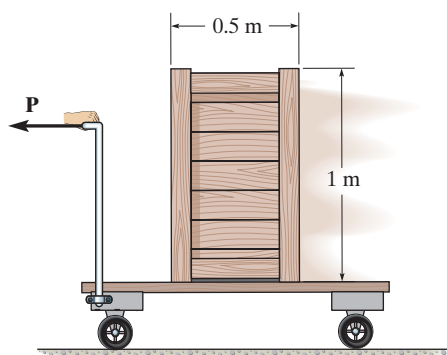
Probs. 17–27/28

17–29. Determine the shortest time possible for the rear-wheel drive, 2-Mg truck to achieve a speed of 16 m/s with a constant acceleration starting from rest. The coefficient of static friction between the wheels and the road surface is $\mu_s = 0.8$. The front wheels are free to roll. Neglect the mass of the wheels.



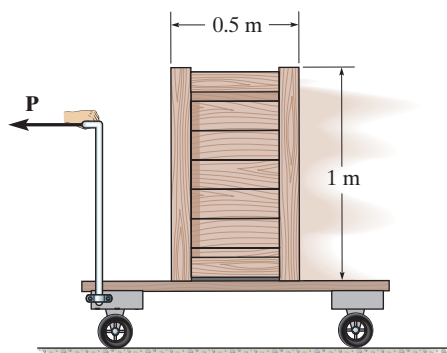
Prob. 17–29

17–30. The 150-kg uniform crate rests on the 10-kg cart. Determine the maximum force P that can be applied to the handle without causing the crate to slip or tip on the cart. The coefficient of static friction between the crate and cart is $\mu_s = 0.2$.



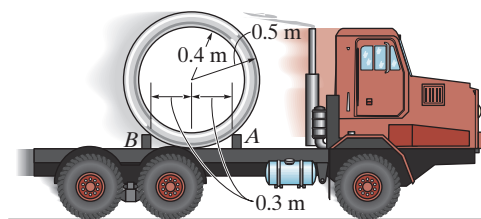
Prob. 17–30

17–31. The 150-kg uniform crate rests on the 10-kg cart. Determine the maximum force P that can be applied to the handle without causing the crate to tip on the cart. Slipping does not occur.



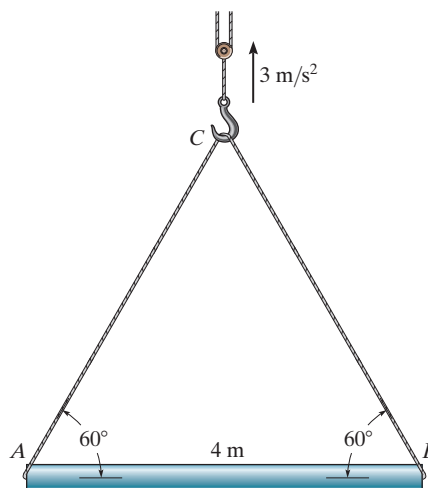
Prob. 17–31

***17–32.** The pipe has a mass of 460 kg and is held in place on the truck bed using the two boards A and B . Determine the greatest acceleration of the truck so that the pipe begins to lose contact at A and the bed of the truck and starts to pivot about B . Assume board B will not slip on the bed of the truck, and the pipe is smooth. Also, what force does board B exert on the pipe during the acceleration?



Prob. 17–32

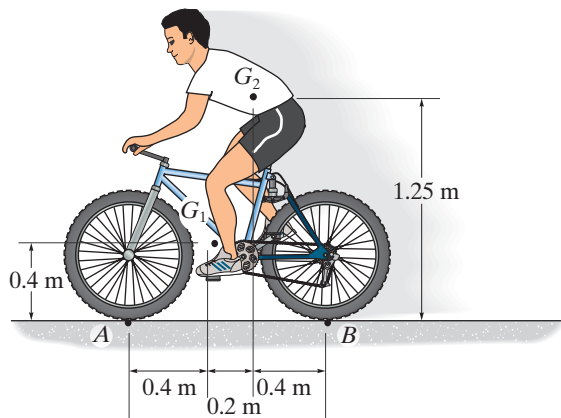
17–33. The uniform girder AB has a mass of 8 Mg. Determine the internal axial, shear, and bending-moment loadings at the center of the girder if a crane gives it an upward acceleration of 3 m/s^2 .



Prob. 17–33

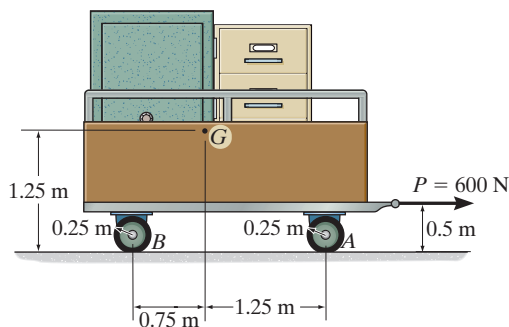
17-34. The mountain bike has a mass of 40 kg with center of mass at point G_1 , while the rider has a mass of 60 kg with center of mass at point G_2 . Determine the maximum deceleration when the brake is applied to the front wheel, without causing the rear wheel B to leave the road. Assume that the front wheel does not slip. Neglect the mass of all the wheels.

17-35. The mountain bike has a mass of 40 kg with center of mass at point G_1 , while the rider has a mass of 60 kg with center of mass at point G_2 . When the brake is applied to the front wheel, it causes the bike to decelerate at a constant rate of 3 m/s^2 . Determine the normal reaction the road exerts on the front and rear wheels. Assume that the rear wheel is free to roll. Neglect the mass of all the wheels.



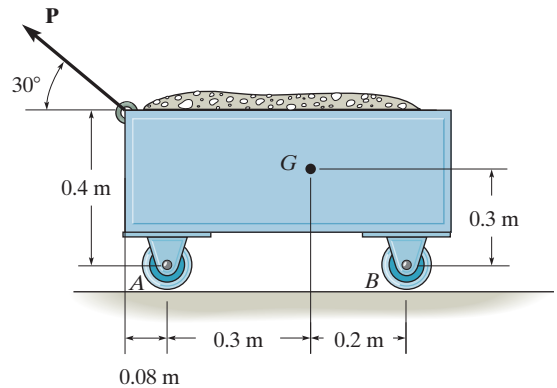
Probs. 17-34/35

***17-36.** The trailer with its load has a mass of 150-kg and a center of mass at G . If it is subjected to a horizontal force of $P = 600 \text{ N}$, determine the trailer's acceleration and the normal force on the pair of wheels at A and at B . The wheels are free to roll and have negligible mass.



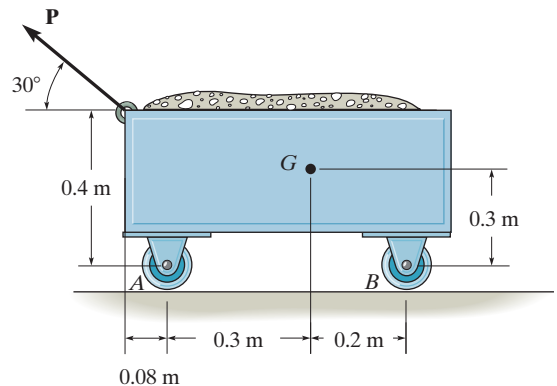
Prob. 17-36

17-37. A force of $P = 300 \text{ N}$ is applied to the 60-kg cart. Determine the reactions at both the wheels at A and both the wheels at B . Also, what is the acceleration of the cart? The mass center of the cart is at G .



Prob. 17-37

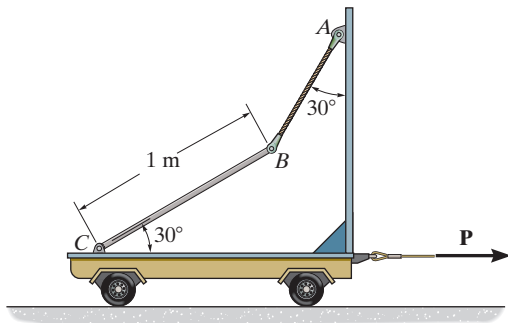
17-38. Determine the largest force P that can be applied to the 60-kg cart, without causing one of the wheel reactions, either at A or at B , to be zero. Also, what is the acceleration of the cart? The mass center of the cart is at G .



Prob. 17-38

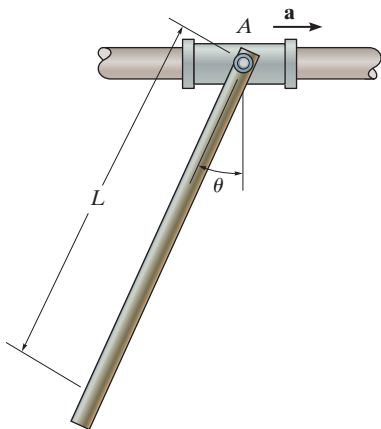
17–39. If the cart's mass is 30 kg and it is subjected to a horizontal force of $P = 90$ N, determine the tension in cord AB and the horizontal and vertical components of reaction on end C of the uniform 15-kg rod BC .

***17–40.** If the cart's mass is 30 kg, determine the horizontal force P that should be applied to the cart so that the cord AB just becomes slack. The uniform rod BC has a mass of 15 kg.



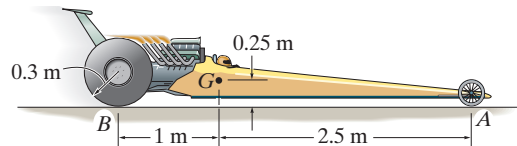
Probs. 17–39/40

17–41. The uniform bar of mass m is pin connected to the collar, which slides along the smooth horizontal rod. If the collar is given a constant acceleration of $\mathbf{a}$, determine the bar's inclination angle θ . Neglect the collar's mass.



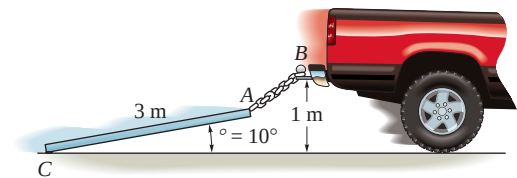
Prob. 17–41

17–42. The dragster has a mass of 1500 kg and a center of mass at G . If the coefficient of kinetic friction between the rear wheels and the pavement is $\mu_k = 0.6$, determine if it is possible for the driver to lift the front wheels, A , off the ground while the rear drive wheels are slipping. Neglect the mass of the wheels and assume that the front wheels are free to roll.



Probs. 17–42/43

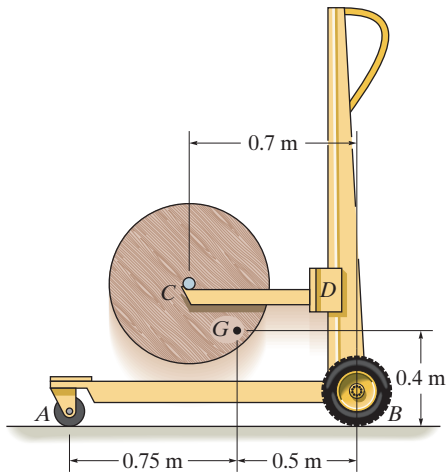
***17–44.** The pipe has a length of 3 m and a mass of 500 kg. It is attached to the back of the truck using a 0.6-m-long chain AB . If the coefficient of kinetic friction at C is $\mu_k = 0.4$, determine the acceleration of the truck if the angle $\theta = 10^\circ$ with the road as shown.



Prob. 17–44

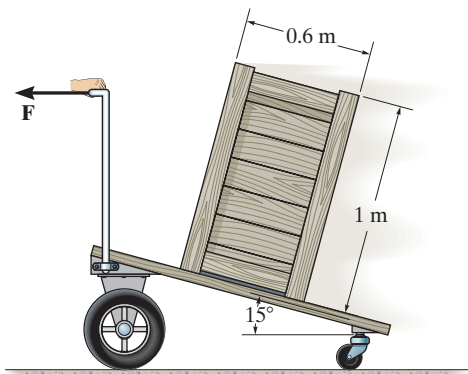
17-45. The lift truck has a mass of 70 kg and mass center at G . If it lifts the 120-kg spool with an acceleration of 3 m/s^2 , determine the reactions of each of the four wheels. The loading is symmetric. Neglect the mass of the movable arm CD .

17-46. The lift truck has a mass of 70 kg and mass center at G . Determine the largest upward acceleration of the 120-kg spool so that no reaction of the wheels on the ground exceeds 600 N.



Probs. 17-45/46

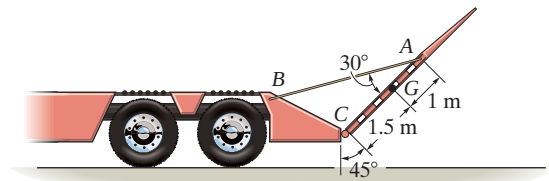
17-47. The uniform crate has a mass of 50 kg and rests on the cart having an inclined surface. Determine the smallest acceleration that will cause the crate either to tip or slip relative to the cart. What is the magnitude of this acceleration? The coefficient of static friction between the crate and cart is $\mu_s = 0.5$.



Prob. 17-47

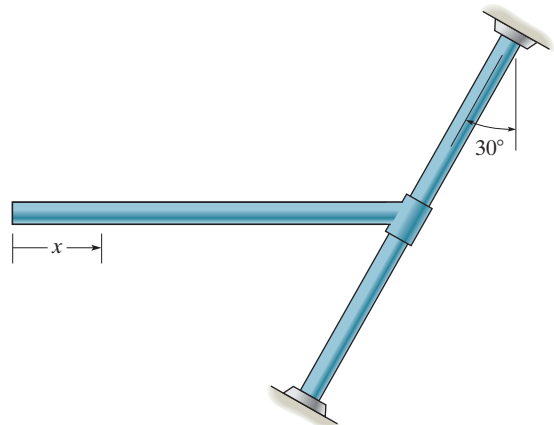
***17-48.** The drop gate at the end of the trailer has a mass of 1.25 Mg and mass center at G . If it is supported by the cable AB and hinge at C , determine the tension in the cable when the truck begins to accelerate at 5 m/s^2 . Also, what are the horizontal and vertical components of reaction at the hinge C ?

17-49. The drop gate at the end of the trailer has a mass of 1.25 Mg and mass center at G . If it is supported by the cable AB and hinge at C , determine the maximum deceleration of the truck so that the gate does not begin to rotate forward. What are the horizontal and vertical components of reaction at the hinge C ?



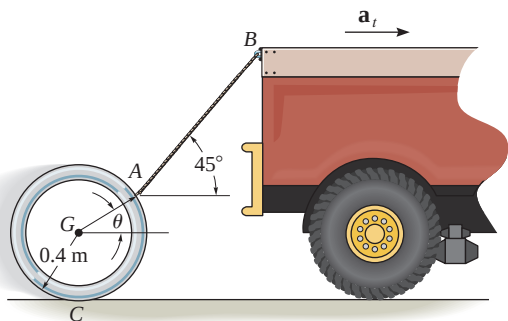
Probs. 17-48/49

17-50. The bar has a weight per length w and is supported by the smooth collar. If it is released from rest, determine the internal normal force, shear force, and bending moment in the bar as a function of x .



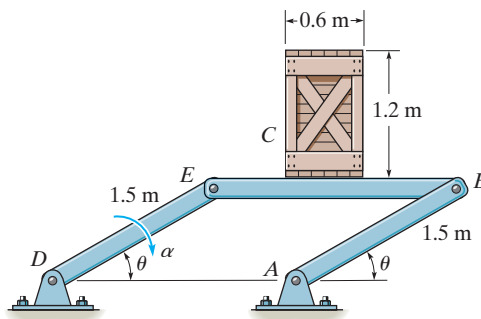
Prob. 17-50

17-51. The pipe has a mass of 800 kg and is being towed behind the truck. If the acceleration of the truck is $a_t = 0.5 \text{ m/s}^2$, determine the angle θ and the tension in the cable. The coefficient of kinetic friction between the pipe and the ground is $\mu_k = 0.1$.



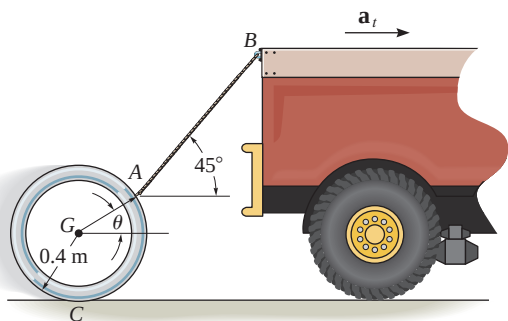
Prob. 17-51

17-53. The 100-kg uniform crate C rests on the elevator floor where the coefficient of static friction is $\mu_s = 0.4$. Determine the largest initial angular acceleration α , starting from rest at $\theta = 90^\circ$, without causing the crate to slip. No tipping occurs.



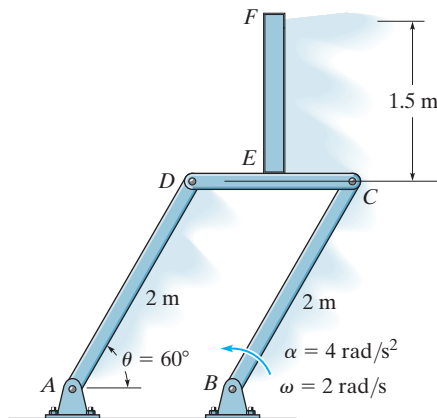
Prob. 17-53

***17-52.** The pipe has a mass of 800 kg and is being towed behind a truck. If the angle $\theta = 30^\circ$, determine the acceleration of the truck and the tension in the cable. The coefficient of kinetic friction between the pipe and the ground is $\mu_k = 0.1$.



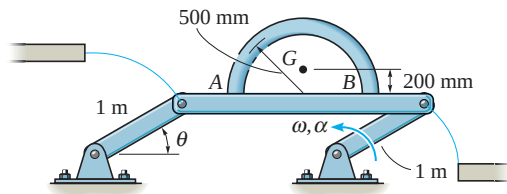
Prob. 17-52

17-54. The two uniform 4-kg bars DC and EF are fixed (welded) together at E . Determine the normal force N_E , shear force V_E , and moment M_E , which DC exerts on EF at E if at the instant $\theta = 60^\circ$. BC has an angular velocity $\omega = 2 \text{ rad/s}$ and an angular acceleration $\alpha = 4 \text{ rad/s}^2$ as shown.



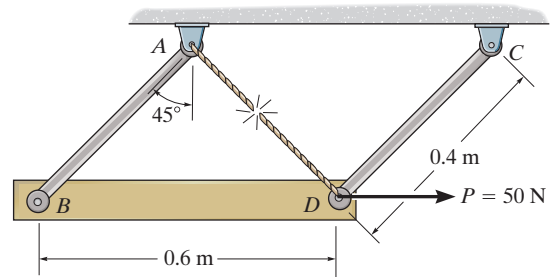
Prob. 17-54

17–55. The arched pipe has a mass of 80 kg and rests on the surface of the platform for which the coefficient of static friction is $\mu_s = 0.3$. Determine the greatest angular acceleration α of the platform, starting from rest when $\theta = 45^\circ$, without causing the pipe to slip on the platform.



Prob. 17–55

***17–56.** Determine the force developed in the links and the acceleration of the bar's mass center immediately after the cord fails. Neglect the mass of links AB and CD. The uniform bar has a mass of 20 kg.



Prob. 17–56

17.4 Equations of Motion: Rotation about a Fixed Axis

Consider the rigid body (or slab) shown in Fig. 17–13a, which is constrained to rotate in the vertical plane about a fixed axis perpendicular to the page and passing through the pin at O . The angular velocity and angular acceleration are caused by the external force and couple moment system acting on the body. Because the body's center of mass G moves around a *circular path*, the acceleration of this point is best represented by its tangential and normal components. The *tangential component of acceleration* has a magnitude of $(a_G)_t = \alpha r_G$ and must act in a *direction* which is *consistent* with the body's angular acceleration α . The *magnitude of the normal component of acceleration* is $(a_G)_n = \omega^2 r_G$. This component is *always directed* from point G to O , regardless of the rotational sense of ω .

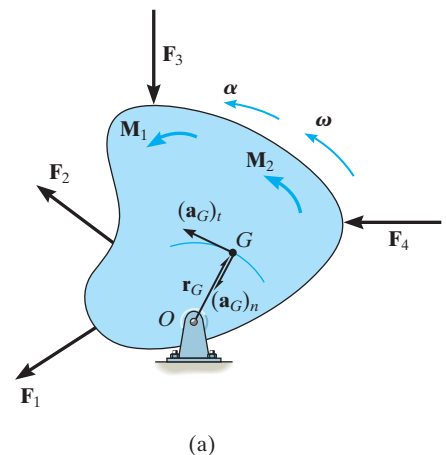
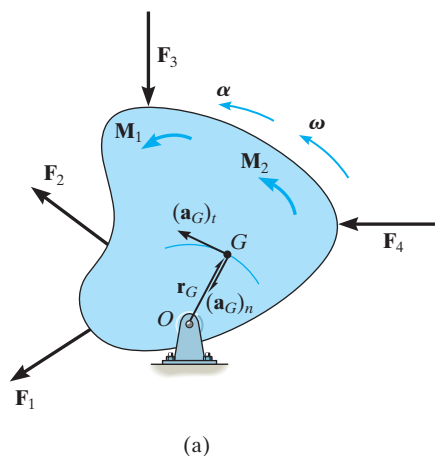


Fig. 17–13



The free-body and kinetic diagrams for the body are shown in Fig. 17-13b. The two components $m(\mathbf{a}_G)_t$ and $m(\mathbf{a}_G)_n$, shown on the kinetic diagram, are associated with the tangential and normal components of acceleration of the body's mass center. The $I_G \alpha$ vector acts in the same *direction* as α and has a *magnitude* of $I_G \alpha$, where I_G is the body's moment of inertia calculated about an axis which is perpendicular to the page and passes through G . From the derivation given in Sec. 17.2, the equations of motion which apply to the body can be written in the form

$$\begin{aligned}\Sigma F_n &= m(a_G)_n = m\omega^2 r_G \\ \Sigma F_t &= m(a_G)_t = m\alpha r_G \\ \Sigma M_G &= I_G \alpha\end{aligned}\quad (17-14)$$

The moment equation can be replaced by a moment summation about any arbitrary point P on or off the body provided one accounts for the moments $\Sigma(\mathcal{M}_k)_P$ produced by $I_G \alpha$, $m(\mathbf{a}_G)_t$, and $m(\mathbf{a}_G)_n$ about the point.

Moment Equation About Point O. Often it is convenient to sum moments about the pin at O in order to eliminate the *unknown* force $\mathbf{F}_O$. From the kinetic diagram, Fig. 17-13b, this requires

$$\zeta + \Sigma M_O = \Sigma(\mathcal{M}_k)_O; \quad \Sigma M_O = r_G m(a_G)_t + I_G \alpha \quad (17-15)$$

Note that the moment of $m(\mathbf{a}_G)_n$ is not included here since the line of action of this vector passes through O . Substituting $(a_G)_t = r_G \alpha$, we may rewrite the above equation as $\zeta + \Sigma M_O = (I_G + mr_G^2)\alpha$. From the parallel-axis theorem, $I_O = I_G + md^2$, and therefore the term in parentheses represents the *moment of inertia of the body about the fixed axis of rotation passing through O*.* Consequently, we can write the three equations of motion for the body as

$$\begin{aligned}\Sigma F_n &= m(a_G)_n = m\omega^2 r_G \\ \Sigma F_t &= m(a_G)_t = m\alpha r_G \\ \Sigma M_O &= I_O \alpha\end{aligned}\quad (17-16)$$

When using these equations, remember that “ $I_O \alpha$ ” accounts for the “moment” of *both* $m(\mathbf{a}_G)_t$ and $I_G \alpha$ about point O , Fig. 17-13b. In other words, $\Sigma M_O = \Sigma(\mathcal{M}_k)_O = I_O \alpha$, as indicated by Eqs. 17-15 and 17-16.

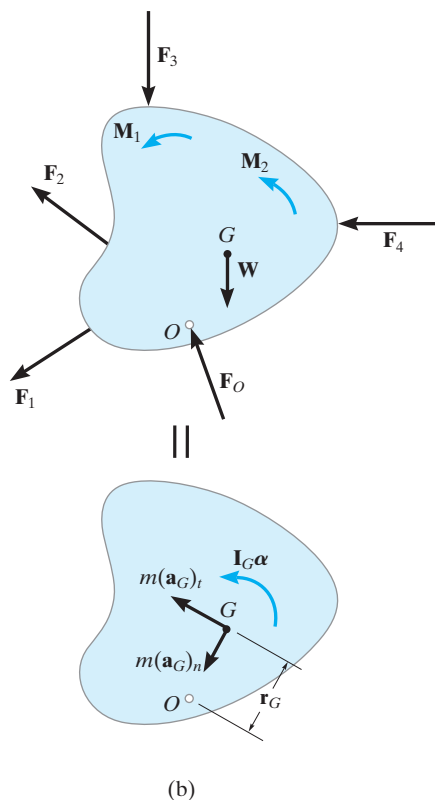


Fig. 17-13 (cont.)

*The result $\Sigma M_O = I_O \alpha$ can also be obtained *directly* from Eq. 17-6 by selecting point P to coincide with O , realizing that $(a_P)_x = (a_P)_y = 0$.

Procedure for Analysis

Kinetic problems which involve the rotation of a body about a fixed axis can be solved using the following procedure.

Free-Body Diagram.

- Establish the inertial n, t coordinate system and specify the direction and sense of the accelerations $(\mathbf{a}_G)_n$ and $(\mathbf{a}_G)_t$ and the angular acceleration α of the body. Recall that $(\mathbf{a}_G)_t$ must act in a direction which is in accordance with the rotational sense of α , whereas $(\mathbf{a}_G)_n$ always acts toward the axis of rotation, point O .
- Draw the free-body diagram to account for all the external forces and couple moments that act on the body.
- Determine the moment of inertia I_G or I_O .
- Identify the unknowns in the problem.
- If it is decided that the rotational equation of motion $\Sigma M_P = \Sigma (\mathcal{M}_k)_P$ is to be used, i.e., P is a point other than G or O , then consider drawing the kinetic diagram in order to help “visualize” the “moments” developed by the components $m(\mathbf{a}_G)_n$, $m(\mathbf{a}_G)_t$, and $I_G \alpha$ when writing the terms for the moment sum $\Sigma (\mathcal{M}_k)_P$.

Equations of Motion.

- Apply the three equations of motion in accordance with the established sign convention.
- If moments are summed about the body’s mass center, G , then $\Sigma M_G = I_G \alpha$, since $(m\mathbf{a}_G)_t$ and $(m\mathbf{a}_G)_n$ create no moment about G .
- If moments are summed about the pin support O on the axis of rotation, then $(m\mathbf{a}_G)_n$ creates no moment about O , and it can be shown that $\Sigma M_O = I_O \alpha$.

Kinematics.

- Use kinematics if a complete solution cannot be obtained strictly from the equations of motion.
- If the angular acceleration is variable, use

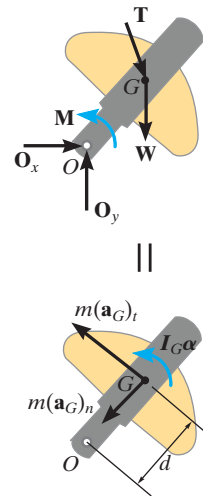
$$\alpha = \frac{d\omega}{dt} \quad \alpha d\theta = \omega d\omega \quad \omega = \frac{d\theta}{dt}$$

- If the angular acceleration is constant, use

$$\omega = \omega_0 + \alpha_c t$$

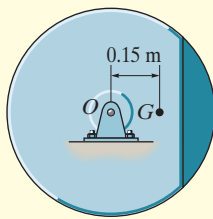
$$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha_c t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha_c(\theta - \theta_0)$$

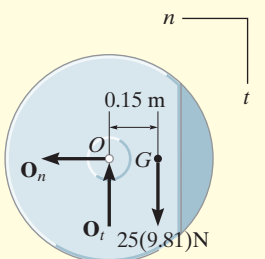


The crank on the oil-pumping rig undergoes rotation about a fixed axis which is caused by a driving torque $\mathbf{M}$ of the motor. The loadings shown on the free-body diagram cause the effects shown on the kinetic diagram. If moments are summed about the mass center, G , then $\Sigma M_G = I_G \alpha$. However, if moments are summed about point O , noting that $(a_G)_t = \alpha d$, then $\Sigma + \Sigma M_O = I_G \alpha + m(a_G)_t d + m(a_G)_n(0) = (I_G + md^2)\alpha = I_O \alpha$.

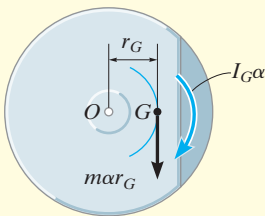
EXAMPLE 17.8



(a)



||



(b)

Fig. 17-14

The unbalanced 25-kg flywheel shown in Fig. 17-14a has a radius of gyration of $k_G = 0.18$ m about an axis passing through its mass center G . If it is released from rest, determine the horizontal and vertical components of reaction at the pin O .

SOLUTION

Free-Body and Kinetic Diagrams. Since G moves in a circular path, it will have both normal and tangential components of acceleration. Also, since α , which is caused by the flywheel's weight, acts clockwise, the tangential component of acceleration must act downward. Why? Since $\omega = 0$, only $m(a_G)_t = mar_G$ and $I_G\alpha$ are shown on the kinetic diagram in Fig. 17-14b. Here, the moment of inertia about G is

$$I_G = mk_G^2 = (25 \text{ kg})(0.18 \text{ m})^2 = 0.81 \text{ kg} \cdot \text{m}^2$$

The three unknowns are O_n , O_t , and α .

Equations of Motion.

$$\leftarrow \Sigma F_n = m\omega^2 r_G; \quad O_n = 0 \quad \text{Ans.}$$

$$+\downarrow \Sigma F_t = mar_G; \quad -O_t + 25(9.81) \text{ N} = (25 \text{ kg})(\alpha)(0.15 \text{ m}) \quad (1)$$

$$\curvearrowright \Sigma M_G = I_G\alpha; \quad O_t(0.15 \text{ m}) = (0.81 \text{ kg} \cdot \text{m}^2)\alpha$$

Solving,

$$\alpha = 26.8 \text{ rad/s}^2 \quad O_t = 144.7 \text{ N} \quad \text{Ans.}$$

Moments can also be summed about point O in order to eliminate O_n and O_t and thereby obtain a *direct solution* for α , Fig. 17-14b. This can be done in one of *two* ways.

$$\curvearrowright \Sigma M_O = \Sigma (\mathcal{M}_k)_O;$$

$$[25(9.81) \text{ N}](0.15 \text{ m}) = (0.81 \text{ kg} \cdot \text{m}^2)\alpha + [(25 \text{ kg})\alpha(0.15 \text{ m})](0.15 \text{ m})$$

$$245.25 \text{ N}(0.15 \text{ m}) = 1.3725\alpha \quad (2)$$

If $\Sigma M_O = I_O\alpha$ is applied, then by the parallel-axis theorem the moment of inertia of the flywheel about O is

$$I_O = I_G + mr_G^2 = 0.81 + (25)(0.15)^2 = 1.3725 \text{ kg} \cdot \text{m}^2$$

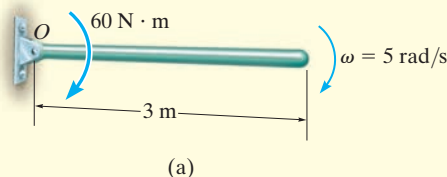
Hence,

$$\curvearrowright \Sigma M_O = I_O\alpha; \quad (245.25 \text{ N})(0.15 \text{ m}) = (1.3725 \text{ kg} \cdot \text{m}^2)\alpha$$

which is the same as Eq. 2. Solving for α and substituting into Eq. 1 yields the answer for O_t obtained previously.

EXAMPLE 17.9

At the instant shown in Fig. 17-15*a*, the 20-kg slender rod has an angular velocity of $\omega = 5 \text{ rad/s}$. Determine the angular acceleration and the horizontal and vertical components of reaction of the pin on the rod at this instant.

**SOLUTION**

Free-Body and Kinetic Diagrams. Fig. 17-15*b*. As shown on the kinetic diagram, point *G* moves around a circular path and so it has two components of acceleration. It is important that the tangential component $a_t = \alpha r_G$ act downward since it must be in accordance with the rotational sense of α . The three unknowns are O_n , O_t , and α .

Equation of Motion.

$$\begin{aligned} \pm \Sigma F_n &= m\omega^2 r_G; & O_n &= (20 \text{ kg})(5 \text{ rad/s})^2(1.5 \text{ m}) \\ + \downarrow \Sigma F_t &= m\alpha r_G; & -O_t + 20(9.81) \text{ N} &= (20 \text{ kg})(\alpha)(1.5 \text{ m}) \\ \curvearrowright \Sigma M_G &= I_G \alpha; & O_t(1.5 \text{ m}) + 60 \text{ N} \cdot \text{m} &= \left[\frac{1}{12}(20 \text{ kg})(3 \text{ m})^2 \right] \alpha \end{aligned}$$

Solving

$$O_n = 750 \text{ N} \quad O_t = 19.05 \text{ N} \quad \alpha = 5.90 \text{ rad/s}^2 \quad \text{Ans.}$$

A more direct solution to this problem would be to sum moments about point *O* to eliminate O_n and O_t and obtain a *direct solution* for α . Here,

$$\begin{aligned} \curvearrowright \Sigma M_O &= \Sigma (\mathcal{M}_k)_O; & 60 \text{ N} \cdot \text{m} + 20(9.81) \text{ N}(1.5 \text{ m}) &= \\ & \left[\frac{1}{12}(20 \text{ kg})(3 \text{ m})^2 \right] \alpha + [20 \text{ kg}(\alpha)(1.5 \text{ m})](1.5 \text{ m}) \\ & \alpha = 5.90 \text{ rad/s}^2 & \text{Ans.} \end{aligned}$$

Also, since $I_O = \frac{1}{3}ml^2$ for a slender rod, we can apply

$$\begin{aligned} \curvearrowright \Sigma M_O &= I_O \alpha; & 60 \text{ N} \cdot \text{m} + 20(9.81) \text{ N}(1.5 \text{ m}) &= \left[\frac{1}{3}(20 \text{ kg})(3 \text{ m})^2 \right] \alpha \\ & \alpha = 5.90 \text{ rad/s}^2 & \text{Ans.} \end{aligned}$$

NOTE: By comparison, the last equation provides the simplest solution for α and *does not* require use of the kinetic diagram.

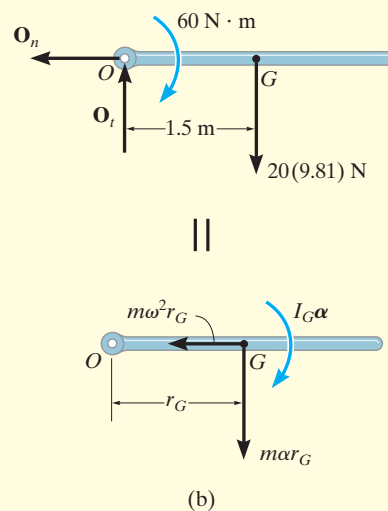
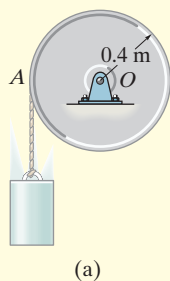
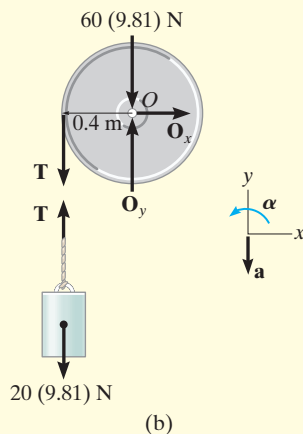


Fig. 17-15

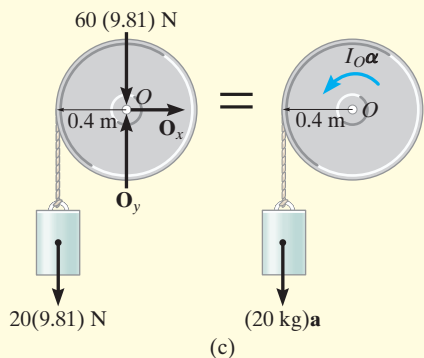
EXAMPLE 17.10



(a)



(b)



(c)

Fig. 17-16

The drum shown in Fig. 17-16a has a mass of 60 kg and a radius of gyration $k_O = 0.25$ m. A cord of negligible mass is wrapped around the periphery of the drum and attached to a block having a mass of 20 kg. If the block is released, determine the drum's angular acceleration.

SOLUTION I

Free-Body Diagram. Here we will consider the drum and block separately, Fig. 17-16b. Assuming the block accelerates *downward* at a , it creates a *counterclockwise* angular acceleration α of the drum. The moment of inertia of the drum is

$$I_O = mk_O^2 = (60 \text{ kg})(0.25 \text{ m})^2 = 3.75 \text{ kg} \cdot \text{m}^2$$

There are five unknowns, namely O_x , O_y , T , a , and α .

Equations of Motion. Applying the translational equations of motion $\Sigma F_x = m(a_G)_x$ and $\Sigma F_y = m(a_G)_y$ to the drum is of no consequence to the solution, since these equations involve the unknowns O_x and O_y . Thus, for the drum and block, respectively,

$$\zeta + \Sigma M_O = I_O \alpha; \quad T(0.4 \text{ m}) = (3.75 \text{ kg} \cdot \text{m}^2) \alpha \quad (1)$$

$$+\uparrow \Sigma F_y = m(a_G)_y; \quad -20(9.81) \text{ N} + T = -(20 \text{ kg})a \quad (2)$$

Kinematics. Since the point of contact A between the cord and drum has a tangential component of acceleration a , Fig. 17-16a, then

$$\zeta + a = a r; \quad a = \alpha(0.4 \text{ m}) \quad (3)$$

Solving the above equations,

$$T = 106 \text{ N} \quad a = 4.52 \text{ m/s}^2$$

$$\alpha = 11.3 \text{ rad/s}^2 \quad \text{Ans.}$$

SOLUTION II

Free-Body and Kinetic Diagrams. The cable tension T can be eliminated from the analysis by considering the drum and block as a *single system*, Fig. 17-16c. The kinetic diagram is shown since moments will be summed about point O.

Equations of Motion. Using Eq. 3 and applying the moment equation about O to eliminate the unknowns O_x and O_y , we have

$$\begin{aligned} \zeta + \Sigma M_O &= \Sigma (\mathcal{M}_k)_O; \quad [20(9.81) \text{ N}](0.4 \text{ m}) = \\ &= (3.75 \text{ kg} \cdot \text{m}^2) \alpha + [20 \text{ kg}(\alpha 0.4 \text{ m})](0.4 \text{ m}) \\ \alpha &= 11.3 \text{ rad/s}^2 \quad \text{Ans.} \end{aligned}$$

NOTE: If the block were *removed* and a force of 20(9.81) N were applied to the cord, show that $\alpha = 20.9 \text{ rad/s}^2$. This value is larger since the block has an inertia, or resistance to acceleration.

EXAMPLE 17.11

The slender rod shown in Fig. 17–17a has a mass m and length l and is released from rest when $\theta = 0^\circ$. Determine the horizontal and vertical components of force which the pin at A exerts on the rod at the instant $\theta = 90^\circ$.

SOLUTION

Free-Body and Kinetic Diagrams. The free-body diagram for the rod in the general position θ is shown in Fig. 17–17b. For convenience, the force components at A are shown acting in the n and t directions. Note that α acts clockwise and so $(\mathbf{a}_G)_t$ acts in the $+t$ direction.

The moment of inertia of the rod about point A is $I_A = \frac{1}{3}ml^2$.

Equations of Motion. Moments will be summed about A in order to eliminate A_n and A_t .

$$+\curvearrowright \Sigma F_n = m\omega^2 r_G; \quad A_n - mg \sin \theta = m\omega^2(l/2) \quad (1)$$

$$+\swarrow \Sigma F_t = m\alpha r_G; \quad A_t + mg \cos \theta = m\alpha(l/2) \quad (2)$$

$$\zeta + \Sigma M_A = I_A \alpha; \quad mg \cos \theta(l/2) = \left(\frac{1}{3}ml^2\right)\alpha \quad (3)$$

Kinematics. For a given angle θ there are four unknowns in the above three equations: A_n , A_t , ω , and α . As shown by Eq. 3, α is *not constant*; rather, it depends on the position θ of the rod. The necessary fourth equation is obtained using kinematics, where α and ω can be related to θ by the equation

$$(\zeta +) \quad \omega d\omega = \alpha d\theta \quad (4)$$

Note that the positive clockwise direction for this equation *agrees* with that of Eq. 3. This is important since we are seeking a simultaneous solution.

In order to solve for ω at $\theta = 90^\circ$, eliminate α from Eqs. 3 and 4, which yields

$$\omega d\omega = (1.5g/l) \cos \theta d\theta$$

Since $\omega = 0$ at $\theta = 0^\circ$, we have

$$\int_0^\omega \omega d\omega = (1.5g/l) \int_{0^\circ}^{90^\circ} \cos \theta d\theta$$

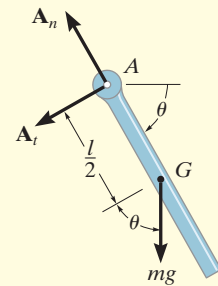
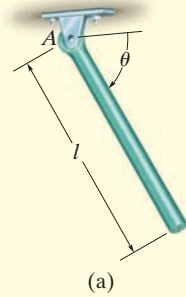
$$\omega^2 = 3g/l$$

Substituting this value into Eq. 1 with $\theta = 90^\circ$ and solving Eqs. 1 to 3 yields

$$\alpha = 0$$

$$A_t = 0 \quad A_n = 2.5 mg \quad \text{Ans.}$$

NOTE: If $\Sigma M_A = \Sigma (\mathcal{M}_k)_A$ is used, one must account for the moments of $I_G \alpha$ and $m(\mathbf{a}_G)_t$ about A .



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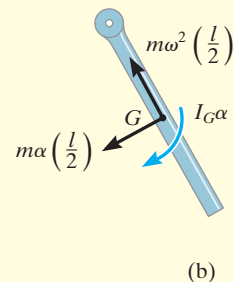
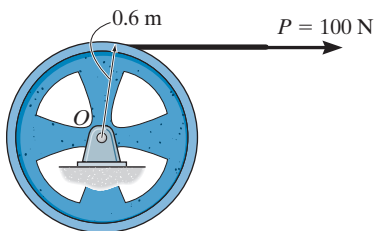


Fig. 17–17

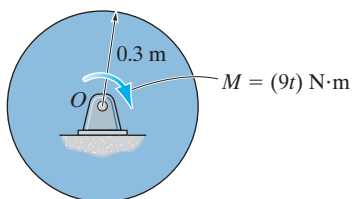
FUNDAMENTAL PROBLEMS

F17-7. The 100-kg wheel has a radius of gyration about its center O of $k_O = 500$ mm. If the wheel starts from rest, determine its angular velocity in $t = 3$ s.



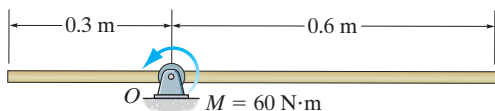
Prob. F17-7

F17-8. The 50-kg disk is subjected to the couple moment of $M = (9t) \text{ N}\cdot\text{m}$, where t is in seconds. Determine the angular velocity of the disk when $t = 4$ s starting from rest.



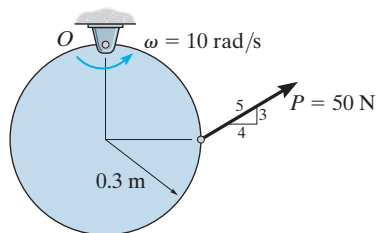
Prob. F17-8

F17-9. At the instant shown, the uniform 30-kg slender rod has a counterclockwise angular velocity of $\omega = 6 \text{ rad/s}$. Determine the tangential and normal components of reaction of pin O on the rod and the angular acceleration of the rod at this instant.



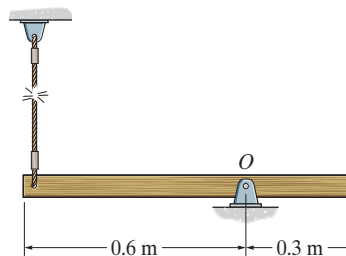
Prob. F17-9

F17-10. At the instant shown, the 30-kg disk has a counterclockwise angular velocity of $\omega = 10 \text{ rad/s}$. Determine the tangential and normal components of reaction of the pin O on the disk and the angular acceleration of the disk at this instant.



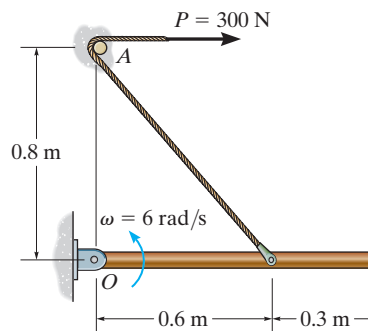
Prob. F17-10

F17-11. The uniform slender rod has a mass of 15 kg. Determine the horizontal and vertical components of reaction at the pin O , and the angular acceleration of the rod just after the cord is cut.



Prob. F17-11

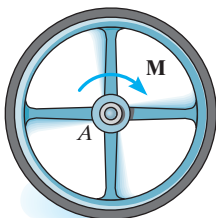
F17-12. The uniform 30-kg slender rod is being pulled by the cord that passes over the small smooth peg at A . If the rod has a counterclockwise angular velocity of $\omega = 6 \text{ rad/s}$ at the instant shown, determine the tangential and normal components of reaction at the pin O and the angular acceleration of the rod.



Prob. F17-12

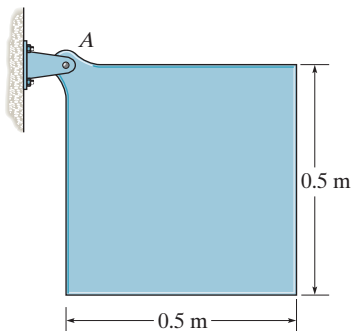
PROBLEMS

17-57. The 10-kg wheel has a radius of gyration $k_A = 200$ mm. If the wheel is subjected to a moment $M = (5t)$ N · m, where t is in seconds, determine its angular velocity when $t = 3$ s starting from rest. Also, compute the reactions which the fixed pin A exerts on the wheel during the motion.



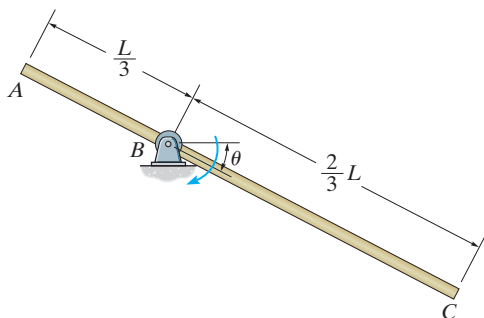
Prob. 17-57

17-58. The uniform 24-kg plate is released from rest at the position shown. Determine its initial angular acceleration and the horizontal and vertical reactions at the pin A .



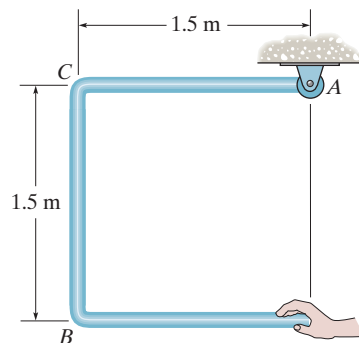
Prob. 17-58

17-59. The uniform slender rod has a mass m . If it is released from rest when $\theta = 0^\circ$, determine the magnitude of the reactive force exerted on it by pin B when $\theta = 90^\circ$.



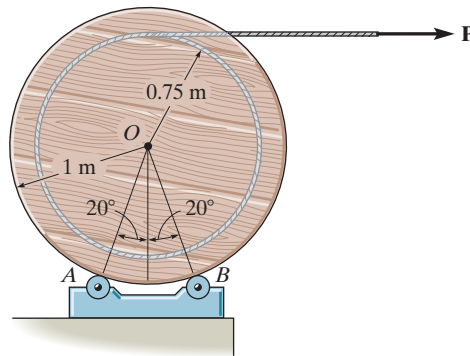
Prob. 17-59

***17-60.** The bent rod has a mass of 2 kg/m. If it is released from rest in the position shown, determine its initial angular acceleration and the horizontal and vertical components of reaction at A .



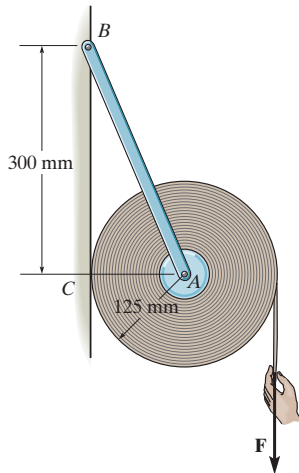
Prob. 17-60

17-61. If a horizontal force of $P = 100$ N is applied to the 300-kg reel of cable, determine its initial angular acceleration. The reel rests on rollers at A and B and has a radius of gyration of $k_O = 0.6$ m.



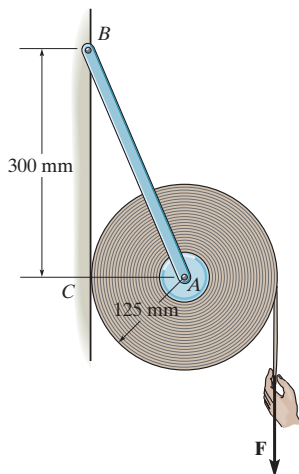
Prob. 17-61

17–62. The 20-kg roll of paper has a radius of gyration $k_A = 90$ mm about an axis passing through point A . It is pin supported at both ends by two brackets AB . If the roll rests against a wall for which the coefficient of kinetic friction is $\mu_k = 0.2$ and a vertical force $F = 30$ N is applied to the end of the paper, determine the angular acceleration of the roll as the paper unrolls.



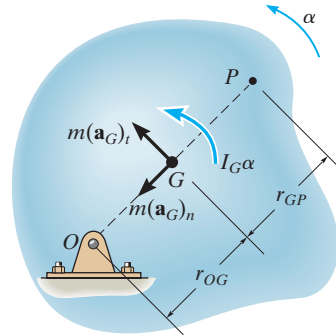
Prob. 17–62

17–63. The 20-kg roll of paper has a radius of gyration $k_A = 90$ mm about an axis passing through point A . It is pin supported at both ends by two brackets AB . If the roll rests against a wall for which the coefficient of kinetic friction is $\mu_k = 0.2$, determine the constant vertical force F that must be applied to the roll to pull off 1 m of paper in $t = 3$ s starting from rest. Neglect the mass of paper that is removed.



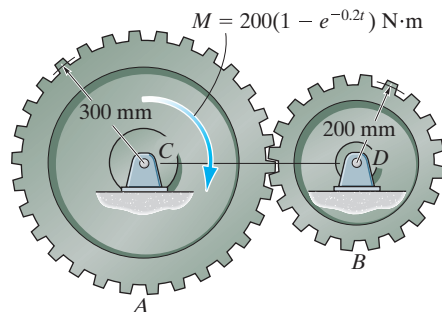
Prob. 17–63

***17–64.** The kinetic diagram representing the general rotational motion of a rigid body about a fixed axis passing through O is shown in the figure. Show that $I_G \alpha$ may be eliminated by moving the vectors $m(\mathbf{a}_G)_t$ and $m(\mathbf{a}_G)_n$ to point P , located a distance $r_{GP} = k_G^2 / r_{OG}$ from the center of mass G of the body. Here k_G represents the radius of gyration of the body about an axis passing through G . The point P is called the *center of percussion* of the body.



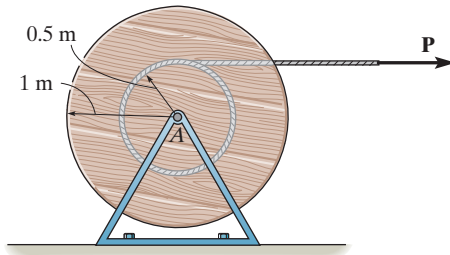
Prob. 17–64

17–65. Gears A and B have a mass of 50 kg and 15 kg, respectively. Their radii of gyration about their respective centers of mass are $k_C = 250$ mm and $k_D = 150$ mm. If a torque of $M = 200(1 - e^{-0.2t})$ N·m, where t is in seconds, is applied to gear A , determine the angular velocity of both gears when $t = 3$ s, starting from rest.



Prob. 17–65

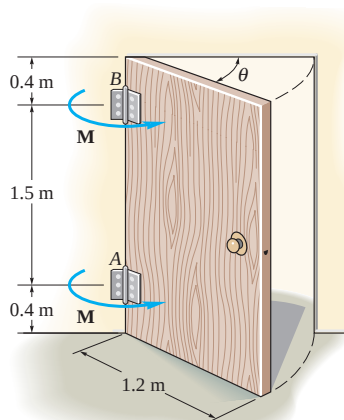
17–66. The reel of cable has a mass of 400 kg and a radius of gyration of $k_A = 0.75$ m. Determine its angular velocity when $t = 2$ s, starting from rest, if the force $\mathbf{P} = (20t^2 + 80)$ N, when t is in seconds. Neglect the mass of the unwound cable, and assume it is always at a radius of 0.5 m.



Prob. 17–66

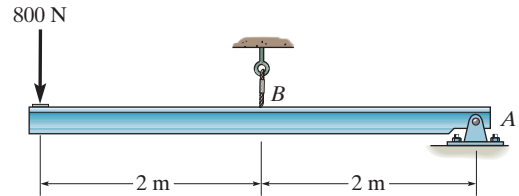
17–67. The door will close automatically using torsional springs mounted on the hinges. Each spring has a stiffness $k = 50$ N·m/rad so that the torque on each hinge is $M = (50\theta)$ N·m, where θ is measured in radians. If the door is released from rest when it is open at $\theta = 90^\circ$, determine its angular velocity at the instant $\theta = 0^\circ$. For the calculation, treat the door as a thin plate having a mass of 70 kg.

***17–68.** The door will close automatically using torsional springs mounted on the hinges. If the torque on each hinge is $M = k\theta$, where θ is measured in radians, determine the required torsional stiffness k so that the door will close ($\theta = 0^\circ$) with an angular velocity $\omega = 2$ rad/s when it is released from rest at $\theta = 90^\circ$. For the calculation, treat the door as a thin plate having a mass of 70 kg.



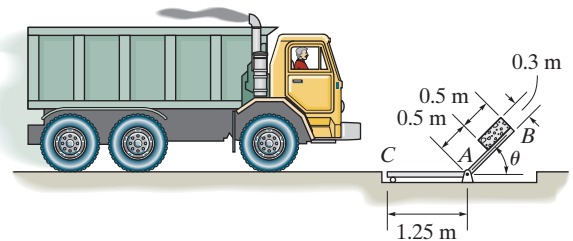
Probs. 17–67/68

17–69. If the cord at B suddenly fails, determine the horizontal and vertical components of the initial reaction at the pin A , and the angular acceleration of the 120-kg beam. Treat the beam as a uniform slender rod.



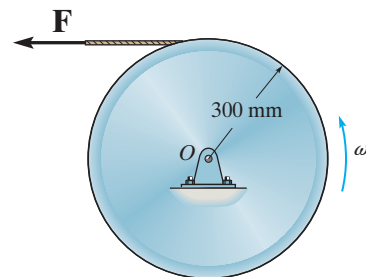
Prob. 17–69

17–70. The device acts as a pop-up barrier to prevent the passage of a vehicle. It consists of a 100-kg steel plate AC and a 200-kg counterweight solid concrete block located as shown. Determine the moment of inertia of the plate and block about the hinged axis through A . Neglect the mass of the supporting arms AB . Also, determine the initial angular acceleration of the assembly when it is released from rest at $\theta = 45^\circ$.



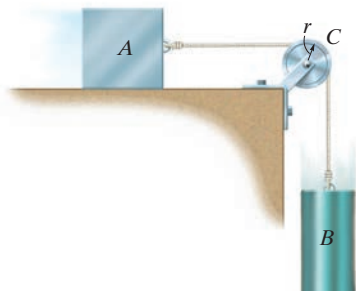
Prob. 17–70

17–71. A cord is wrapped around the outer surface of the 8-kg disk. If a force of $F = (\frac{1}{4}\theta^2)$ N, where θ is in radians, is applied to the cord, determine the disk's angular acceleration when it has turned 5 revolutions. The disk has an initial angular velocity of $\omega_0 = 1$ rad/s.



Prob. 17–71

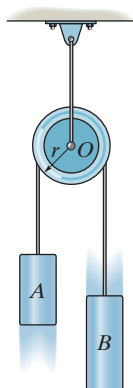
***17-72.** Block A has a mass m and rests on a surface having a coefficient of kinetic friction μ_k . The cord attached to A passes over a pulley at C and is attached to a block B having a mass $2m$. If B is released, determine the acceleration of A . Assume that the cord does not slip over the pulley. The pulley can be approximated as a thin disk of radius r and mass $\frac{1}{4}m$. Neglect the mass of the cord.



Prob. 17-72

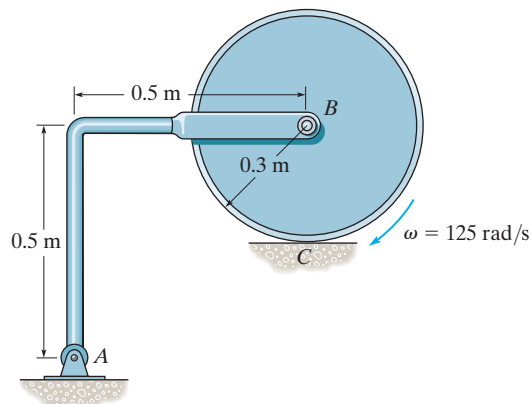
17 **17-73.** The two blocks A and B have a mass of 5 kg and 10 kg, respectively. If the pulley can be treated as a disk of mass 3 kg and radius 0.15 m, determine the acceleration of block A . Neglect the mass of the cord and any slipping on the pulley.

17-74. The two blocks A and B have a mass m_A and m_B , respectively, where $m_B > m_A$. If the pulley can be treated as a disk of mass M , determine the acceleration of block A . Neglect the mass of the cord and any slipping on the pulley.



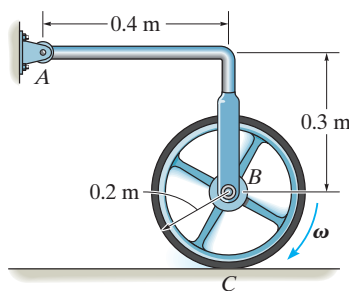
Probs. 17-73/74

17-75. The 30-kg disk is originally spinning at $\omega = 125$ rad/s. If it is placed on the ground, for which the coefficient of kinetic friction is $\mu_C = 0.5$, determine the time required for the motion to stop. What are the horizontal and vertical components of force which the member AB exerts on the pin at A during this time? Neglect the mass of AB .



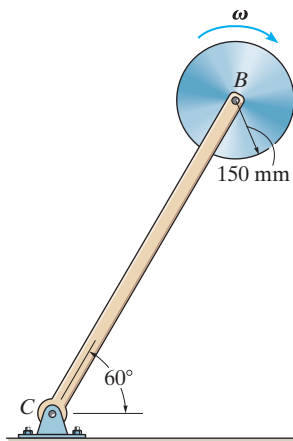
Prob. 17-75

***17-76.** The wheel has a mass of 25 kg and a radius of gyration $k_B = 0.15$ m. It is originally spinning at $\omega = 40$ rad/s. If it is placed on the ground, for which the coefficient of kinetic friction is $\mu_C = 0.5$, determine the time required for the motion to stop. What are the horizontal and vertical components of reaction which the pin at A exerts on AB during this time? Neglect the mass of AB .



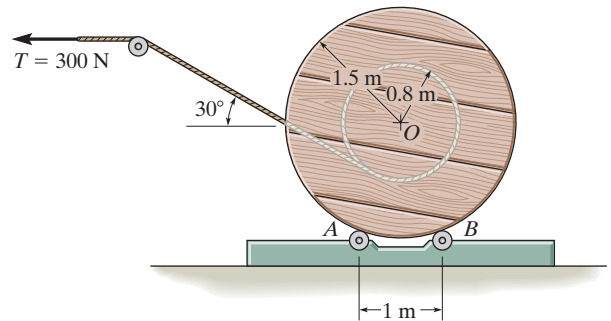
Prob. 17-76

17-77. The disk has a mass of 20 kg and is originally spinning at the end of the strut with an angular velocity of $\omega = 60 \text{ rad/s}$. If it is then placed against the wall, for which the coefficient of kinetic friction is $\mu_k = 0.3$ determine the time required for the motion to stop. What is the force in strut BC during this time?



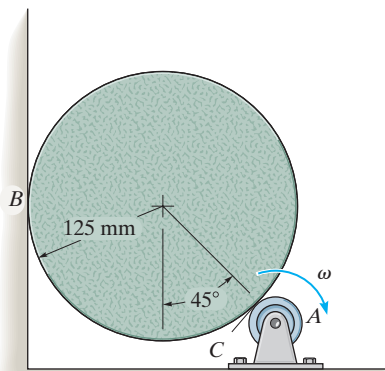
Prob. 17-77

17-79. Cable is unwound from a spool supported on small rollers at A and B by exerting a force $T = 300 \text{ N}$ on the cable. Compute the time needed to unravel 5 m of cable from the spool if the spool and cable have a total mass of 600 kg and a radius of gyration of $k_O = 1.2 \text{ m}$. For the calculation, neglect the mass of the cable being unwound and the mass of the rollers at A and B . The rollers turn with no friction.



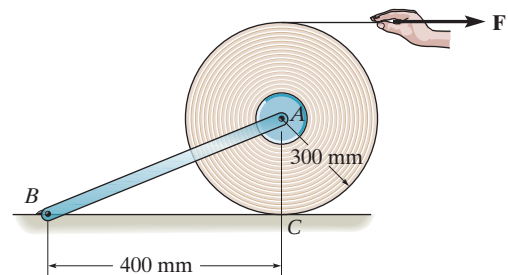
Prob. 17-79

17-78. The 5-kg cylinder is initially at rest when it is placed in contact with the wall B and the rotor at A . If the rotor always maintains a constant clockwise angular velocity $\omega = 6 \text{ rad/s}$, determine the initial angular acceleration of the cylinder. The coefficient of kinetic friction at the contacting surfaces B and C is $\mu_k = 0.2$.



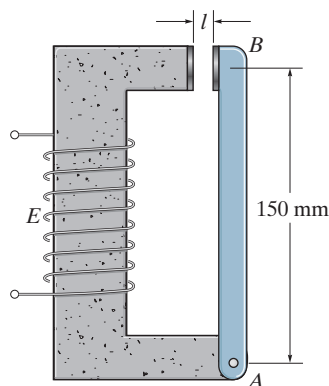
Prob. 17-78

***17-80.** The 20-kg roll of paper has a radius of gyration $k_A = 120 \text{ mm}$ about an axis passing through point A . It is pin supported at both ends by two brackets AB . The roll rests on the floor, for which the coefficient of kinetic friction is $\mu_k = 0.2$. If a horizontal force $F = 60 \text{ N}$ is applied to the end of the paper, determine the initial angular acceleration of the roll as the paper unrolls.



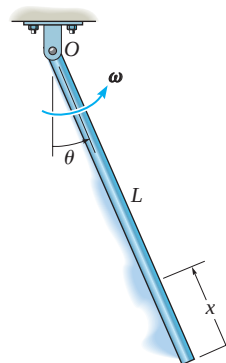
Prob. 17-80

17–81. The armature (slender rod) AB has a mass of 0.2 kg and can pivot about the pin at A . Movement is controlled by the electromagnet E , which exerts a horizontal attractive force on the armature at B of $F_B = (0.2(10^{-3})l^2) \text{ N}$, where l in meters is the gap between the armature and the magnet at any instant. If the armature lies in the horizontal plane, and is originally at rest, determine the speed of the contact at B the instant $l = 0.01 \text{ m}$. Originally $l = 0.02 \text{ m}$.



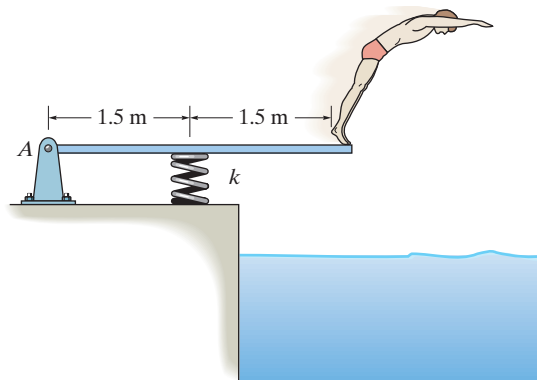
Prob. 17–81

17–83. The bar has a weight per length of w . If it is rotating in the vertical plane at a constant rate ω about point O , determine the internal normal force, shear force, and moment as a function of x and θ .



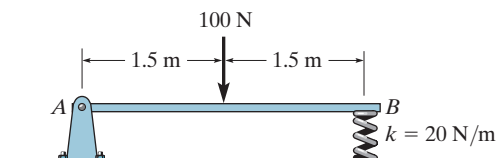
Prob. 17–83

***17–84.** Determine the angular acceleration of the 25-kg diving board and the horizontal and vertical components of reaction at the pin A the instant the man jumps off. Assume that the board is uniform and rigid, and that at the instant he jumps off the spring is compressed a maximum amount of 200 mm , $\omega = 0$, and the board is horizontal. Take $k = 7 \text{ kN/m}$.



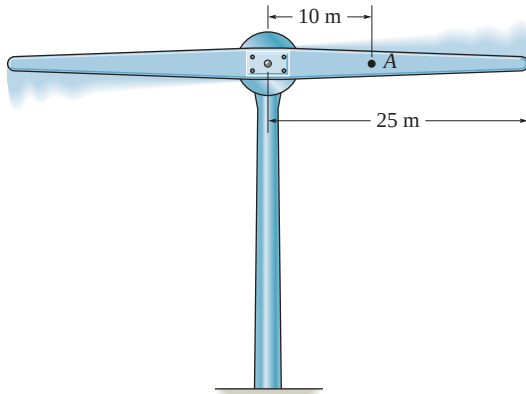
Prob. 17–84

17–82. The 4-kg slender rod is initially supported horizontally by a spring at B and pin at A . Determine the angular acceleration of the rod and the acceleration of the rod's mass center at the instant the 100-N force is applied.



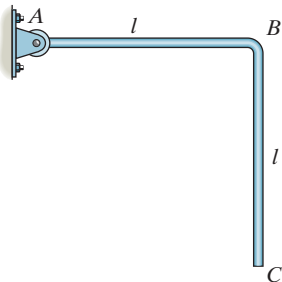
Prob. 17–82

17–85. The lightweight turbine consists of a rotor which is powered from a torque applied at its center. At the instant the rotor is horizontal it has an angular velocity of 15 rad/s and a clockwise angular acceleration of 8 rad/s^2 . Determine the internal normal force, shear force, and moment at a section through A . Assume the rotor is a 50-m -long slender rod, having a mass of 3 kg/m .



Prob. 17–85

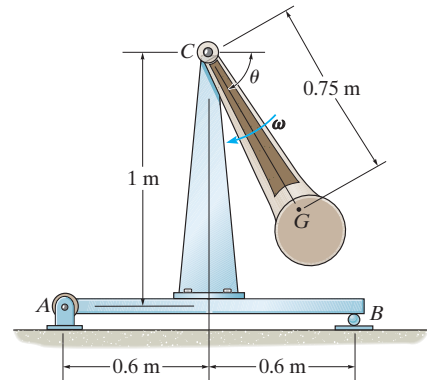
17–86. The two-bar assembly is released from rest in the position shown. Determine the initial bending moment at the fixed joint B . Each bar has a mass m and length l .



Prob. 17–86

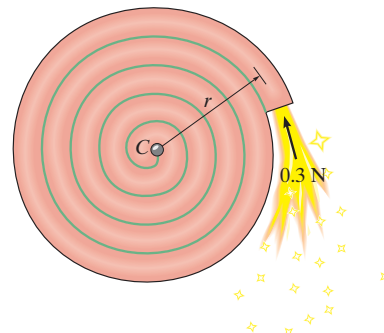
17–87. The 100-kg pendulum has a center of mass at G and a radius of gyration about G of $k_G = 250 \text{ mm}$. Determine the horizontal and vertical components of reaction on the beam by the pin A and the normal reaction of the roller B at the instant $\theta = 90^\circ$ when the pendulum is rotating at $\omega = 8 \text{ rad/s}$. Neglect the weight of the beam and the support.

***17–88.** The 100-kg pendulum has a center of mass at G and a radius of gyration about G of $k_G = 250 \text{ mm}$. Determine the horizontal and vertical components of reaction on the beam by the pin A and the normal reaction of the roller B at the instant $\theta = 0^\circ$ when the pendulum is rotating at $\omega = 4 \text{ rad/s}$. Neglect the weight of the beam and the support.



Probs. 17–87/88

17–89. The “Catherine wheel” is a firework that consists of a coiled tube of powder which is pinned at its center. If the powder burns at a constant rate of 20 g/s such that the exhaust gases always exert a force having a constant magnitude of 0.3 N , directed tangent to the wheel, determine the angular velocity of the wheel when 75% of the mass is burned off. Initially, the wheel is at rest and has a mass of 100 g and a radius of $r = 75 \text{ mm}$. For the calculation, consider the wheel to always be a thin disk.



Prob. 17–89

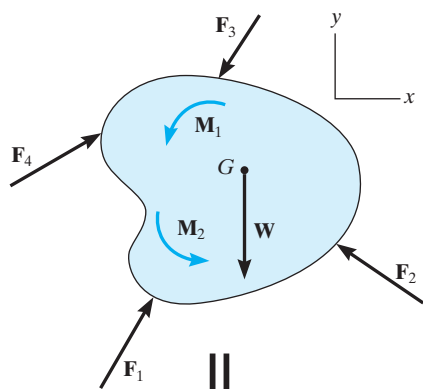
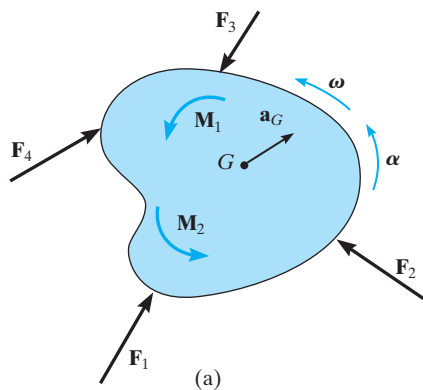


Fig. 17-18

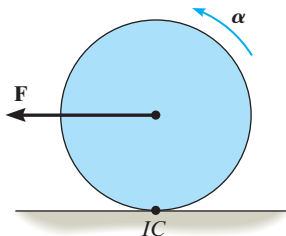


Fig. 17-19

17.5 Equations of Motion: General Plane Motion

The rigid body (or slab) shown in Fig. 17–18a is subjected to general plane motion caused by the externally applied force and couple-moment system. The free-body and kinetic diagrams for the body are shown in Fig. 17–18b. If an x and y inertial coordinate system is established as shown, the three equations of motion are

$$\begin{aligned}\Sigma F_x &= m(a_G)_x \\ \Sigma F_y &= m(a_G)_y \\ \Sigma M_G &= I_G \alpha\end{aligned}\quad (17-17)$$

In some problems it may be convenient to sum moments about a point P other than G in order to eliminate as many unknown forces as possible from the moment summation. When used in this more general case, the three equations of motion are

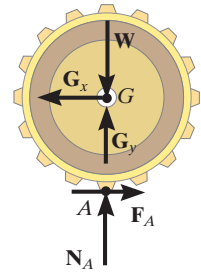
$$\begin{aligned}\Sigma F_x &= m(a_G)_x \\ \Sigma F_y &= m(a_G)_y \\ \Sigma M_P &= \Sigma (\mathcal{M}_k)_P\end{aligned}\quad (17-18)$$

Here $\Sigma (\mathcal{M}_k)_P$ represents the moment sum of $I_G \alpha$ and $m \mathbf{a}_G$ (or its components) about P as determined by the data on the kinetic diagram.

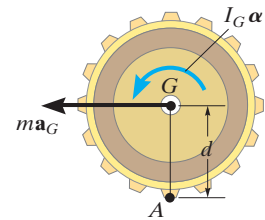
Moment Equation About the IC. There is a particular type of problem that involves a uniform disk, or body of circular shape, that rolls on a rough surface *without slipping*, Fig. 17–19. If we sum the moments about the instantaneous center of zero velocity, then $\Sigma (\mathcal{M}_k)_{IC}$ becomes $I_{IC} \alpha$, so that

$$\Sigma M_{IC} = I_{IC} \alpha \quad (17-19)$$

This result compares with $\Sigma M_O = I_O \alpha$, which is used for a body pinned at point O , Eq. 17–16. See Prob. 17–90.



II



Procedure for Analysis

Kinetic problems involving general plane motion of a rigid body can be solved using the following procedure.

Free-Body Diagram.

- Establish the x, y inertial coordinate system and draw the free-body diagram for the body.
- Specify the direction and sense of the acceleration of the mass center, $\mathbf{a}_G$, and the angular acceleration α of the body.
- Determine the moment of inertia I_G .
- Identify the unknowns in the problem.
- If it is decided that the rotational equation of motion $\Sigma M_P = \Sigma (\mathcal{M}_k)_P$ is to be used, then consider drawing the kinetic diagram in order to help “visualize” the “moments” developed by the components $m(\mathbf{a}_G)_x$, $m(\mathbf{a}_G)_y$, and $I_G \alpha$ when writing the terms in the moment sum $\Sigma (\mathcal{M}_k)_P$.

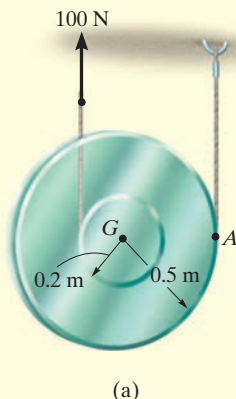
Equations of Motion.

- Apply the three equations of motion in accordance with the established sign convention.
- When friction is present, there is the possibility for motion with no slipping or tipping. Each possibility for motion should be considered.

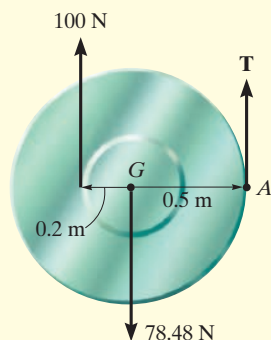
Kinematics.

- Use kinematics if a complete solution cannot be obtained strictly from the equations of motion.
- If the body's motion is *constrained* due to its supports, additional equations may be obtained by using $\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$, which relates the accelerations of any two points A and B on the body.
- When a wheel, disk, cylinder, or ball *rolls without slipping*, then $a_G = ar$.

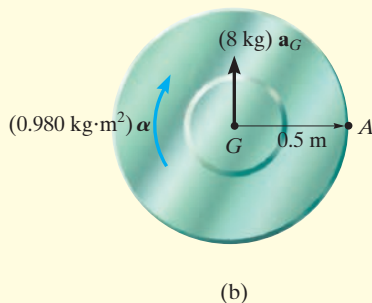
As the soil compactor, or “sheep’s foot roller” moves forward, the roller has general plane motion. The forces shown on its free-body diagram cause the effects shown on the kinetic diagram. If moments are summed about the mass center, G , then $\Sigma M_G = I_G \alpha$. However, if moments are summed about point A (the IC) then $\zeta + \Sigma M_A = I_G \alpha + (ma_G)d = I_A \alpha$.

EXAMPLE 17.12

(a)



||



(b)

Fig. 17-20

Determine the angular acceleration of the spool in Fig. 17-20a. The spool has a mass of 8 kg and a radius of gyration of $k_G = 0.35$ m. The cords of negligible mass are wrapped around its inner hub and outer rim.

SOLUTION I

Free-Body and Kinetic Diagrams. Fig. 17-20b. The 100-N force causes $\mathbf{a}_G$ to act upward. Also, α acts clockwise, since the spool winds around the cord at A.

There are three unknowns T , a_G , and α . The moment of inertia of the spool about its mass center is

$$I_G = mk_G^2 = 8 \text{ kg}(0.35 \text{ m})^2 = 0.980 \text{ kg} \cdot \text{m}^2$$

Equations of Motion.

$$+\uparrow \Sigma F_y = m(a_G)_y; \quad T + 100 \text{ N} - 78.48 \text{ N} = (8 \text{ kg})a_G \quad (1)$$

$$\curvearrowright + \Sigma M_G = I_G \alpha; \quad 100 \text{ N}(0.2 \text{ m}) - T(0.5 \text{ m}) = (0.980 \text{ kg} \cdot \text{m}^2)\alpha \quad (2)$$

Kinematics. A complete solution is obtained if kinematics is used to relate a_G to α . In this case the spool “rolls without slipping” on the cord at A. Hence, we can use the results of Example 16.4 or 16.15 so that,

$$(\curvearrowright +) a_G = \alpha r; \quad a_G = \alpha (0.5 \text{ m}) \quad (3)$$

Solving Eqs. 1 to 3, we have

$$\alpha = 10.3 \text{ rad/s}^2 \quad \text{Ans.}$$

$$a_G = 5.16 \text{ m/s}^2$$

$$T = 19.8 \text{ N}$$

SOLUTION II

Equations of Motion. We can eliminate the unknown T by summing moments about point A. From the free-body and kinetic diagrams Figs. 17-20b and 17-20c, we have

$$\begin{aligned} \curvearrowright + \Sigma M_A &= \Sigma (\mathcal{M}_k)_A; & 100 \text{ N}(0.7 \text{ m}) - 78.48 \text{ N}(0.5 \text{ m}) \\ &= (0.980 \text{ kg} \cdot \text{m}^2)\alpha + [(8 \text{ kg})a_G](0.5 \text{ m}) \end{aligned}$$

Using Eq. (3),

$$\alpha = 10.3 \text{ rad/s}^2 \quad \text{Ans.}$$

SOLUTION III

Equations of Motion. The simplest way to solve this problem is to realize that point A is the IC for the spool. Then Eq. 17-19 applies.

$$\begin{aligned} \curvearrowright + \Sigma M_A &= I_A \alpha; & (100 \text{ N})(0.7 \text{ m}) - (78.48 \text{ N})(0.5 \text{ m}) \\ &= [0.980 \text{ kg} \cdot \text{m}^2 + (8 \text{ kg})(0.5 \text{ m})^2]\alpha \\ \alpha &= 10.3 \text{ rad/s}^2 \end{aligned}$$

EXAMPLE 17.13

The 25-kg wheel shown in Fig. 17-21a has a radius of gyration $k_G = 0.2$ m. If a $50 \text{ N} \cdot \text{m}$ couple moment is applied to the wheel, determine the acceleration of its mass center G . The coefficients of static and kinetic friction between the wheel and the plane at A are $\mu_s = 0.3$ and $\mu_k = 0.25$, respectively.

SOLUTION

Free-Body Diagram. By inspection of Fig. 17-21b, it is seen that the couple moment causes the wheel to have a clockwise angular acceleration of α . As a result, the acceleration of the mass center, $\mathbf{a}_G$, is directed to the right. The moment of inertia is

$$I_G = mk_G^2 = 25 \text{ kg} (0.2 \text{ m})^2 = 1.0 \text{ kg} \cdot \text{m}^2$$

The unknowns are N_A , F_A , a_G , and α .

Equations of Motion.

$$\rightarrow \Sigma F_x = m(a_G)_x; \quad F_A = (25 \text{ kg}) a_G \quad (1)$$

$$+\uparrow \Sigma F_y = m(a_G)_y; \quad N_A - 25(9.81) \text{ N} = 0 \quad (2)$$

$$\zeta + \Sigma M_G = I_G \alpha; \quad 50 \text{ N} \cdot \text{m} - 0.36 \text{ m}(F_A) = (1.0 \text{ kg} \cdot \text{m}^2) \alpha \quad (3)$$

A fourth equation is needed for a complete solution.

Kinematics (No Slipping). If this assumption is made, then

$$(\zeta +) \quad a_G = (0.36 \text{ m}) \alpha \quad (4)$$

Solving Eqs. 1 to 4,

$$\begin{aligned} N_A &= 245.25 \text{ N} & F_A &= 106.1 \text{ N} \\ \alpha &= 11.79 \text{ rad/s}^2 & a_G &= 4.245 \text{ m/s}^2 \end{aligned}$$

This solution requires that no slipping occurs, i.e., $F_A \leq \mu_s N_A$. However, since $106.1 \text{ N} > 0.3(25)(9.81) \text{ N} = 73.6 \text{ N}$, the wheel slips as it rolls.

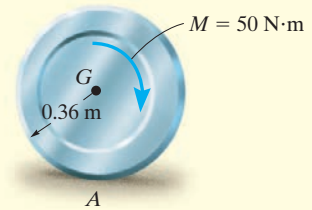
(Slipping). Equation 4 is not valid, and so $F_A = \mu_k N_A$, or

$$F_A = 0.25 N_A \quad (5)$$

Solving Eqs. 1 to 3 and 5 yields

$$\begin{aligned} N_A &= 245.25 \text{ N} & F_A &= 61.31 \text{ N} \\ \alpha &= 27.93 \text{ rad/s}^2 \\ a_G &= 2.45 \text{ m/s}^2 \rightarrow \end{aligned}$$

Ans.



(a)

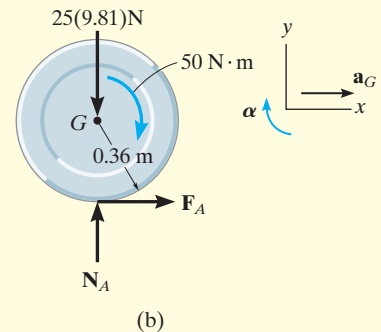


Fig. 17-21

EXAMPLE 17.14

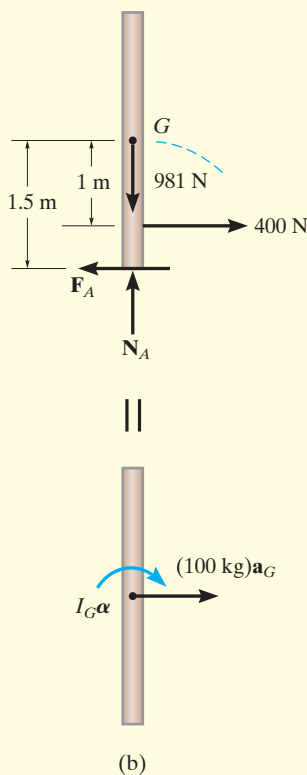
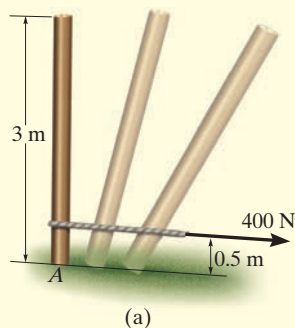


Fig. 17-22

The uniform slender pole shown in Fig. 17-22a has a mass of 100 kg. If the coefficients of static and kinetic friction between the end of the pole and the surface are $\mu_s = 0.3$, and $\mu_k = 0.25$, respectively, determine the pole's angular acceleration at the instant the 400-N horizontal force is applied. The pole is originally at rest.

SOLUTION

Free-Body and Kinetic Diagrams. Figure 17-22b. The path of motion of the mass center G will be along an unknown curved path having a radius of curvature ρ , which is initially on a vertical line. However, there is no normal or y component of acceleration since the pole is originally at rest, i.e., $\mathbf{v}_G = \mathbf{0}$, so that $(a_G)_y = v_G^2/\rho = 0$. We will assume the mass center accelerates to the right and that the pole has a clockwise angular acceleration of α . The unknowns are N_A , F_A , a_G , and α .

Equation of Motion.

$$\rightarrow \Sigma F_x = m(a_G)_x; \quad 400 \text{ N} - F_A = (100 \text{ kg})a_G \quad (1)$$

$$+\uparrow \Sigma F_y = m(a_G)_y; \quad N_A - 981 \text{ N} = 0 \quad (2)$$

$$\curvearrowright \Sigma M_G = I_G \alpha; \quad F_A(1.5 \text{ m}) - (400 \text{ N})(1 \text{ m}) = \left[\frac{1}{12}(100 \text{ kg})(3 \text{ m})^2\right]\alpha \quad (3)$$

A fourth equation is needed for a complete solution.

Kinematics (No Slipping). With this assumption, point A acts as a “pivot” so that α is clockwise, then a_G is directed to the right.

$$a_G = \alpha r_{AG}; \quad a_G = (1.5 \text{ m})\alpha \quad (4)$$

Solving Eqs. 1 to 4 yields

$$N_A = 981 \text{ N} \quad F_A = 300 \text{ N}$$

$$a_G = 1 \text{ m/s}^2 \quad \alpha = 0.667 \text{ rad/s}^2$$

The assumption of no slipping requires $F_A \leq \mu_s N_A$. However, $300 \text{ N} > 0.3(981 \text{ N}) = 294 \text{ N}$ and so the pole slips at A .

(Slipping). For this case Eq. 4 does *not* apply. Instead the frictional equation $F_A = \mu_k N_A$ must be used. Hence,

$$F_A = 0.25N_A \quad (5)$$

Solving Eqs. 1 to 3 and 5 simultaneously yields

$$N_A = 981 \text{ N} \quad F_A = 245 \text{ N} \quad a_G = 1.55 \text{ m/s}^2$$

$$\alpha = -0.428 \text{ rad/s}^2 = 0.428 \text{ rad/s}^2 \curvearrowright \quad \text{Ans.}$$

EXAMPLE 17.15

The uniform 50-kg bar in Fig. 17–23*a* is held in the equilibrium position by cords AC and BD . Determine the tension in BD and the angular acceleration of the bar immediately after AC is cut.

SOLUTION

Free-Body and Kinetic Diagrams. Fig. 17–23*b*. There are four unknowns, T_B , $(a_G)_x$, $(a_G)_y$, and α .

Equations of Motion.

$$\begin{aligned} \rightarrow \Sigma F_x &= m(a_G)_x; & 0 &= 50 \text{ kg } (a_G)_x \\ & & (a_G)_x &= 0 \\ + \uparrow \Sigma F_y &= m(a_G)_y; & T_B - 50(9.81) \text{ N} &= -50 \text{ kg } (a_G)_y \quad (1) \\ \zeta + \Sigma M_G &= I_G \alpha; & T_B(1.5 \text{ m}) &= \left[\frac{1}{12}(50 \text{ kg})(3 \text{ m})^2 \right] \alpha \quad (2) \end{aligned}$$

Kinematics. Since the bar is at rest just after the cable is cut, then its angular velocity and the velocity of point B at this instant are equal to zero. Thus $(a_B)_n = v_B^2/\rho_{BD} = 0$. Therefore, $\mathbf{a}_B$ only has a tangential component, which is directed along the x axis, Fig. 17–23*c*. Applying the relative acceleration equation to points G and B ,

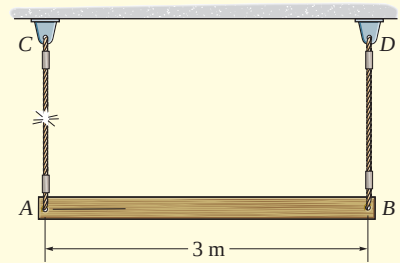
$$\begin{aligned} \mathbf{a}_G &= \mathbf{a}_B + \boldsymbol{\alpha} \times \mathbf{r}_{G/B} - \omega^2 \mathbf{r}_{G/B} \\ -(a_G)_y \mathbf{j} &= a_B \mathbf{i} + (\alpha \mathbf{k}) \times (-1.5 \mathbf{i}) - 0 \\ -(a_G)_y \mathbf{j} &= a_B \mathbf{i} - 1.5 \alpha \mathbf{j} \end{aligned}$$

Equating the $\mathbf{i}$ and $\mathbf{j}$ components of both sides of this equation,

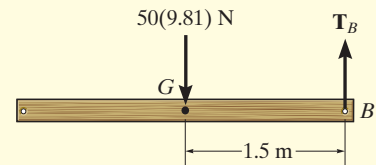
$$\begin{aligned} 0 &= a_B \\ (a_G)_y &= 1.5 \alpha \quad (3) \end{aligned}$$

Solving Eqs. (1) through (3) yields

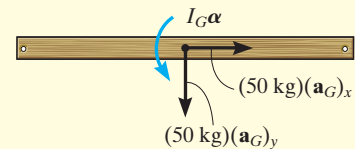
$$\begin{aligned} \alpha &= 4.905 \text{ rad/s}^2 && \text{Ans.} \\ T_B &= 123 \text{ N} && \text{Ans.} \\ (a_G)_y &= 7.36 \text{ m/s}^2 \end{aligned}$$



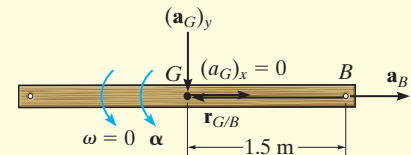
(a)



||



(b)

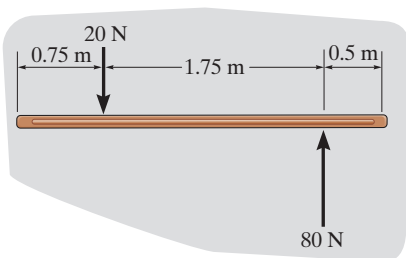


(c)

Fig. 17–23

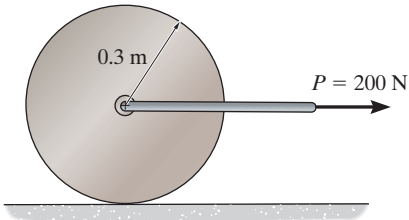
FUNDAMENTAL PROBLEMS

F17-13. The uniform 60-kg slender bar is initially at rest on a smooth *horizontal plane* when the forces are applied. Determine the acceleration of the bar's mass center and the angular acceleration of the bar at this instant.



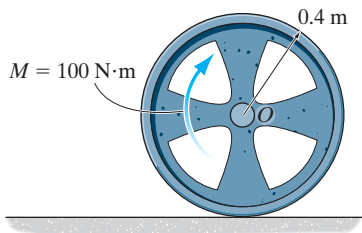
Prob. F17-13

F17-14. The 100-kg cylinder rolls without slipping on the horizontal plane. Determine the acceleration of its mass center and its angular acceleration.



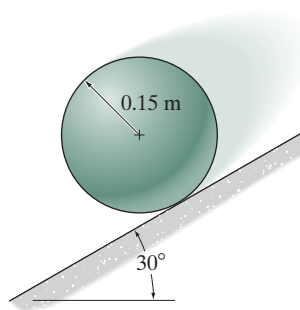
Prob. F17-14

F17-15. The 20-kg wheel has a radius of gyration about its center O of $k_O = 300$ mm. When the wheel is subjected to the couple moment, it slips as it rolls. Determine the angular acceleration of the wheel and the acceleration of the wheel's center O . The coefficient of kinetic friction between the wheel and the plane is $\mu_k = 0.5$.



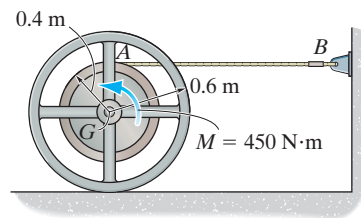
Prob. F17-15

F17-16. The 20-kg sphere rolls down the inclined plane without slipping. Determine the angular acceleration of the sphere and the acceleration of its mass center.



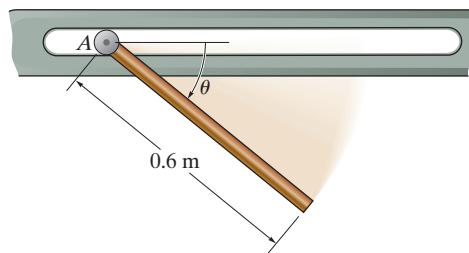
Prob. F17-16

F17-17. The 200-kg spool has a radius of gyration about its mass center of $k_G = 300$ mm. If the couple moment is applied to the spool and the coefficient of kinetic friction between the spool and the ground is $\mu_k = 0.2$, determine the angular acceleration of the spool, the acceleration of G and the tension in the cable.



Prob. F17-17

F17-18. The 12-kg slender rod is pinned to a small roller A that slides freely along the slot. If the rod is released from rest at $\theta = 0^\circ$, determine the angular acceleration of the rod and the acceleration of the roller immediately after the release.

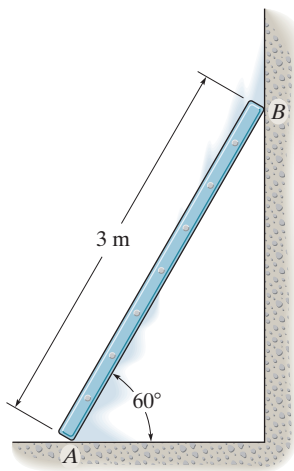


Prob. F17-18

PROBLEMS

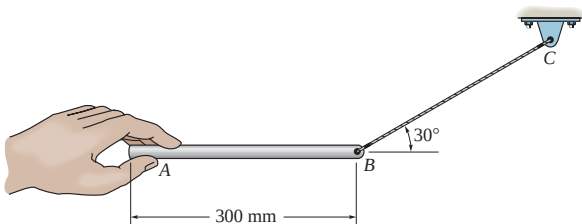
17-90. If the disk in Fig. 17-19 *rolls without slipping*, show that when moments are summed about the instantaneous center of zero velocity, IC , it is possible to use the moment equation $\Sigma M_{IC} = I_{IC} \alpha$, where I_{IC} represents the moment of inertia of the disk calculated about the instantaneous axis of zero velocity.

17-91. The slender 12-kg bar has a clockwise angular velocity of $\omega = 2$ rad/s when it is in the position shown. Determine its angular acceleration and the normal reactions of the smooth surface A and B at this instant.



Prob. 17-91

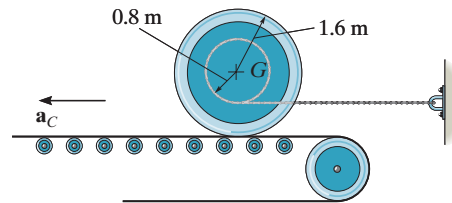
***17-92.** The 2-kg slender bar is supported by cord BC and then released from rest at A . Determine the initial angular acceleration of the bar and the tension in the cord.



Prob. 17-92

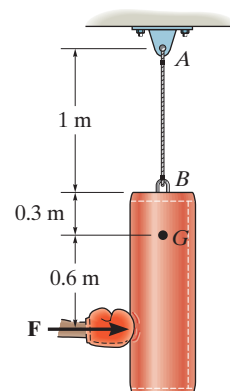
17-93. The spool has a mass of 500 kg and a radius of gyration $k_G = 1.30$ m. It rests on the surface of a conveyor belt for which the coefficient of static friction is $\mu_s = 0.5$ and the coefficient of kinetic friction is $\mu_k = 0.4$. If the conveyor accelerates at $a_C = 1$ m/s², determine the initial tension in the wire and the angular acceleration of the spool. The spool is originally at rest.

17-94. The spool has a mass of 500 kg and a radius of gyration $k_G = 1.30$ m. It rests on the surface of a conveyor belt for which the coefficient of static friction is $\mu_s = 0.5$. Determine the greatest acceleration a_C of the conveyor so that the spool will not slip. Also, what are the initial tension in the wire and the angular acceleration of the spool? The spool is originally at rest.



Probs. 17-93/94

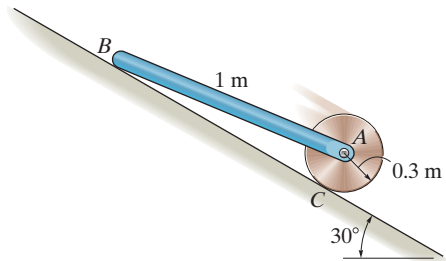
17-95. The 20-kg punching bag has a radius of gyration about its center of mass G of $k_G = 0.4$ m. If it is initially at rest and is subjected to a horizontal force $F = 30$ N, determine the initial angular acceleration of the bag and the tension in the supporting cable AB .



Prob. 17-95

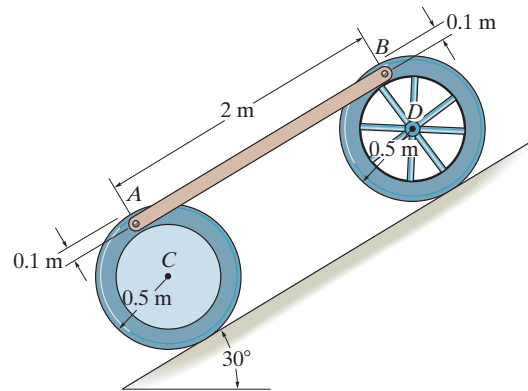
***17-96.** The assembly consists of an 8-kg disk and a 10-kg bar which is pin connected to the disk. If the system is released from rest, determine the angular acceleration of the disk. The coefficients of static and kinetic friction between the disk and the inclined plane are $\mu_s = 0.6$ and $\mu_k = 0.4$, respectively. Neglect friction at B .

17-97. Solve Prob. 17-96 if the bar is removed. The coefficients of static and kinetic friction between the disk and inclined plane are $\mu_s = 0.15$ and $\mu_k = 0.1$, respectively.



Probs. 17-96/97

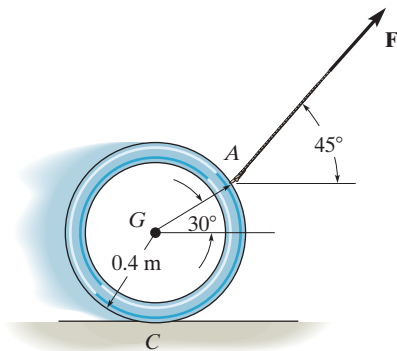
***17-100.** Wheel C has a mass of 60 kg and a radius of gyration of 0.4 m, whereas wheel D has a mass of 40 kg and a radius of gyration of 0.35 m. Determine the angular acceleration of each wheel at the instant shown. Neglect the mass of the link and assume that the assembly does not slip on the plane.



Prob. 17-100

17-98. A force of $F = 10$ N is applied to the 10-kg ring as shown. If slipping does not occur, determine the ring's initial angular acceleration, and the acceleration of its mass center, G . Neglect the thickness of the ring.

17-99. If the coefficient of static friction at C is $\mu_s = 0.3$, determine the largest force F that can be applied to the 5-kg ring, without causing it to slip. Neglect the thickness of the ring.

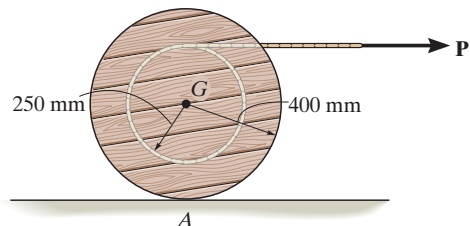


Probs. 17-98/99

17-101. The spool has a mass of 100 kg and a radius of gyration of $k_G = 0.3$ m. If the coefficients of static and kinetic friction at A are $\mu_s = 0.2$ and $\mu_k = 0.15$, respectively, determine the angular acceleration of the spool if $P = 50$ N.

17-102. Solve Prob. 17-101 if the cord and force $P = 50$ N are directed vertically upwards.

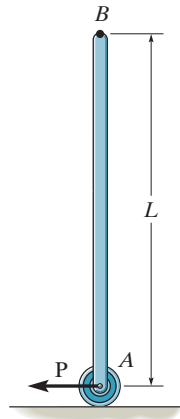
17-103. The spool has a mass of 100 kg and a radius of gyration $k_G = 0.3$ m. If the coefficients of static and kinetic friction at A are $\mu_s = 0.2$ and $\mu_k = 0.15$, respectively, determine the angular acceleration of the spool if $P = 600$ N.



Probs. 17-101/102/103

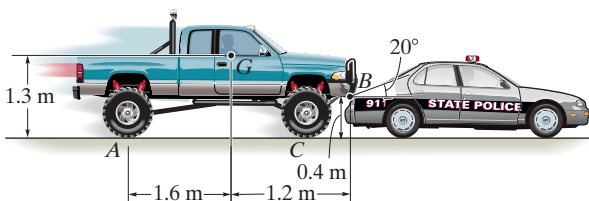
***17–104.** The uniform bar of mass m and length L is balanced in the vertical position when the horizontal force $\mathbf{P}$ is applied to the roller at A . Determine the bar's initial angular acceleration and the acceleration of its top point B .

17–105. Solve Prob. 17–104 if the roller is removed and the coefficient of kinetic friction at the ground is μ_k .



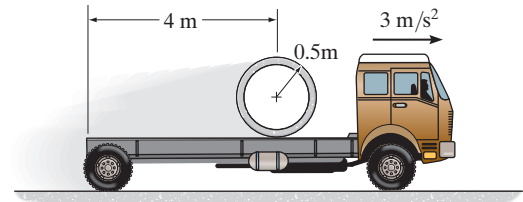
Probs. 17–104/105

17–106. A “lifted” truck can become a road hazard since the bumper is high enough to ride up a standard car in the event the car is rear-ended. As a model of this case consider the truck to have a mass of 2.70 Mg, a mass center G , and a radius of gyration about G of $k_G = 1.45$ m. Determine the horizontal and vertical components of acceleration of the mass center G , and the angular acceleration of the truck, at the moment its front wheels at C have just left the ground and its smooth front bumper begins to ride up the back of the stopped car so that point B has a velocity of $v_B = 8$ m/s at 20° from the horizontal. Assume the wheels are free to roll, and neglect the size of the wheels and the deformation of the material.



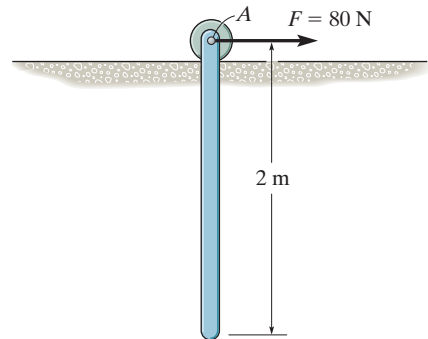
Prob. 17–106

17–107. The 500-kg concrete culvert has a mean radius of 0.5 m. If the truck has an acceleration of 3 m/s^2 , determine the culvert's angular acceleration. Assume that the culvert does not slip on the truck bed, and neglect its thickness.



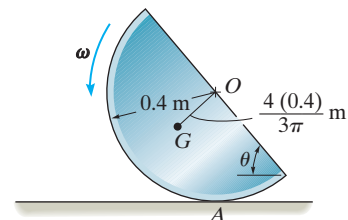
Prob. 17–107

***17–108.** The 12-kg uniform bar is supported by a roller at A . If a horizontal force of $F = 80$ N is applied to the roller, determine the acceleration of the center of the roller at the instant the force is applied. Neglect the weight and the size of the roller.



Prob. 17–108

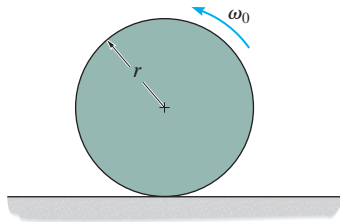
17–109. The semicircular disk having a mass of 10 kg is rotating at $\omega = 4 \text{ rad/s}$ at the instant $\theta = 60^\circ$. If the coefficient of static friction at A is $\mu_s = 0.5$, determine if the disk slips at this instant.



Prob. 17–109

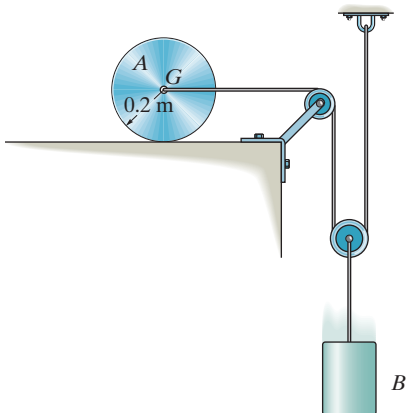
17-110. The uniform disk of mass m is rotating with an angular velocity of ω_0 when it is placed on the floor. Determine the initial angular acceleration of the disk and the acceleration of its mass center. The coefficient of kinetic friction between the disk and the floor is μ_k .

17-111. The uniform disk of mass m is rotating with an angular velocity of ω_0 when it is placed on the floor. Determine the time before it starts to roll without slipping. What is the angular velocity of the disk at this instant? The coefficient of kinetic friction between the disk and the floor is μ_k .



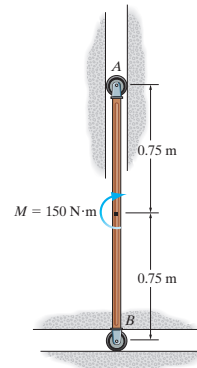
Probs. 17-110/111

***17-112.** The 20-kg disk A is attached to the 10-kg block B using the cable and pulley system shown. If the disk rolls without slipping, determine its angular acceleration and the acceleration of the block when they are released. Also, what is the tension in the cable? Neglect the mass of the pulleys.



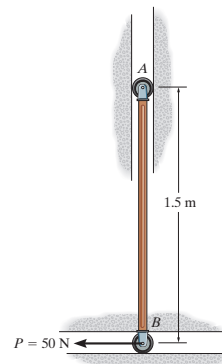
Prob. 17-112

17-113. The 30-kg uniform slender rod AB rests in the position shown when the couple moment of $M = 150 \text{ N}\cdot\text{m}$ is applied. Determine the initial angular acceleration of the rod. Neglect the mass of the rollers.



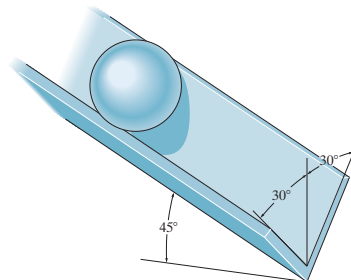
Prob. 17-113

17-114. The 30-kg slender rod AB rests in the position shown when the horizontal force $P = 50 \text{ N}$ is applied. Determine the initial angular acceleration of the rod. Neglect the mass of the rollers.



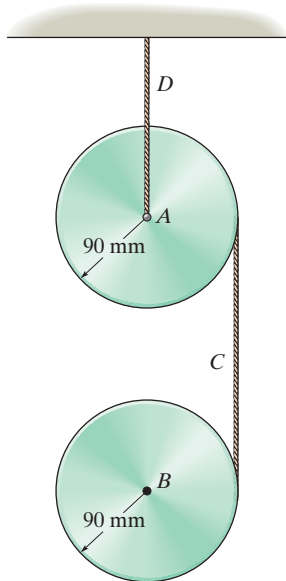
Prob. 17-114

17-115. The solid ball of radius r and mass m rolls without slipping down the 60° trough. Determine its angular acceleration.



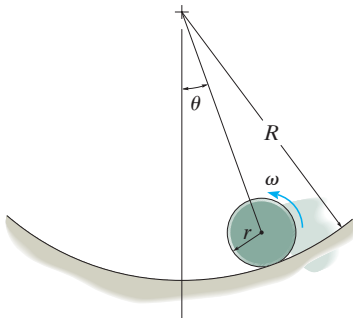
Prob. 17-115

***17–116.** A cord is wrapped around each of the two 10-kg disks. If they are released from rest determine the angular acceleration of each disk and the tension in the cord C . Neglect the mass of the cord.



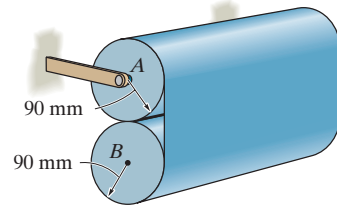
Prob. 17–116

17–117. The disk of mass m and radius r rolls without slipping on the circular path. Determine the normal force which the path exerts on the disk and the disk's angular acceleration if at the instant shown the disk has an angular velocity of ω .



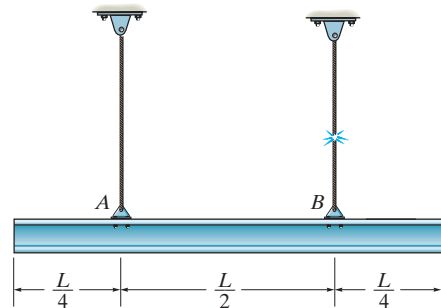
Prob. 17–117

17–118. A long strip of paper is wrapped into two rolls, each having a mass of 8 kg. Roll A is pin supported about its center whereas roll B is not centrally supported. If B is brought into contact with A and released from rest, determine the initial tension in the paper between the rolls and the angular acceleration of each roll. For the calculation, assume the rolls to be approximated by cylinders.



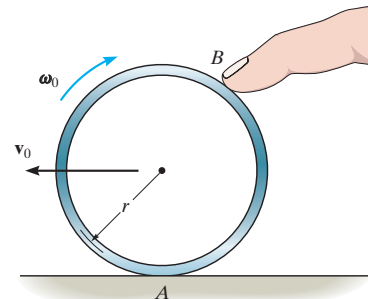
Prob. 17–118

17–119. The uniform beam has a weight W . If it is originally at rest while being supported at A and B by cables, determine the tension in cable A if cable B suddenly fails. Assume the beam is a slender rod.



Prob. 17–119

***17–120.** By pressing down with the finger at B , a thin ring having a mass m is given an initial velocity v_0 and a backspin ω_0 when the finger is released. If the coefficient of kinetic friction between the table and the ring is μ_k , determine the distance the ring travels forward before backspinning stops.



Prob. 17–120

CONCEPTUAL PROBLEMS

C17-1. The truck is used to pull the heavy container. To be most effective at providing traction to the rear wheels at A , is it best to keep the container where it is or place it at the front of the trailer? Use appropriate numerical values to explain your answer.



Prob. C17-1

C17-3. How can you tell the driver is accelerating this SUV? To explain your answer, draw the free-body and kinetic diagrams. Here power is supplied to the rear wheels. Would the photo look the same if power were supplied to the front wheels? Will the accelerations be the same? Use appropriate numerical values to explain your answers.



Prob. C17-3

C17-2. The tractor is about to tow the plane to the right. Is it possible for the driver to cause the front wheel of the plane to lift off the ground as he accelerates the tractor? Draw the free-body and kinetic diagrams and explain algebraically (letters) if and how this might be possible.



Prob. C17-2

C17-4. Here is something you should not try at home, at least not without wearing a helmet! Draw the free-body and kinetic diagrams and show what the rider must do to maintain this position. Use appropriate numerical values to explain your answer.



Prob. C17-4

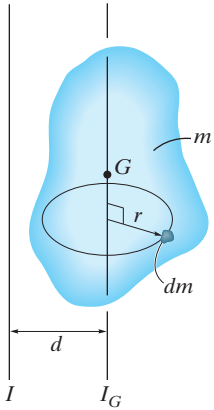
CHAPTER REVIEW

Moment of Inertia

The moment of inertia is a measure of the resistance of a body to a change in its angular velocity. It is defined by $I = \int r^2 dm$ and will be different for each axis about which it is computed.

Many bodies are composed of simple shapes. If this is the case, then tabular values of I can be used, such as the ones given on the inside back cover of this book. To obtain the moment of inertia of a composite body about any specified axis, the moment of inertia of each part is determined about the axis and the results are added together. Doing this often requires use of the parallel-axis theorem.

$$I = I_G + md^2$$



Planar Equations of Motion

The equations of motion define the translational, and rotational motion of a rigid body. In order to account for all of the terms in these equations, a free-body diagram should always accompany their application, and for some problems, it may also be convenient to draw the kinetic diagram which shows $m\mathbf{a}_G$ and $I_G\boldsymbol{\alpha}$.

$$\Sigma F_x = m(a_G)_x$$

$$\Sigma F_y = m(a_G)_y$$

$$\Sigma M_G = 0$$

Rectilinear translation

$$\Sigma F_n = m(a_G)_n$$

$$\Sigma F_t = m(a_G)_t$$

$$\Sigma M_G = 0$$

Curvilinear translation

$$\Sigma F_n = m(a_G)_n = m\omega^2 r_G$$

$$\Sigma F_t = m(a_G)_t = m\alpha r_G$$

$$\Sigma M_G = I_G\alpha \text{ or } \Sigma M_O = I_O\alpha$$

Rotation About a Fixed Axis

$$\Sigma F_x = m(a_G)_x$$

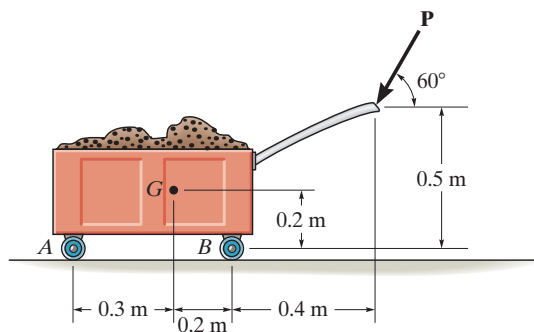
$$\Sigma F_y = m(a_G)_y$$

$$\Sigma M_G = I_G\alpha \text{ or } \Sigma M_P = \Sigma (\mathcal{M}_k)_P$$

General Plane Motion

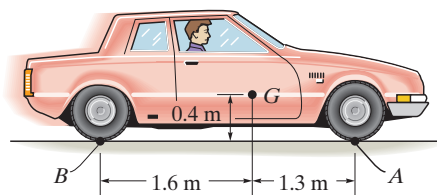
REVIEW PROBLEMS

R17-1. The handcart has a mass of 200 kg and center of mass at G . Determine the normal reactions at *each* of the wheels at A and B if a force $P = 50$ N is applied to the handle. Neglect the mass and rolling resistance of the wheels.



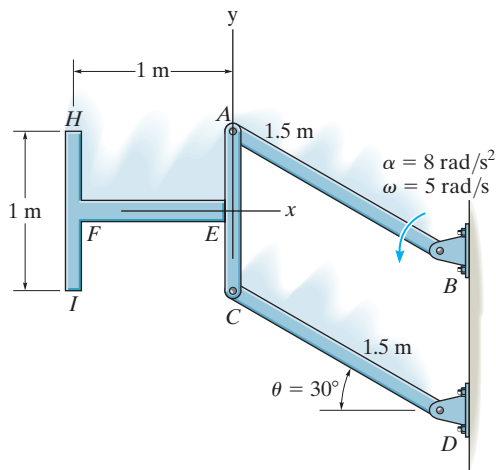
Prob. R17-1

R17-3. The car has a mass of 1.50 Mg and a mass center at G . Determine the maximum acceleration it can have if power is supplied only to the rear wheels. Neglect the mass of the wheels in the calculation, and assume that the wheels that do not receive power are free to roll. Also, assume that slipping of the powered wheels occurs, where the coefficient of kinetic friction is $\mu_k = 0.3$.



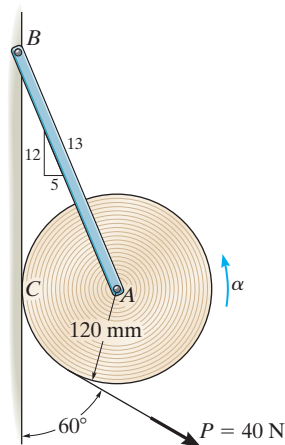
Prob. R17-3

R17-2. The two 1.5-kg rods EF and HI are fixed (welded) to the link AC at E . Determine the internal axial force E_x , shear force E_y , and moment M_E , which the bar AC exerts on FE at E if at the instant $\theta = 30^\circ$, link AB has an angular velocity $\omega = 5$ rad/s and an angular acceleration $\alpha = 8$ rad/s² as shown.



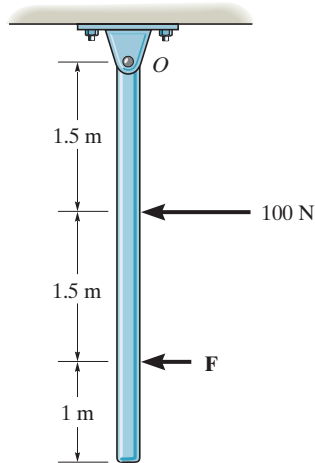
Prob. R17-2

R17-4. A 20-kg roll of paper, originally at rest, is pin-supported at its ends to bracket AB . The roll rests against a wall for which the coefficient of kinetic friction at C is $\mu_C = 0.3$. If a force of 40 N is applied uniformly to the end of the sheet, determine the initial angular acceleration of the roll and the tension in the bracket as the paper unwraps. For the calculation, treat the roll as a cylinder.



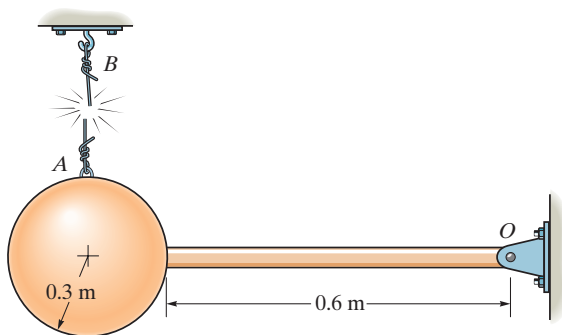
Prob. R17-4

R17-5. At the instant shown, two forces act on the 15-kg slender rod which is pinned at O . Determine the magnitude of force $\mathbf{F}$ and the initial angular acceleration of the rod so that the horizontal reaction which the *pin exerts on the rod* is 25 N directed to the right.



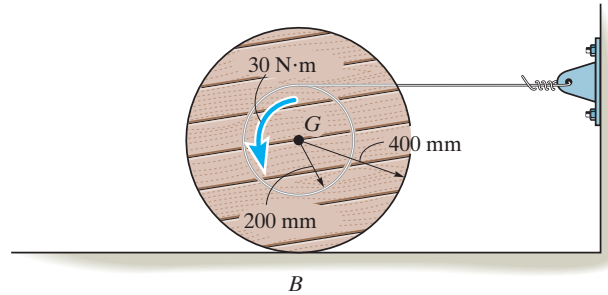
Prob. R17-5

R17-6. The pendulum consists of a 15-kg sphere and a 5-kg slender rod. Compute the reaction at the pin O just after the cord AB is cut.



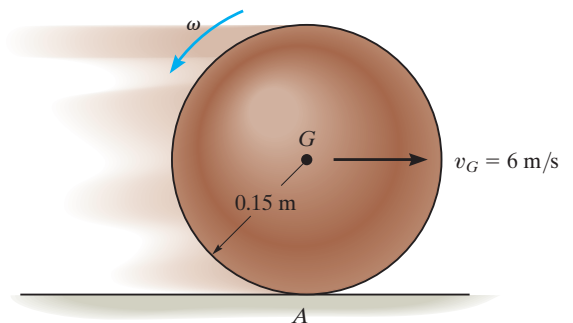
Prob. R17-6

R17-7. The spool and wire wrapped around its core have a mass of 20 kg and a centroidal radius of gyration $k_G = 250$ mm. If the coefficient of kinetic friction at the ground is $\mu_B = 0.1$, determine the angular acceleration of the spool when the 30-N·m couple moment is applied.



Prob. R17-7

R17-8. Determine the backspin ω which should be given to the 9-kg ball so that when its center is given an initial horizontal velocity $v_G = 6$ m/s it stops spinning and translating at the same instant. The coefficient of kinetic friction is $\mu_A = 0.3$.



Prob. R17-8

Chapter 18



(© Arinahabich/Fotolia)

Roller coasters must be able to coast over loops and through turns, and have enough energy to do so safely. Accurate calculation of this energy must account for the size of the car as it moves along the track.

Planar Kinetics of a Rigid Body: Work and Energy

CHAPTER OBJECTIVES

- To develop formulations for the kinetic energy of a body, and define the various ways a force and couple do work.
- To apply the principle of work and energy to solve rigid-body planar kinetic problems that involve force, velocity, and displacement.
- To show how the conservation of energy can be used to solve rigid-body planar kinetic problems.



Video Solutions are available for selected questions in this chapter.

18.1 Kinetic Energy

In this chapter we will apply work and energy methods to solve planar motion problems involving force, velocity, and displacement. But first it will be necessary to develop a means of obtaining the body's kinetic energy when the body is subjected to translation, rotation about a fixed axis, or general plane motion.

To do this we will consider the rigid body shown in Fig. 18–1, which is represented here by a *slab* moving in the inertial x – y reference plane. An arbitrary i th particle of the body, having a mass dm , is located a distance r from the arbitrary point P . If at the *instant* shown the particle has a velocity $\mathbf{v}_i$, then the particle's kinetic energy is $T_i = \frac{1}{2} dm v_i^2$.

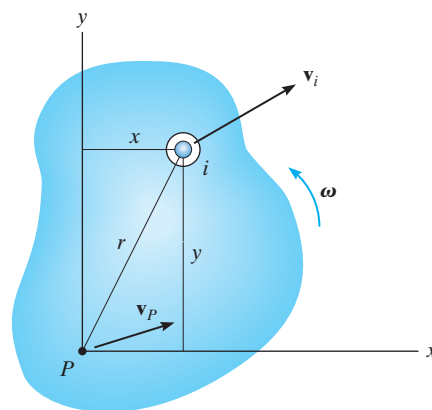


Fig. 18–1

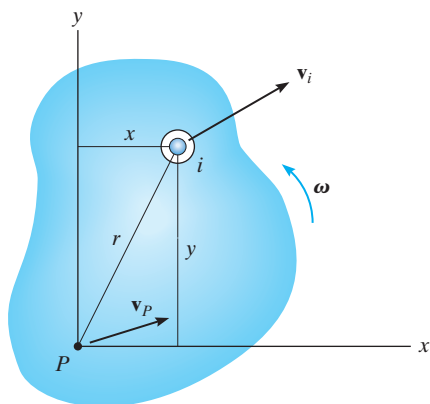


Fig. 18-1 (repeated)

The kinetic energy of the entire body is determined by writing similar expressions for each particle of the body and integrating the results, i.e.,

$$T = \frac{1}{2} \int_m dm v_i^2$$

This equation may also be expressed in terms of the velocity of point P . If the body has an angular velocity ω , then from Fig. 18-1 we have

$$\begin{aligned} \mathbf{v}_i &= \mathbf{v}_P + \mathbf{v}_{i/P} \\ &= (v_P)_x \mathbf{i} + (v_P)_y \mathbf{j} + \omega \mathbf{k} \times (x\mathbf{i} + y\mathbf{j}) \\ &= [(v_P)_x - \omega y] \mathbf{i} + [(v_P)_y + \omega x] \mathbf{j} \end{aligned}$$

The square of the magnitude of $\mathbf{v}_i$ is thus

$$\begin{aligned} \mathbf{v}_i \cdot \mathbf{v}_i &= v_i^2 = [(v_P)_x - \omega y]^2 + [(v_P)_y + \omega x]^2 \\ &= (v_P)_x^2 - 2(v_P)_x \omega y + \omega^2 y^2 + (v_P)_y^2 + 2(v_P)_y \omega x + \omega^2 x^2 \\ &= v_P^2 - 2(v_P)_x \omega y + 2(v_P)_y \omega x + \omega^2 r^2 \end{aligned}$$

Substituting this into the equation of kinetic energy yields

$$T = \frac{1}{2} \left(\int_m dm \right) v_P^2 - (v_P)_x \omega \left(\int_m y dm \right) + (v_P)_y \omega \left(\int_m x dm \right) + \frac{1}{2} \omega^2 \left(\int_m r^2 dm \right)$$

The first integral on the right represents the entire mass m of the body. Since $\bar{y}m = \int y dm$ and $\bar{x}m = \int x dm$, the second and third integrals locate the body's center of mass G with respect to P . The last integral represents the body's moment of inertia I_P , computed about the z axis passing through point P . Thus,

$$T = \frac{1}{2} m v_P^2 - (v_P)_x \omega \bar{y}m + (v_P)_y \omega \bar{x}m + \frac{1}{2} I_P \omega^2 \quad (18-1)$$

As a special case, if point P coincides with the mass center G of the body, then $\bar{y} = \bar{x} = 0$, and therefore

$$T = \frac{1}{2} m v_G^2 + \frac{1}{2} I_G \omega^2 \quad (18-2)$$

Both terms on the right side are *always positive*, since v_G and ω are squared. The first term represents the translational kinetic energy, referenced from the mass center, and the second term represents the body's rotational kinetic energy about the mass center.

Translation. When a rigid body of mass m is subjected to either rectilinear or curvilinear *translation*, Fig. 18–2, the kinetic energy due to rotation is zero, since $\boldsymbol{\omega} = \mathbf{0}$. The kinetic energy of the body is therefore

$$T = \frac{1}{2}mv_G^2 \quad (18-3)$$

Rotation about a Fixed Axis. When a rigid body *rotates about a fixed axis* passing through point O , Fig. 18–3, the body has both *translational* and *rotational* kinetic energy so that

$$T = \frac{1}{2}mv_G^2 + \frac{1}{2}I_G\omega^2 \quad (18-4)$$

The body's kinetic energy may also be formulated for this case by noting that $v_G = r_G\omega$, so that $T = \frac{1}{2}(I_G + mr_G^2)\omega^2$. By the parallel-axis theorem, the terms inside the parentheses represent the moment of inertia I_O of the body about an axis perpendicular to the plane of motion and passing through point O . Hence,*

$$T = \frac{1}{2}I_O\omega^2 \quad (18-5)$$

From the derivation, this equation will give the same result as Eq. 18–4, since it accounts for *both* the translational and rotational kinetic energies of the body.

General Plane Motion. When a rigid body is subjected to general plane motion, Fig. 18–4, it has an angular velocity $\boldsymbol{\omega}$ and its mass center has a velocity $\mathbf{v}_G$. Therefore, the kinetic energy is

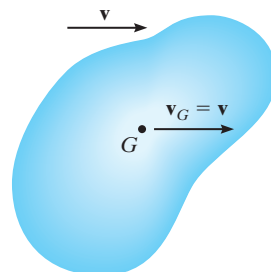
$$T = \frac{1}{2}mv_G^2 + \frac{1}{2}I_G\omega^2 \quad (18-6)$$

This equation can also be expressed in terms of the body's motion about its instantaneous center of zero velocity i.e.,

$$T = \frac{1}{2}I_{IC}\omega^2 \quad (18-7)$$

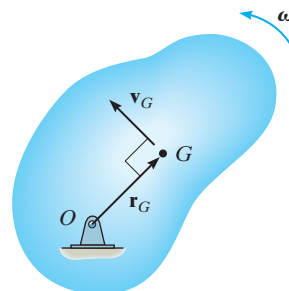
where I_{IC} is the moment of inertia of the body about its instantaneous center. The proof is similar to that of Eq. 18–5. (See Prob. 18–1.)

*The similarity between this derivation and that of $\Sigma M_O = I_O\alpha$, should be noted. Also the same result can be obtained directly from Eq. 18–1 by selecting point P at O , realizing that $v_O = 0$.



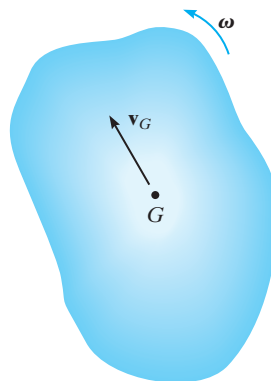
Translation

Fig. 18–2



Rotation About a Fixed Axis

Fig. 18–3



General Plane Motion

Fig. 18–4



The total kinetic energy of this soil compactor consists of the kinetic energy of the body or frame of the machine due to its translation, and the translational and rotational kinetic energies of the roller and the wheels due to their general plane motion. Here we exclude the additional kinetic energy developed by the moving parts of the engine and drive train.

System of Bodies. Because energy is a scalar quantity, the total kinetic energy for a system of *connected* rigid bodies is the sum of the kinetic energies of all its moving parts. Depending on the type of motion, the kinetic energy of *each body* is found by applying Eq. 18-2 or the alternative forms mentioned above.

18.2 The Work of a Force

Several types of forces are often encountered in planar kinetics problems involving a rigid body. The work of each of these forces has been presented in Sec. 14.1 and is listed below as a summary.

Work of a Variable Force. If an external force $\mathbf{F}$ acts on a body, the work done by the force when the body moves along the path s , Fig. 18-5, is

$$U_F = \int \mathbf{F} \cdot d\mathbf{r} = \int_s F \cos \theta \, ds \quad (18-8)$$

Here θ is the angle between the “tails” of the force and the differential displacement. The integration must account for the variation of the force’s direction and magnitude.

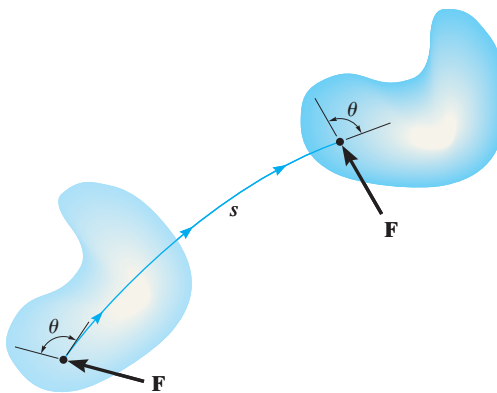


Fig. 18-5

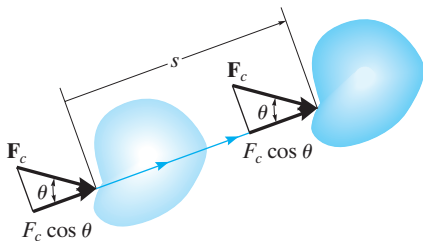


Fig. 18-6

Work of a Constant Force. If an external force $\mathbf{F}_c$ acts on a body, Fig. 18-6, and maintains a constant magnitude F_c and constant direction θ , while the body undergoes a translation s , then the above equation can be integrated, so that the work becomes

$$U_{F_c} = (F_c \cos \theta)s \quad (18-9)$$

Work of a Weight. The weight of a body does work only when the body's center of mass G undergoes a *vertical displacement* Δy . If this displacement is *upward*, Fig. 18–7, the work is negative, since the weight is opposite to the displacement.

$$U_W = -W \Delta y$$

(18–10)

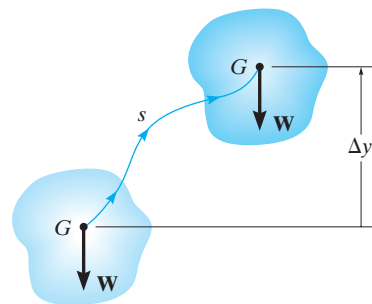


Fig. 18–7

Likewise, if the displacement is *downward* ($-\Delta y$) the work becomes *positive*. In both cases the elevation change is considered to be small so that W , which is caused by gravitation, is constant.

Work of a Spring Force. If a linear elastic spring is attached to a body, the spring force $F_s = ks$ acting *on the body* does work when the spring either stretches or compresses from s_1 to a *farther* position s_2 . In both cases the work will be *negative* since the *displacement of the body* is in the opposite direction to the force, Fig. 18–8. The work is

$$U_s = -\left(\frac{1}{2}ks_2^2 - \frac{1}{2}ks_1^2\right)$$

(18–11)

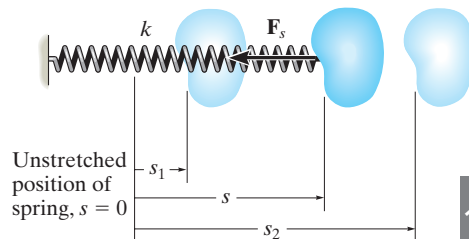


Fig. 18–8

where $|s_2| > |s_1|$.

Forces That Do No Work. There are some external forces that do no work when the body is displaced. These forces act either at *fixed points* on the body, or they have a direction *perpendicular to their displacement*. Examples include the reactions at a pin support about which a body rotates, the normal reaction acting on a body that moves along a fixed surface, and the weight of a body when the center of gravity of the body moves in a *horizontal plane*, Fig. 18–9. A frictional force F_f acting on a round body as it *rolls without slipping* over a rough surface also does no work.* This is because, during any *instant of time* dt , F_f acts at a point on the body which has *zero velocity* (instantaneous center, IC) and so the work done by the force on the point is zero. In other words, the point is not displaced in the direction of the force during this instant. Since F_f contacts successive points for only an instant, the work of F_f will be zero.

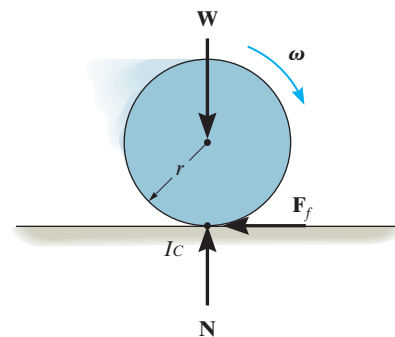


Fig. 18–9

*The work done by a frictional force *when the body slips* is discussed in Sec. 14.3.

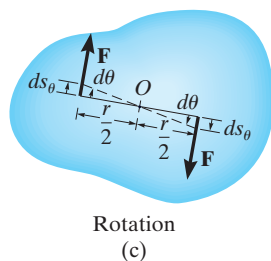
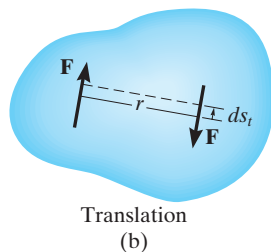
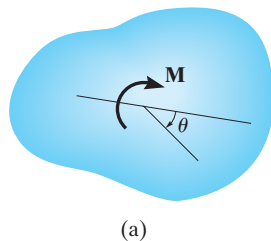


Fig. 18–10

18.3 The Work of a Couple Moment

Consider the body in Fig. 18–10*a*, which is subjected to a couple moment $M = Fr$. If the body undergoes a differential displacement, then the work done by the couple forces can be found by considering the displacement as the sum of a separate translation plus rotation. When the body *translates*, the work of each force is produced only by the *component of displacement* along the line of action of the forces ds_t , Fig. 18–10*b*. Clearly the “positive” work of one force *cancels* the “negative” work of the other. When the body undergoes a differential rotation $d\theta$ about the arbitrary point O , Fig. 18–10*c*, then each force undergoes a displacement $ds_\theta = (r/2) d\theta$ in the direction of the force. Hence, the total work done is

$$\begin{aligned} dU_M &= F\left(\frac{r}{2}d\theta\right) + F\left(\frac{r}{2}d\theta\right) = (Fr) d\theta \\ &= M d\theta \end{aligned}$$

The work is *positive* when $\mathbf{M}$ and $d\theta$ have the *same sense of direction* and *negative* if these vectors are in the *opposite sense*.

When the body rotates in the plane through a finite angle θ measured in radians, from θ_1 to θ_2 , the work of a couple moment is therefore

$$U_M = \int_{\theta_1}^{\theta_2} M d\theta \quad (18-12)$$

If the couple moment $\mathbf{M}$ has a *constant magnitude*, then

$$U_M = M(\theta_2 - \theta_1) \quad (18-13)$$

EXAMPLE 18.1

The bar shown in Fig. 18–11*a* has a mass of 10 kg and is subjected to a couple moment of $M = 50 \text{ N} \cdot \text{m}$ and a force of $P = 80 \text{ N}$, which is always applied perpendicular to the end of the bar. Also, the spring has an unstretched length of 0.5 m and remains in the vertical position due to the roller guide at B . Determine the total work done by all the forces acting on the bar when it has rotated downward from $\theta = 0^\circ$ to $\theta = 90^\circ$.

SOLUTION

First the free-body diagram of the bar is drawn in order to account for all the forces that act on it, Fig. 18–11*b*.

Weight W . Since the weight $10(9.81) \text{ N} = 98.1 \text{ N}$ is displaced downward 1.5 m, the work is

$$U_W = 98.1 \text{ N}(1.5 \text{ m}) = 147.2 \text{ J}$$

Why is the work positive?

Couple Moment M . The couple moment rotates through an angle of $\theta = \pi/2 \text{ rad}$. Hence,

$$U_M = 50 \text{ N} \cdot \text{m}(\pi/2) = 78.5 \text{ J}$$

Spring Force F_s . When $\theta = 0^\circ$ the spring is stretched $(0.75 \text{ m} - 0.5 \text{ m}) = 0.25 \text{ m}$, and when $\theta = 90^\circ$, the stretch is $(2 \text{ m} + 0.75 \text{ m}) - 0.5 \text{ m} = 2.25 \text{ m}$. Thus,

$$U_s = -\left[\frac{1}{2}(30 \text{ N/m})(2.25 \text{ m})^2 - \frac{1}{2}(30 \text{ N/m})(0.25 \text{ m})^2\right] = -75.0 \text{ J}$$

By inspection the spring does negative work on the bar since F_s acts in the opposite direction to displacement. This checks with the result.

Force P . As the bar moves downward, the force is displaced through a distance of $(\pi/2)(3 \text{ m}) = 4.712 \text{ m}$. The work is positive. Why?

$$U_P = 80 \text{ N}(4.712 \text{ m}) = 377.0 \text{ J}$$

Pin Reactions. Forces A_x and A_y do no work since they are not displaced.

Total Work. The work of all the forces when the bar is displaced is thus

$$U = 147.2 \text{ J} + 78.5 \text{ J} - 75.0 \text{ J} + 377.0 \text{ J} = 528 \text{ J} \quad \text{Ans.}$$

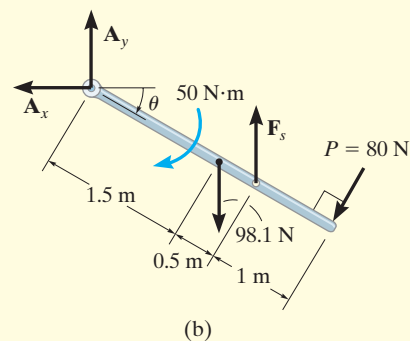
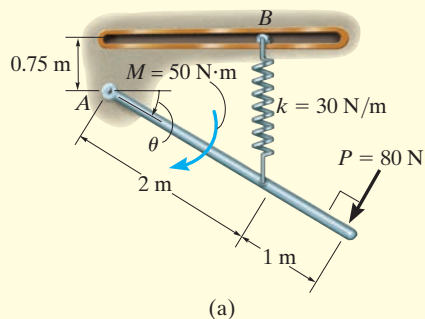


Fig. 18–11



The counterweight on this bascule bridge does positive work as the bridge is lifted and thereby cancels the negative work done by the weight of the bridge.

18.4 Principle of Work and Energy

By applying the principle of work and energy developed in Sec. 14.2 to each of the particles of a rigid body and adding the results algebraically, since energy is a scalar, the principle of work and energy for a rigid body becomes

$$T_1 + \Sigma U_{1-2} = T_2 \quad (18-14)$$

This equation states that the body's initial translational *and* rotational kinetic energy, plus the work done by all the external forces and couple moments acting on the body as the body moves from its initial to its final position, is equal to the body's final translational *and* rotational kinetic energy. Note that the work of the body's *internal forces* does not have to be considered. These forces occur in equal but opposite collinear pairs, so that when the body moves, the work of one force cancels that of its counterpart. Furthermore, since the body is rigid, *no relative movement* between these forces occurs, so that no internal work is done.

When several rigid bodies are pin connected, connected by inextensible cables, or in mesh with one another, Eq. 18-14 can be applied to the *entire system* of connected bodies. In all these cases the internal forces, which hold the various members together, do no work and hence are eliminated from the analysis.



The work of the torque or moment developed by the driving gears on the motors is transformed into kinetic energy of rotation of the drum.

Procedure for Analysis

The principle of work and energy is used to solve kinetic problems that involve *velocity*, *force*, and *displacement*, since these terms are involved in the formulation. For application, it is suggested that the following procedure be used.

Kinetic Energy (Kinematic Diagrams).

- The kinetic energy of a body is made up of two parts. Kinetic energy of translation is referenced to the velocity of the mass center, $T = \frac{1}{2}mv_G^2$, and kinetic energy of rotation is determined using the moment of inertia of the body about the mass center, $T = \frac{1}{2}I_G\omega^2$. In the special case of rotation about a fixed axis (or rotation about the *IC*), these two kinetic energies are combined and can be expressed as $T = \frac{1}{2}I_O\omega^2$, where I_O is the moment of inertia about the axis of rotation.
- *Kinematic diagrams* for velocity may be useful for determining v_G and ω or for establishing a *relationship* between v_G and ω .*

Work (Free-Body Diagram).

- Draw a free-body diagram of the body when it is located at an intermediate point along the path in order to account for all the forces and couple moments which do work on the body as it moves along the path.
- A force does work when it moves through a displacement in the direction of the force.
- Forces that are functions of displacement must be integrated to obtain the work. Graphically, the work is equal to the area under the force-displacement curve.
- The work of a weight is the product of its magnitude and the vertical displacement, $U_W = Wy$. It is positive when the weight moves downwards.
- The work of a spring is of the form $U_s = \frac{1}{2}ks^2$, where k is the spring stiffness and s is the stretch or compression of the spring.
- The work of a couple is the product of the couple moment and the angle in radians through which it rotates, $U_M = M\theta$.
- Since *algebraic addition* of the work terms is required, it is important that the proper sign of each term be specified. Specifically, work is *positive* when the force (couple moment) is in the *same direction* as its displacement (rotation); otherwise, it is negative.

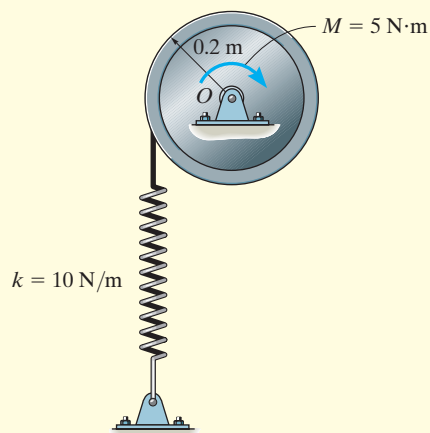
Principle of Work and Energy.

- Apply the principle of work and energy, $T_1 + \Sigma U_{1-2} = T_2$. Since this is a scalar equation, it can be used to solve for only one unknown when it is applied to a single rigid body.

*A brief review of Secs. 16.5 to 16.7 may prove helpful when solving problems, since computations for kinetic energy require a kinematic analysis of velocity.

EXAMPLE 18.2

The 30-kg disk shown in Fig. 18–12a is pin supported at its center. Determine the angle through which it must rotate to attain an angular velocity of 2 rad/s starting from rest. It is acted upon by a constant couple moment $M = 5 \text{ N} \cdot \text{m}$. The spring is originally unstretched and its cord wraps around the rim of the disk.



(a)

SOLUTION

Kinetic Energy. Since the disk rotates about a fixed axis, and it is initially at rest, then

$$T_1 = 0$$

$$T_2 = \frac{1}{2} I_O \omega_2^2 = \frac{1}{2} \left[\frac{1}{2} (30 \text{ kg}) (0.2 \text{ m})^2 \right] (2 \text{ rad/s})^2 = 1.2 \text{ J}$$

Work (Free-Body Diagram). As shown in Fig. 18–12b, the pin reactions O_x and O_y and the weight (294.3 N) do no work, since they are not displaced. The *couple moment*, having a constant magnitude, does positive work $U_M = M\theta$ as the disk rotates through a clockwise angle of θ rad, and the spring does negative work $U_s = -\frac{1}{2} k s^2$.

Principle of Work and Energy.

$$\{T_1\} + \{\Sigma U_{1-2}\} = \{T_2\}$$

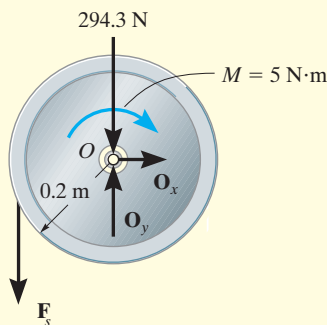
$$\{T_1\} + \left\{ M\theta - \frac{1}{2} k s^2 \right\} = \{T_2\}$$

$$\{0\} + \left\{ (5 \text{ N} \cdot \text{m})\theta - \frac{1}{2} (10 \text{ N/m}) [\theta(0.2 \text{ m})]^2 \right\} = \{1.2 \text{ J}\}$$

$$-0.2\theta^2 + 5\theta - 1.2 = 0$$

Solving this quadratic equation for the smallest positive root,

$$\theta = 0.2423 \text{ rad} = 0.2423 \text{ rad} \left(\frac{180^\circ}{\pi \text{ rad}} \right) = 13.9^\circ \quad \text{Ans.}$$



(b)

Fig. 18–12

EXAMPLE 18.3

The wheel shown in Fig. 18–13a weighs 20 kg and has a radius of gyration $k_G = 0.2$ m about its mass center G . If it is subjected to a clockwise couple moment of $25 \text{ N} \cdot \text{m}$ and rolls from rest without slipping, determine its angular velocity after its center G moves 0.18 m. The spring has a stiffness $k = 150 \text{ N/m}$ and is initially unstretched when the couple moment is applied.

SOLUTION

Kinetic Energy (Kinematic Diagram). Since the wheel is initially at rest,

$$T_1 = 0$$

The kinematic diagram of the wheel when it is in the final position is shown in Fig. 18–13b. The final kinetic energy is determined from

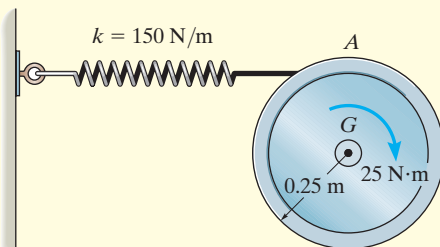
$$\begin{aligned} T_2 &= \frac{1}{2} I_{IC} \omega_2^2 \\ &= \frac{1}{2} \left[20 \text{ kg} (0.2 \text{ m})^2 + (20 \text{ kg})(0.25 \text{ m})^2 \right] \omega_2^2 \\ T_2 &= 1.025 \omega_2^2 \end{aligned}$$

Work (Free-Body Diagram). As shown in Fig. 18–13c, only the spring force $\mathbf{F}_s$ and the couple moment do work. The normal force does not move along its line of action and the frictional force does *no work*, since the wheel does not slip as it rolls.

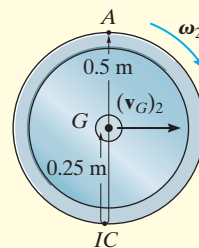
The work of $\mathbf{F}_s$ is found using $U_s = -\frac{1}{2} k s^2$. Here the work is negative since $\mathbf{F}_s$ is in the opposite direction to displacement. Since the wheel does not slip when the center G moves 0.18 m, then the wheel rotates $\theta = s_G / r_{G/IC} = 0.18 \text{ m} / 0.25 \text{ m} = 0.72 \text{ rad}$, Fig. 18–13b. Hence, the spring stretches $s = \theta r_{A/IC} = (0.72 \text{ rad})(0.5 \text{ m}) = 0.36 \text{ m}$.

Principle of Work and Energy.

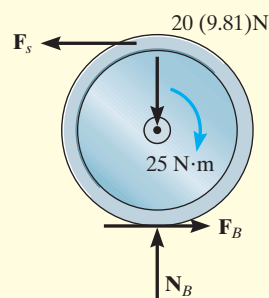
$$\begin{aligned} \{T_1\} + \{\Sigma U_{1-2}\} &= \{T_2\} \\ \{T_1\} + \{M\theta - \frac{1}{2} k s^2\} &= \{T_2\} \\ \{0\} + \left\{ 25 \text{ N} \cdot \text{m} (0.72 \text{ rad}) - \frac{1}{2} (150 \text{ N/m}) (0.36 \text{ m})^2 \right\} &= \{1.025 \omega_2^2 \text{ N} \cdot \text{m}\} \\ \omega_2 &= 2.84 \text{ rad/s} \quad \text{Ans.} \end{aligned}$$



(a)



(b)

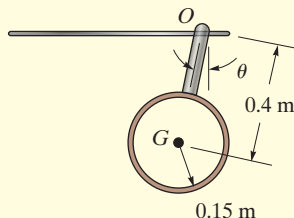


(c)

Fig. 18–13

EXAMPLE 18.4

The 700-kg pipe is equally suspended from the two tines of the fork lift shown in the photo. It is undergoing a swinging motion such that when $\theta = 30^\circ$ it is momentarily at rest. Determine the normal and frictional forces acting on each tine which are needed to support the pipe at the instant $\theta = 0^\circ$. Measurements of the pipe and the suspender are shown in Fig. 18–14a. Neglect the mass of the suspender and the thickness of the pipe.



(a)

Fig. 18–14**SOLUTION**

We must use the equations of motion to find the forces on the tines since these forces do no work. Before doing this, however, we will apply the principle of work and energy to determine the angular velocity of the pipe when $\theta = 0^\circ$.

Kinetic Energy (Kinematic Diagram). Since the pipe is originally at rest, then

$$T_1 = 0$$

The final kinetic energy may be computed with reference to either the fixed point O or the center of mass G . For the calculation we will consider the pipe to be a thin ring so that $I_G = mr^2$. If point G is considered, we have

$$\begin{aligned} T_2 &= \frac{1}{2}m(v_G)_2^2 + \frac{1}{2}I_G\omega_2^2 \\ &= \frac{1}{2}(700 \text{ kg})[(0.4 \text{ m})\omega_2]^2 + \frac{1}{2}[700 \text{ kg}(0.15 \text{ m})^2]\omega_2^2 \\ &= 63.875\omega_2^2 \end{aligned}$$

If point O is considered then the parallel-axis theorem must be used to determine I_O . Hence,

$$\begin{aligned} T_2 &= \frac{1}{2}I_O\omega_2^2 = \frac{1}{2}[700 \text{ kg}(0.15 \text{ m})^2 + 700 \text{ kg}(0.4 \text{ m})^2]\omega_2^2 \\ &= 63.875\omega_2^2 \end{aligned}$$

Work (Free-Body Diagram). Fig. 18–14*b*. The normal and frictional forces on the tines do no work since they do not move as the pipe swings. The weight does positive work since the weight moves downward through a vertical distance $\Delta y = 0.4 \text{ m} - 0.4 \cos 30^\circ \text{ m} = 0.05359 \text{ m}$.

Principle of Work and Energy.

$$\begin{aligned}\{T_1\} + \{\Sigma U_{1-2}\} &= \{T_2\} \\ \{0\} + \{700(9.81) \text{ N}(0.05359 \text{ m})\} &= \{63.875\omega_2^2\} \\ \omega_2 &= 2.400 \text{ rad/s}\end{aligned}$$

Equations of Motion. Referring to the free-body and kinetic diagrams shown in Fig. 18–14*c*, and using the result for ω_2 , we have

$$\begin{aligned}\leftarrow \Sigma F_t &= m(a_G)_t; & F_T &= (700 \text{ kg})(a_G)_t \\ + \uparrow \Sigma F_n &= m(a_G)_n; & N_T - 700(9.81) \text{ N} &= (700 \text{ kg})(2.400 \text{ rad/s})^2(0.4 \text{ m}) \\ \curvearrowright + \Sigma M_O &= I_O\alpha; & 0 &= [(700 \text{ kg})(0.15 \text{ m})^2 + (700 \text{ kg})(0.4 \text{ m})^2]\alpha\end{aligned}$$

Since $(a_G)_t = (0.4 \text{ m})\alpha$, then

$$\begin{aligned}\alpha &= 0, & (a_G)_t &= 0 \\ F_T &= 0 \\ N_T &= 8.480 \text{ kN}\end{aligned}$$

There are two tines used to support the load, therefore

$$\begin{aligned}F'_T &= 0 & \text{Ans.} \\ N'_T &= \frac{8.480 \text{ kN}}{2} = 4.24 \text{ kN} & \text{Ans.}\end{aligned}$$

NOTE: Due to the swinging motion the tines are subjected to a *greater* normal force than would be the case if the load were static, in which case $N'_T = 700(9.81) \text{ N}/2 = 3.43 \text{ kN}$.

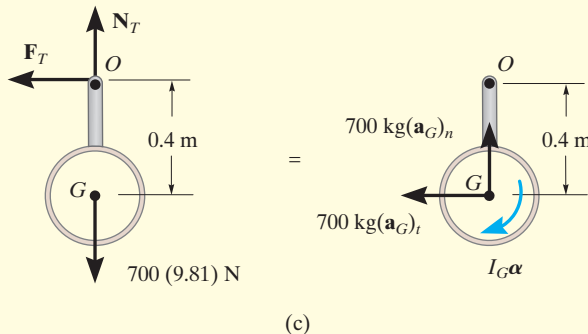
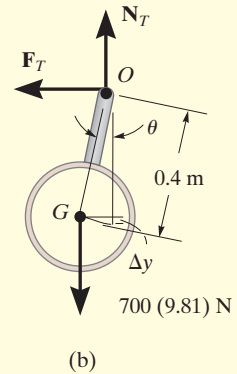
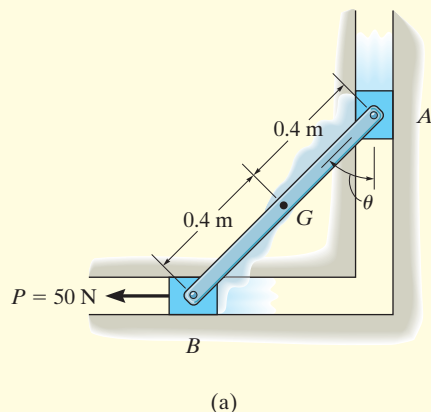
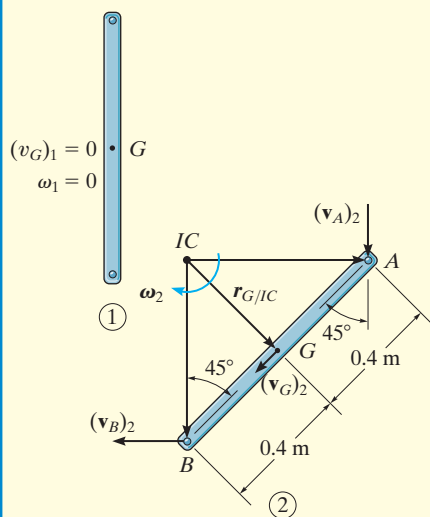


Fig. 18–14

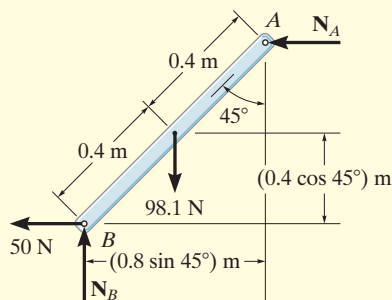
EXAMPLE 18.5



(a)



(b)



(c)

Fig. 18-15

The 10-kg rod shown in Fig. 18-15a is constrained so that its ends move along the grooved slots. The rod is initially at rest when $\theta = 0^\circ$. If the slider block at B is acted upon by a horizontal force $P = 50$ N, determine the angular velocity of the rod at the instant $\theta = 45^\circ$. Neglect friction and the mass of blocks A and B.

SOLUTION

Why can the principle of work and energy be used to solve this problem?

Kinetic Energy (Kinematic Diagrams). Two kinematic diagrams of the rod, when it is in the initial position 1 and final position 2, are shown in Fig. 18-15b. When the rod is in position 1, $T_1 = 0$ since $(\mathbf{v}_G)_1 = \boldsymbol{\omega}_1 = \mathbf{0}$. In position 2 the angular velocity is ω_2 and the velocity of the mass center is $(\mathbf{v}_G)_2$. Hence, the kinetic energy is

$$\begin{aligned} T_2 &= \frac{1}{2}m(v_G)_2^2 + \frac{1}{2}I_G\omega_2^2 \\ &= \frac{1}{2}(10 \text{ kg})(v_G)_2^2 + \frac{1}{2}\left[\frac{1}{12}(10 \text{ kg})(0.8 \text{ m})^2\right]\omega_2^2 \\ &= 5(v_G)_2^2 + 0.2667(\omega_2)^2 \end{aligned}$$

The two unknowns $(v_G)_2$ and ω_2 can be related from the instantaneous center of zero velocity for the rod. Fig. 18-15b. It is seen that as A moves downward with a velocity $(\mathbf{v}_A)_2$, B moves horizontally to the left with a velocity $(\mathbf{v}_B)_2$. Knowing these directions, the IC is located as shown in the figure. Hence,

$$\begin{aligned} (v_G)_2 &= r_{G/IC}\omega_2 = (0.4 \tan 45^\circ \text{ m})\omega_2 \\ &= 0.4\omega_2 \end{aligned}$$

Therefore,

$$T_2 = 0.8\omega_2^2 + 0.2667\omega_2^2 = 1.0667\omega_2^2$$

Of course, we can also determine this result using $T_2 = \frac{1}{2}I_{IC}\omega_2^2$.

Work (Free-Body Diagram). Fig. 18-15c. The normal forces $\mathbf{N}_A$ and $\mathbf{N}_B$ do no work as the rod is displaced. Why? The 98.1-N weight is displaced a vertical distance of $\Delta y = (0.4 - 0.4 \cos 45^\circ) \text{ m}$; whereas the 50-N force moves a horizontal distance of $s = (0.8 \sin 45^\circ) \text{ m}$. Both of these forces do positive work. Why?

Principle of Work and Energy.

$$\begin{aligned} \{T_1\} + \{\Sigma U_{1-2}\} &= \{T_2\} \\ \{T_1\} + \{W \Delta y + Ps\} &= \{T_2\} \\ \{0\} + \{98.1 \text{ N}(0.4 \text{ m} - 0.4 \cos 45^\circ \text{ m}) + 50 \text{ N}(0.8 \sin 45^\circ \text{ m})\} \\ &= \{1.0667\omega_2^2 \text{ J}\} \end{aligned}$$

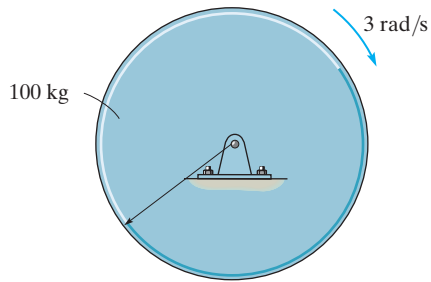
Solving for ω_2 gives

$$\omega_2 = 6.11 \text{ rad/s} \curvearrowright$$

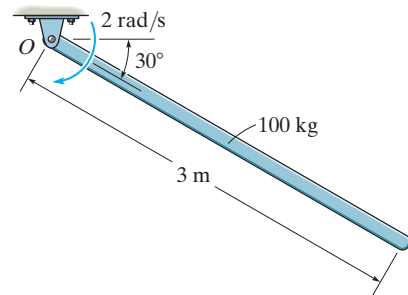
Ans.

PRELIMINARY PROBLEM

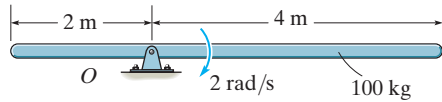
P18-1. Determine the kinetic energy of the 100-kg object.



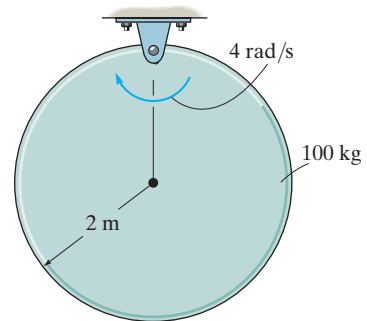
(a)



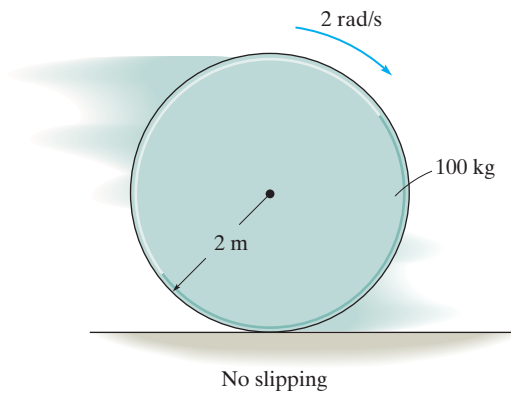
(d)



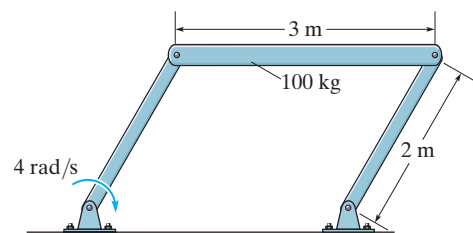
(b)



(e)



(c)

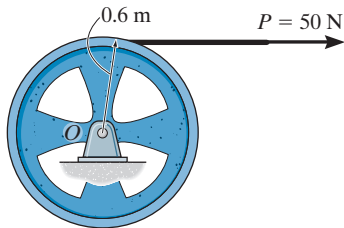


(f)

Prob. P18-1

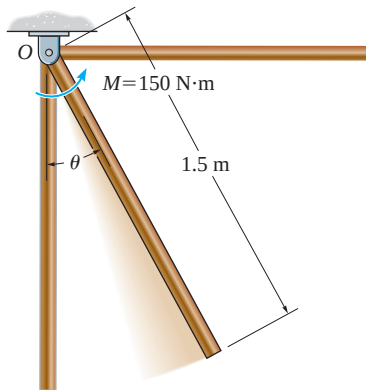
FUNDAMENTAL PROBLEMS

F18-1. The 80-kg wheel has a radius of gyration about its mass center O of $k_O = 400$ mm. Determine its angular velocity after it has rotated 20 revolutions starting from rest.



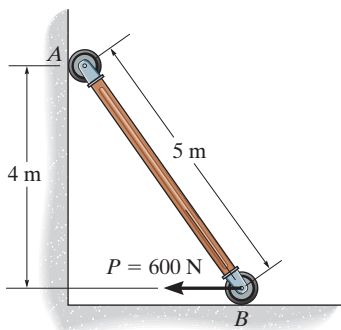
Prob. F18-1

F18-2. The uniform 25-kg slender rod is subjected to a couple moment of $M = 150$ N·m. If the rod is at rest when $\theta = 0^\circ$, determine its angular velocity when $\theta = 90^\circ$.



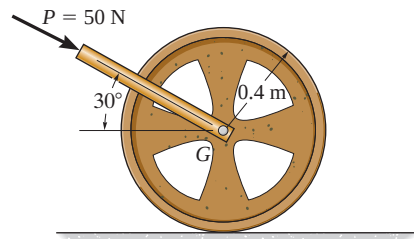
Prob. F18-2

F18-3. The uniform 50-kg slender rod is at rest in the position shown when $P = 600$ N is applied. Determine the angular velocity of the rod when the rod reaches the vertical position.



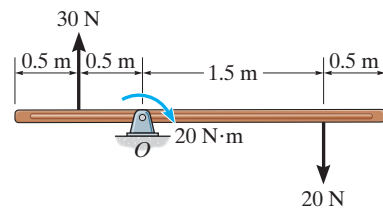
Prob. F18-3

F18-4. The 50-kg wheel is subjected to a force of 50 N. If the wheel starts from rest and rolls without slipping, determine its angular velocity after it has rotated 10 revolutions. The radius of gyration of the wheel about its mass center G is $k_G = 0.3$ m.



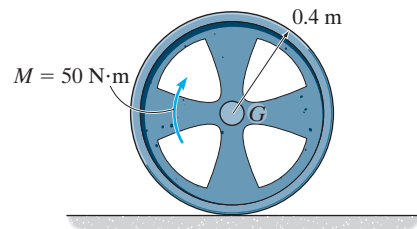
Prob. F18-4

F18-5. If the uniform 30-kg slender rod starts from rest at the position shown, determine its angular velocity after it has rotated 4 revolutions. The forces remain perpendicular to the rod.



Prob. F18-5

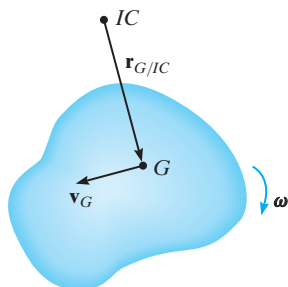
F18-6. The 20-kg wheel has a radius of gyration about its center G of $k_G = 300$ mm. When it is subjected to a couple moment of $M = 50$ N·m, it rolls without slipping. Determine the angular velocity of the wheel after its mass center G has traveled through a distance of $s_G = 20$ m, starting from rest.



Prob. F18-6

PROBLEMS

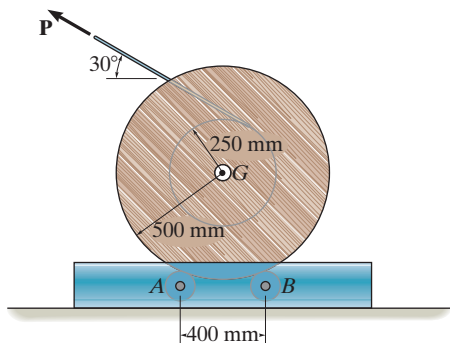
18-1. At a given instant the body of mass m has an angular velocity ω and its mass center has a velocity $\mathbf{v}_G$. Show that its kinetic energy can be represented as $T = \frac{1}{2}I_{IC}\omega^2$, where I_{IC} is the moment of inertia of the body determined about the instantaneous axis of zero velocity, located a distance $r_{G/IC}$ from the mass center as shown.



Prob. 18-1

18-2. A force of $P = 20$ N is applied to the cable, which causes the 175-kg reel to turn since it is resting on the two rollers A and B of the dispenser. Determine the angular velocity of the reel after it has made two revolutions starting from rest. Neglect the mass of the rollers and the mass of the cable. The radius of gyration of the reel about its center axis is $k_G = 0.42$ m.

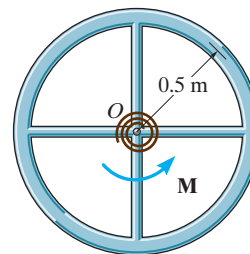
18-3. A force of $P = 20$ N is applied to the cable, which causes the 175-kg reel to turn without slipping on the two rollers A and B of the dispenser. Determine the angular velocity of the reel after it has made two revolutions starting from rest. Neglect the mass of the cable. Each roller can be considered as an 18-kg cylinder, having a radius of 0.1 m. The radius of gyration of the reel about its center axis is $k_G = 0.42$ m.



Probs. 18-2/3

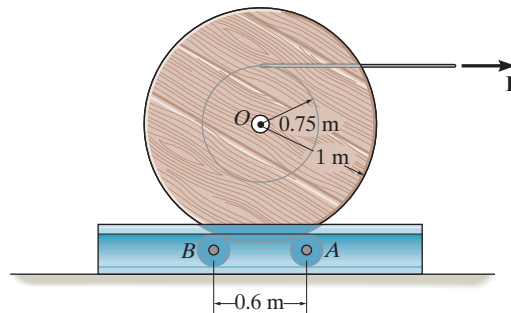
***18-4.** The wheel is made from a 5-kg thin ring and two 2-kg slender rods. If the torsional spring attached to the wheel's center has a stiffness $k = 2$ N·m/rad, and the wheel is rotated until the torque $M = 25$ N·m is developed, determine the maximum angular velocity of the wheel if it is released from rest.

18-5. The wheel is made from a 5-kg thin ring and two 2-kg slender rods. If the torsional spring attached to the wheel's center has a stiffness $k = 2$ N·m/rad, so that the torque on the center of the wheel is $M = (2\theta)$ N·m, where θ is in radians, determine the maximum angular velocity of the wheel if it is rotated two revolutions and then released from rest.



Probs. 18-4/5

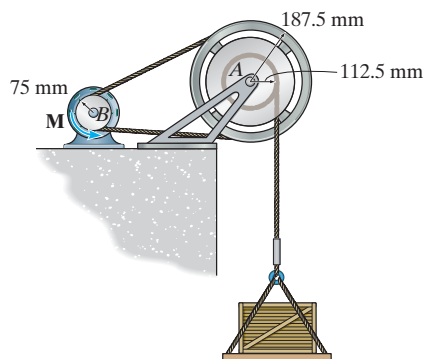
18-6. A force of $P = 60$ N is applied to the cable, which causes the 200-kg reel to turn since it is resting on the two rollers A and B of the dispenser. Determine the angular velocity of the reel after it has made two revolutions starting from rest. Neglect the mass of the rollers and the mass of the cable. Assume the radius of gyration of the reel about its center axis remains constant at $k_O = 0.6$ m.



Prob. 18-6

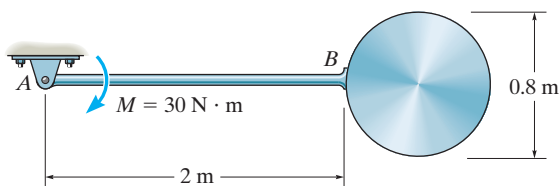
18-7. The wheel and the attached reel have a combined mass of 25 kg and a radius of gyration about their center of $k_A = 150$ mm. If pulley B attached to the motor is subjected to a torque of $M = 60(2 - e^{-0.1\theta})$ N·m, where θ is in radians, determine the velocity of the 100-kg crate after it has moved upwards a distance of 1.5 m, starting from rest. Neglect the mass of pulley B .

***18-8.** The wheel and the attached reel have a combined mass of 25 kg and a radius of gyration about their center of $k_A = 150$ mm. If pulley B attached to the motor is subjected to a torque of $M = 75$ N·m, determine the velocity of the 100-kg crate after the pulley has turned 5 revolutions. Neglect the mass of the pulley.



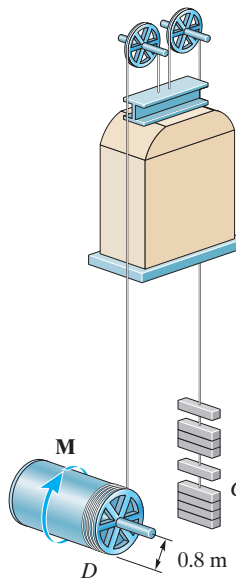
Probs. 18-7/8

18-9. The pendulum consists of a 10-kg uniform disk and a 3-kg uniform slender rod. If it is released from rest in the position shown, determine its angular velocity when it rotates clockwise 90° .



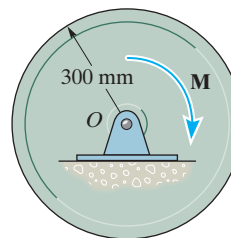
Prob. 18-9

18-10. A motor supplies a constant torque $M = 6$ kN·m to the winding drum that operates the elevator. If the elevator has a mass of 900 kg, the counterweight C has a mass of 200 kg, and the winding drum has a mass of 600 kg and radius of gyration about its axis of $k = 0.6$ m, determine the speed of the elevator after it rises 5 m starting from rest. Neglect the mass of the pulleys.



Prob. 18-10

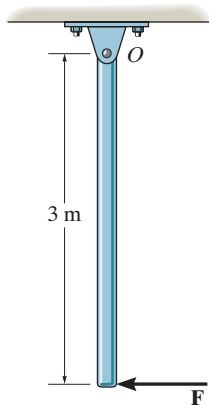
18-11. The disk, which has a mass of 20 kg, is subjected to the couple moment of $M = (2\theta + 4)$ N·m, where θ is in radians. If it starts from rest, determine its angular velocity when it has made two revolutions.



Prob. 18-11

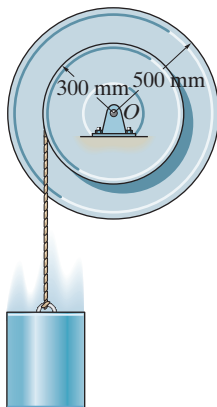
***18–12.** The 10-kg uniform slender rod is suspended at rest when the force of $F = 150$ N is applied to its end. Determine the angular velocity of the rod when it has rotated 90° clockwise from the position shown. The force is always perpendicular to the rod.

18–13. The 10-kg uniform slender rod is suspended at rest when the force of $F = 150$ N is applied to its end. Determine the angular velocity of the rod when it has rotated 180° clockwise from the position shown. The force is always perpendicular to the rod.



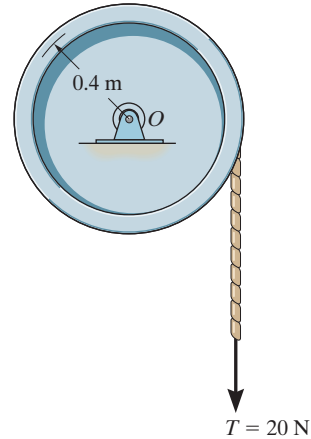
Probs. 18–12/13

18–14. The spool has a mass of 40 kg and a radius of gyration of $k_O = 0.3$ m. If the 10-kg block is released from rest, determine the distance the block must fall in order for the spool to have an angular velocity $\omega = 15$ rad/s. Also, what is the tension in the cord while the block is in motion? Neglect the mass of the cord.



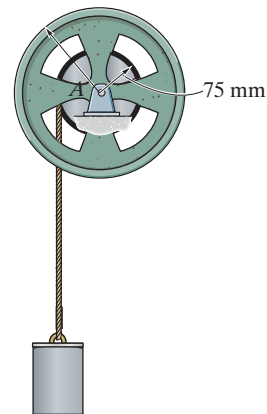
Prob. 18–14

18–15. The force of $T = 20$ N is applied to the cord of negligible mass. Determine the angular velocity of the 20-kg wheel when it has rotated 4 revolutions starting from rest. The wheel has a radius of gyration of $k_O = 0.3$ m.



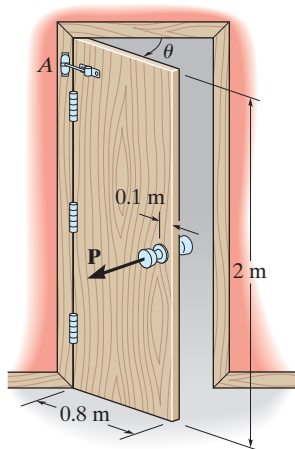
Prob. 18–15

***18–16.** Determine the velocity of the 50-kg cylinder after it has descended a distance of 2 m. Initially, the system is at rest. The reel has a mass of 25 kg and a radius of gyration about its center of mass A of $k_A = 125$ mm.



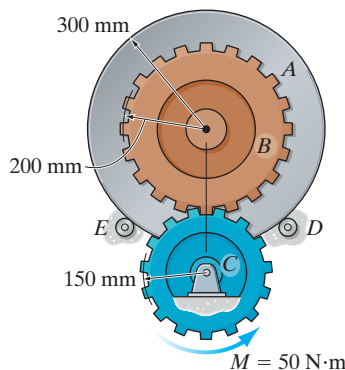
Prob. 18–16

18-17. The uniform door has a mass of 20 kg and can be treated as a thin plate having the dimensions shown. If it is connected to a torsional spring at A , which has a stiffness of $k = 80 \text{ N} \cdot \text{m}/\text{rad}$, determine the required initial twist of the spring in radians so that the door has an angular velocity of 12 rad/s when it closes at $\theta = 0^\circ$ after being opened at $\theta = 90^\circ$ and released from rest. *Hint:* For a torsional spring $M = k\theta$, where k is the stiffness and θ is the angle of twist.



Prob. 18-17

***18-20.** Gear B is rigidly attached to drum A and is supported by two small rollers at E and D . Gear B is in mesh with gear C and is subjected to a torque of $M = 50 \text{ N} \cdot \text{m}$. Determine the angular velocity of the drum after C has rotated 10 revolutions, starting from rest. Gear B and the drum have 100 kg and a radius of gyration about their rotating axis of 250 mm. Gear C has a mass of 30 kg and a radius of gyration about its rotating axis of 125 mm.

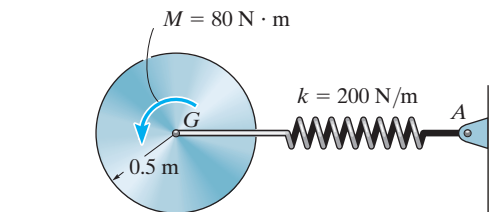


Prob. 18-20

18

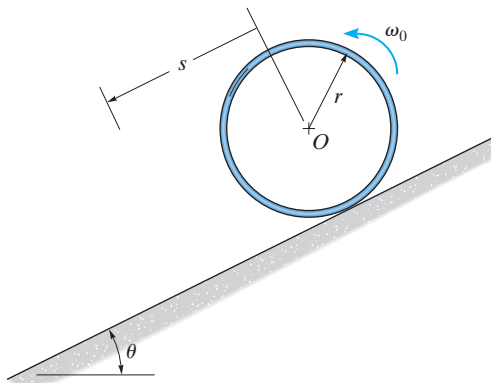
18-18. The 30-kg disk is originally at rest, and the spring is unstretched. A couple moment of $M = 80 \text{ N} \cdot \text{m}$ is then applied to the disk as shown. Determine its angular velocity when its mass center G has moved 0.5 m along the plane. The disk rolls without slipping.

18-19. The 30-kg disk is originally at rest, and the spring is unstretched. A couple moment $M = 80 \text{ N} \cdot \text{m}$ is then applied to the disk as shown. Determine how far the center of mass of the disk travels along the plane before it momentarily stops. The disk rolls without slipping.



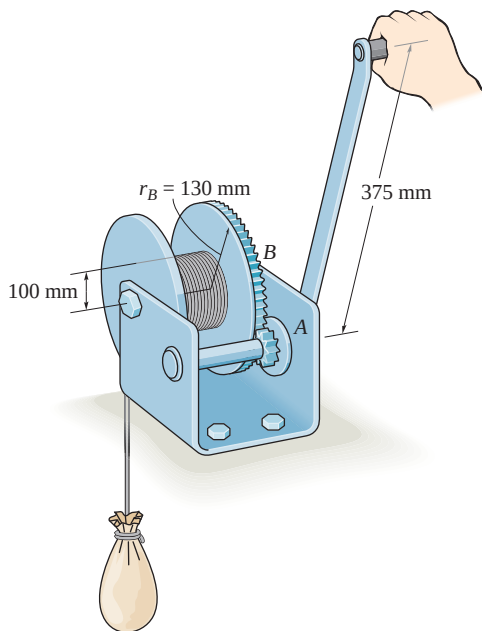
Probs. 18-18/19

18-21. The center O of the thin ring of mass m is given an angular velocity of ω_0 . If the ring rolls without slipping, determine its angular velocity after it has traveled a distance of s down the plane. Neglect its thickness.



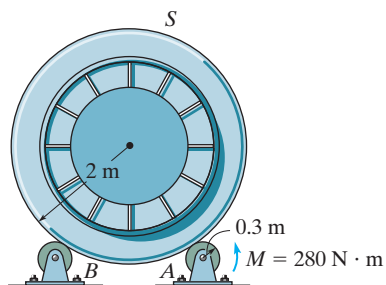
Prob. 18-21

18–22. The hand winch is used to lift the 50-kg load. Determine the work required to rotate the handle five revolutions, starting and ending at rest. The gear at A has a radius of 20 mm.



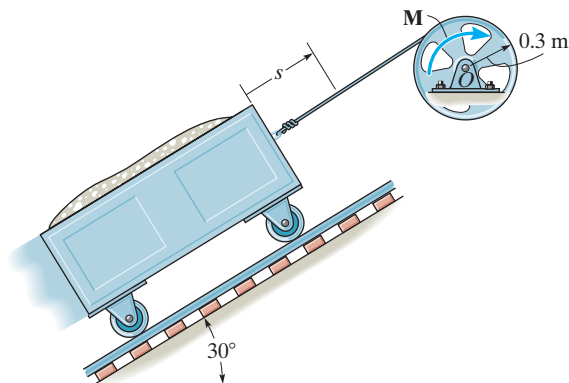
Prob. 18–22

18–23. The rotary screen S is used to wash limestone. When empty it has a mass of 800 kg and a radius of gyration of $k_G = 1.75$ m. Rotation is achieved by applying a torque of $M = 280$ N·m about the drive wheel at A . If no slipping occurs at A and the supporting wheel at B is free to roll, determine the angular velocity of the screen after it has rotated 5 revolutions. Neglect the mass of A and B .



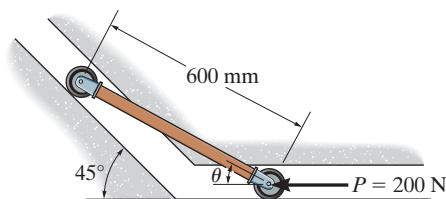
Prob. 18–23

***18–24.** The wheel has a mass of 100 kg and a radius of gyration of $k_O = 0.2$ m. A motor supplies a torque $M = (40\theta + 900)$ N·m, where θ is in radians, about the drive shaft at O . Determine the speed of the loading car, which has a mass of 300 kg, after it travels $s = 4$ m. Initially the car is at rest when $s = 0$ and $\theta = 0^\circ$. Neglect the mass of the attached cable and the mass of the car's wheels.



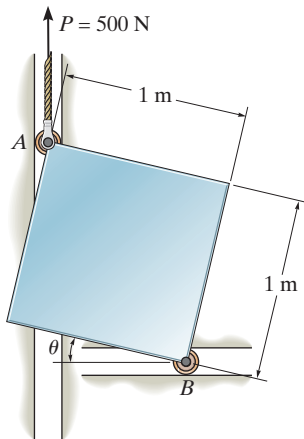
Prob. 18–24

18–25. If $P = 200$ N and the 15-kg uniform slender rod starts from rest at $\theta = 0^\circ$, determine the rod's angular velocity at the instant just before $\theta = 45^\circ$.



Prob. 18–25

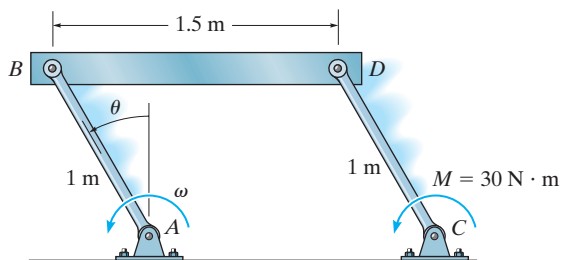
18-26. If corner A of the 60-kg plate is subjected to a vertical force of $P = 500$ N, and the plate is released from rest when $\theta = 0^\circ$, determine the angular velocity of the plate when $\theta = 45^\circ$.



Prob. 18-26

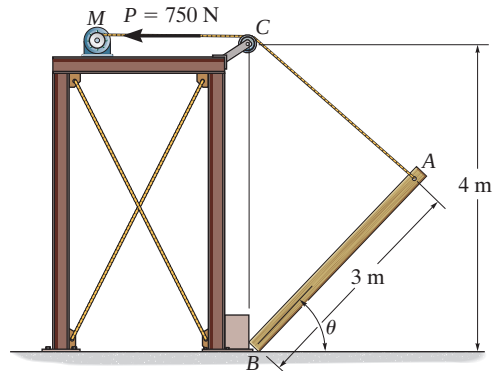
18-27. The linkage consists of two 6-kg rods AB and CD and a 20-kg bar BD . When $\theta = 0^\circ$, rod AB is rotating with an angular velocity $\omega = 2$ rad/s. If rod CD is subjected to a couple moment of $M = 30$ N·m, determine ω_{AB} at the instant $\theta = 90^\circ$.

***18-28.** The linkage consists of two 6-kg rods AB and CD and a 20-kg bar BD . When $\theta = 0^\circ$, rod AB is rotating with an angular velocity $\omega = 2$ rad/s. If rod CD is subjected to a couple moment $M = 30$ N·m, determine ω at the instant $\theta = 45^\circ$.



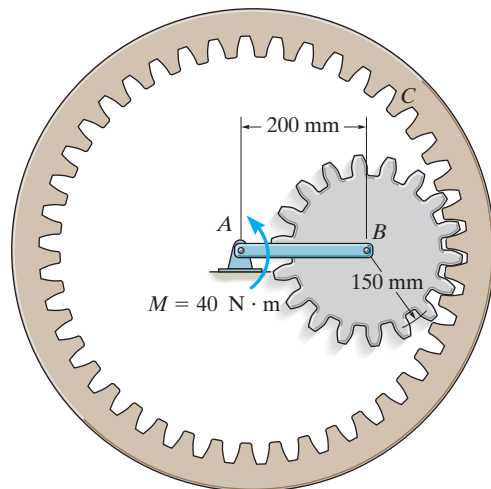
Probs. 18-27/28

18-29. Motor M exerts a constant force of $P = 750$ N on the rope. If the 100-kg post is at rest when $\theta = 0^\circ$, determine the angular velocity of the post at the instant $\theta = 60^\circ$. Neglect the mass of the pulley and its size, and consider the post as a slender rod.



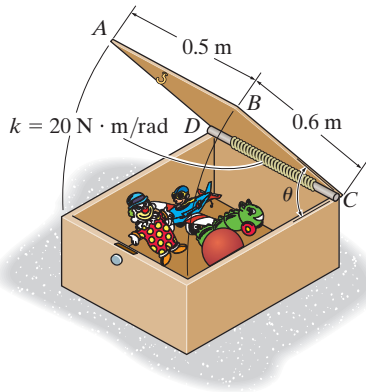
Prob. 18-29

18-30. The link AB is subjected to a couple moment of $M = 40$ N·m. If the ring gear C is fixed, determine the angular velocity of the 15-kg inner gear when the link has made two revolutions starting from rest. Neglect the mass of the link and assume the inner gear is a disk. Motion occurs in the vertical plane.



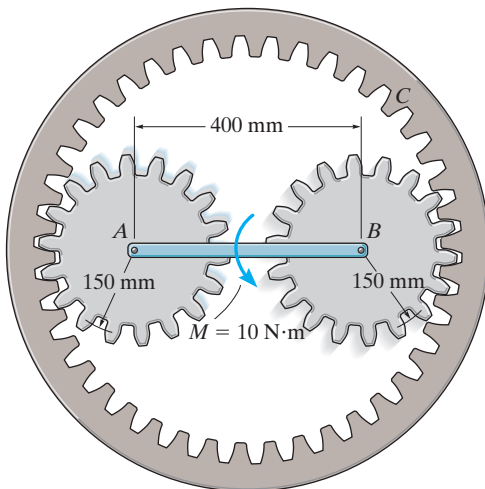
Prob. 18-30

18–31. The 6-kg lid on the box is held in equilibrium by the torsional spring at $\theta = 60^\circ$. If the lid is forced closed, $\theta = 0^\circ$, and then released, determine its angular velocity at the instant it opens to $\theta = 45^\circ$.



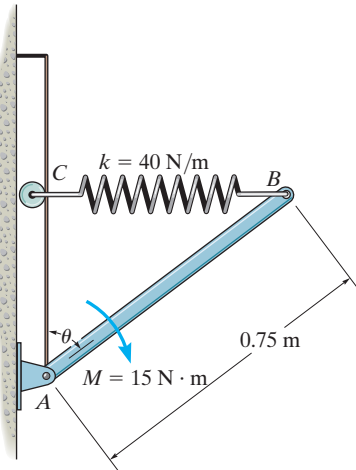
Prob. 18–31

***18–32.** The two 2-kg gears A and B are attached to the ends of a 3-kg slender bar. The gears roll within the fixed ring gear C , which lies in the horizontal plane. If a $10\text{-N}\cdot\text{m}$ torque is applied to the center of the bar as shown, determine the number of revolutions the bar must rotate starting from rest in order for it to have an angular velocity of $\omega_{AB} = 20\text{ rad/s}$. For the calculation, assume the gears can be approximated by thin disks. What is the result if the gears lie in the vertical plane?



Prob. 18–32

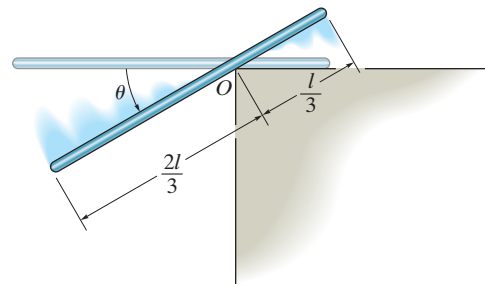
18–33. The 10-kg rod AB is pin connected at A and subjected to a couple moment of $M = 15\text{ N}\cdot\text{m}$. If the rod is released from rest when the spring is unstretched at $\theta = 30^\circ$, determine the rod's angular velocity at the instant $\theta = 60^\circ$. As the rod rotates, the spring always remains horizontal, because of the roller support at C .



Prob. 18–33

18–34. The uniform bar has a mass m and length l . If it is released from rest when $\theta = 0^\circ$, determine its angular velocity as a function of the angle θ before it slips.

18–35. The uniform bar has a mass m and length l . If it is released from rest when $\theta = 0^\circ$, determine the angle θ at which it first begins to slip. The coefficient of static friction at O is $\mu_s = 0.3$.

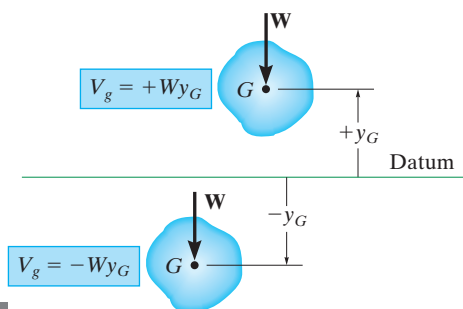


Probs. 18–34/35

18.5 Conservation of Energy

When a force system acting on a rigid body consists only of *conservative forces*, the conservation of energy theorem can be used to solve a problem that otherwise would be solved using the principle of work and energy. This theorem is often easier to apply since the work of a conservative force is *independent of the path* and depends only on the initial and final positions of the body. It was shown in Sec. 14.5 that the work of a conservative force can be expressed as the difference in the body's potential energy measured from an arbitrarily selected reference or datum.

Gravitational Potential Energy. Since the total weight of a body can be considered concentrated at its center of gravity, the *gravitational potential energy* of the body is determined by knowing the height of the body's center of gravity above or below a horizontal datum.



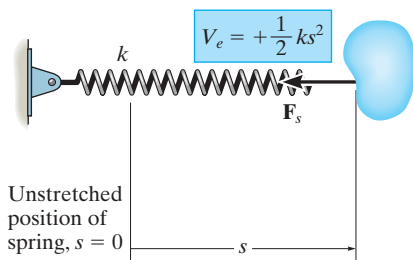
Gravitational potential energy

Fig. 18-16

$$V_g = Wy_G \quad (18-15)$$

Here the potential energy is *positive* when y_G is positive upward, since the weight has the ability to do *positive work* when the body moves back to the datum, Fig. 18-16. Likewise, if G is located *below* the datum ($-y_G$), the gravitational potential energy is *negative*, since the weight does *negative work* when the body returns to the datum.

Elastic Potential Energy. The force developed by an elastic spring is also a conservative force. The *elastic potential energy* which a spring imparts to an attached body when the spring is stretched or compressed from an initial undeformed position ($s = 0$) to a final position s , Fig. 18-17, is



Elastic potential energy

Fig. 18-17

$$V_e = +\frac{1}{2}ks^2 \quad (18-16)$$

In the deformed position, the spring force acting *on the body* always has the ability for doing positive work when the spring returns back to its original undeformed position (see Sec. 14.5).

Conservation of Energy. In general, if a body is subjected to both gravitational and elastic forces, the total *potential energy* can be expressed as a potential function represented as the algebraic sum

$$V = V_g + V_e \quad (18-17)$$

Here measurement of V depends upon the location of the body with respect to the selected datum.

Realizing that the work of conservative forces can be written as a difference in their potential energies, i.e., $(\Sigma U_{1-2})_{\text{cons}} = V_1 - V_2$, Eq. 14-16, we can rewrite the principle of work and energy for a rigid body as

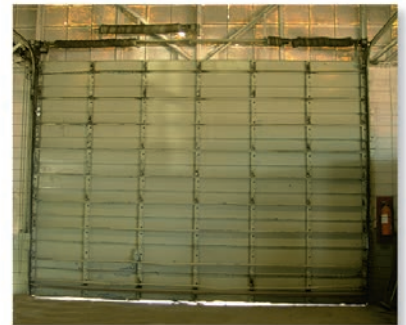
$$T_1 + V_1 + (\Sigma U_{1-2})_{\text{noncons}} = T_2 + V_2 \quad (18-18)$$

Here $(\Sigma U_{1-2})_{\text{noncons}}$ represents the work of the nonconservative forces such as friction. If this term is zero, then

$$T_1 + V_1 = T_2 + V_2 \quad (18-19)$$

This equation is referred to as the conservation of mechanical energy. It states that the *sum* of the potential and kinetic energies of the body remains *constant* when the body moves from one position to another. It also applies to a system of smooth, pin-connected rigid bodies, bodies connected by inextensible cords, and bodies in mesh with other bodies. In all these cases the forces acting at the points of contact are *eliminated* from the analysis, since they occur in equal but opposite collinear pairs and each pair of forces moves through an equal distance when the system undergoes a displacement.

It is important to remember that only problems involving conservative force systems can be solved by using Eq. 18-19. As stated in Sec. 14.5, friction or other drag-resistant forces, which depend on velocity or acceleration, are nonconservative. The work of such forces is transformed into thermal energy used to heat up the surfaces of contact, and consequently this energy is dissipated into the surroundings and may not be recovered. Therefore, problems involving frictional forces can be solved by using either the principle of work and energy written in the form of Eq. 18-18, if it applies, or the equations of motion.



The torsional springs located at the top of the garage door wind up as the door is lowered. When the door is raised, the potential energy stored in the springs is then transferred into gravitational potential energy of the door's weight, thereby making it easy to open.

Procedure for Analysis

The conservation of energy equation is used to solve problems involving *velocity*, *displacement*, and *conservative force systems*. For application it is suggested that the following procedure be used.

Potential Energy.

- Draw two diagrams showing the body located at its initial and final positions along the path.
- If the center of gravity, G , is subjected to a *vertical displacement*, establish a fixed horizontal datum from which to measure the body's gravitational potential energy V_g .
- Data pertaining to the elevation y_G of the body's center of gravity from the datum and the extension or compression of any connecting springs can be determined from the problem geometry and listed on the two diagrams.
- The potential energy is determined from $V = V_g + V_e$. Here $V_g = Wy_G$, which can be positive or negative, and $V_e = \frac{1}{2}ks^2$, which is always positive.

Kinetic Energy.

- The kinetic energy of the body consists of two parts, namely translational kinetic energy, $T = \frac{1}{2}mv_G^2$, and rotational kinetic energy, $T = \frac{1}{2}I_G\omega^2$.
- Kinematic diagrams for velocity may be useful for establishing a *relationship* between v_G and ω .

Conservation of Energy.

- Apply the conservation of energy equation $T_1 + V_1 = T_2 + V_2$.

EXAMPLE 18.6

The 10-kg rod AB shown in Fig. 18–18a is confined so that its ends move in the horizontal and vertical slots. The spring has a stiffness of $k = 800 \text{ N/m}$ and is unstretched when $\theta = 0^\circ$. Determine the angular velocity of AB when $\theta = 0^\circ$, if the rod is released from rest when $\theta = 30^\circ$. Neglect the mass of the slider blocks.

SOLUTION

Potential Energy. The two diagrams of the rod, when it is located at its initial and final positions, are shown in Fig. 18–18b. The datum, used to measure the gravitational potential energy, is placed in line with the rod when $\theta = 0^\circ$.

When the rod is in position 1, the center of gravity G is located *below the datum* so its gravitational potential energy is *negative*. Furthermore, (positive) elastic potential energy is stored in the spring, since it is stretched a distance of $s_1 = (0.4 \sin 30^\circ) \text{ m}$. Thus,

$$\begin{aligned} V_1 &= -Wy_1 + \frac{1}{2}ks_1^2 \\ &= -(98.1 \text{ N})(0.2 \sin 30^\circ \text{ m}) + \frac{1}{2}(800 \text{ N/m})(0.4 \sin 30^\circ \text{ m})^2 = 6.19 \text{ J} \end{aligned}$$

When the rod is in position 2, the potential energy of the rod is zero, since the center of gravity G is located at the datum, and the spring is unstretched, $s_2 = 0$. Thus,

$$V_2 = 0$$

Kinetic Energy. The rod is released from rest from position 1, thus $(\mathbf{v}_G)_1 = \mathbf{0}$, and so

$$T_1 = 0$$

In position 2, the angular velocity is ω_2 and the rod's mass center has a velocity of $(\mathbf{v}_G)_2$. Thus,

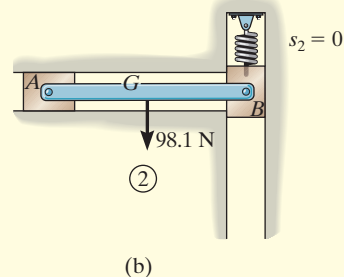
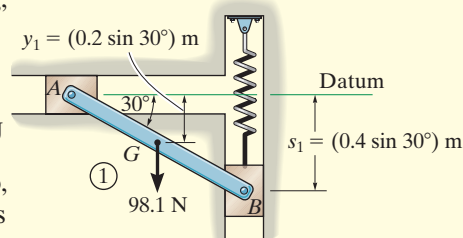
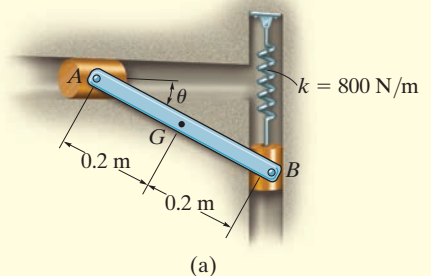
$$\begin{aligned} T_2 &= \frac{1}{2}m(v_G)_2^2 + \frac{1}{2}I_G\omega_2^2 \\ &= \frac{1}{2}(10 \text{ kg})(v_G)_2^2 + \frac{1}{2}\left[\frac{1}{12}(10 \text{ kg})(0.4 \text{ m})^2\right]\omega_2^2 \end{aligned}$$

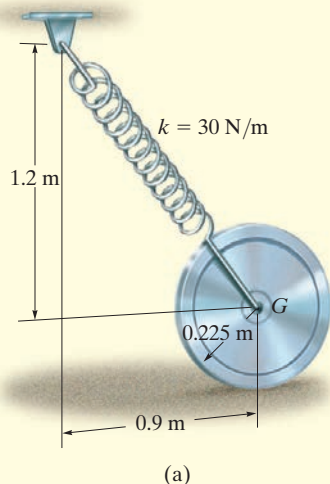
Using *kinematics*, $(\mathbf{v}_G)_2$ can be related to ω_2 as shown in Fig. 18–18c. At the instant considered, the instantaneous center of zero velocity (IC) for the rod is at point A ; hence, $(v_G)_2 = (r_{G/IC})\omega_2 = (0.2 \text{ m})\omega_2$. Substituting into the above expression and simplifying (or using $\frac{1}{2}I_{IC}\omega_2^2$), we get

$$T_2 = 0.2667\omega_2^2$$

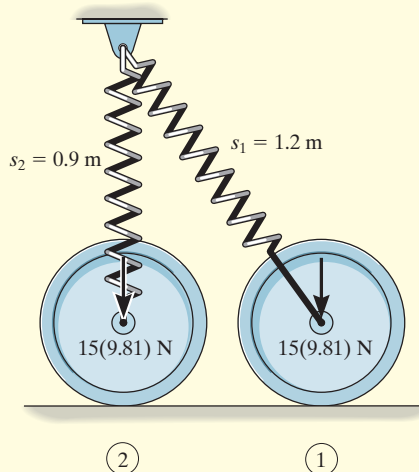
Conservation of Energy.

$$\begin{aligned} \{T_1\} + \{V_1\} &= \{T_2\} + \{V_2\} \\ \{0\} + \{6.19 \text{ J}\} &= \{0.2667\omega_2^2\} + \{0\} \\ \omega_2 &= 4.82 \text{ rad/s} \end{aligned}$$

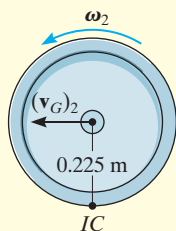
Ans.**Fig. 18–18**

EXAMPLE 18.7

(a)



(b)



(c)

Fig. 18–19

The wheel shown in Fig. 18–19a has a weight of 15 kg and a radius of gyration of $k_G = 0.2$ m. It is attached to a spring which has a stiffness $k = 30$ N/m and an unstretched length of 0.3 m. If the disk is released from rest in the position shown and rolls without slipping, determine its angular velocity at the instant G moves 0.9 m to the left.

SOLUTION

Potential Energy. Two diagrams of the wheel, when it is at the initial and final positions, are shown in Fig. 18–19b. A gravitational datum is not needed here since the weight is not displaced vertically. From the problem geometry the spring is stretched $s_1 = (\sqrt{0.9^2 + 1.2^2} - 0.3) = 1.2$ m in the initial position, and stretched $s_2 = (1.2 - 0.3) = 0.9$ m in the final position. Hence, the positive spring potential energy is

$$V_1 = \frac{1}{2}ks_1^2 = \frac{1}{2}(30 \text{ N/m})(1.2 \text{ m})^2 = 21.6 \text{ N} \cdot \text{m}$$

$$V_2 = \frac{1}{2}ks_2^2 = \frac{1}{2}(30 \text{ N/m})(0.9 \text{ m})^2 = 12.15 \text{ N} \cdot \text{m}$$

Kinetic Energy. The disk is released from rest and so $(\mathbf{v}_G)_1 = \mathbf{0}$, $\omega_1 = 0$. Therefore,

$$T_1 = 0$$

Since the instantaneous center of zero velocity is at the ground, Fig. 18–19c, we have

$$\begin{aligned} T_2 &= \frac{1}{2}I_{IC}\omega_2^2 \\ &= \frac{1}{2}\left[(15 \text{ kg})(0.2 \text{ m})^2 + (15 \text{ kg})(0.225 \text{ m})^2\right]\omega_2^2 \\ &= 0.6797\omega_2^2 \end{aligned}$$

Conservation of Energy.

$$\begin{aligned} \{T_1\} + \{V_1\} &= \{T_2\} + \{V_2\} \\ \{0\} + \{21.6 \text{ N} \cdot \text{m}\} &= \{0.6797\omega_2^2\} + \{12.15 \text{ N} \cdot \text{m}\} \\ \omega_2 &= 3.73 \text{ rad/s} \end{aligned}$$

Ans.

NOTE: If the principle of work and energy were used to solve this problem, then the work of the spring would have to be determined by considering both the change in magnitude and direction of the spring force.

EXAMPLE 18.8

The 10-kg homogeneous disk shown in Fig. 18–20*a* is attached to a uniform 5-kg rod AB . If the assembly is released from rest when $\theta = 60^\circ$, determine the angular velocity of the rod when $\theta = 0^\circ$. Assume that the disk rolls without slipping. Neglect friction along the guide and the mass of the collar at B .

SOLUTION

Potential Energy. Two diagrams for the rod and disk, when they are located at their initial and final positions, are shown in Fig. 18–20*b*. For convenience the datum passes through point A .

When the system is in position 1, only the rod's weight has positive potential energy. Thus,

$$V_1 = W_r y_1 = (49.05 \text{ N})(0.3 \sin 60^\circ \text{ m}) = 12.74 \text{ J}$$

When the system is in position 2, both the weight of the rod and the weight of the disk have zero potential energy. Why? Thus,

$$V_2 = 0$$

Kinetic Energy. Since the entire system is at rest at the initial position,

$$T_1 = 0$$

In the final position the rod has an angular velocity $(\omega_r)_2$ and its mass center has a velocity $(\mathbf{v}_G)_2$, Fig. 18–20*c*. Since the rod is *fully extended* in this position, the disk is momentarily at rest, so $(\omega_d)_2 = 0$ and $(\mathbf{v}_A)_2 = 0$. For the rod $(\mathbf{v}_G)_2$ can be related to $(\omega_r)_2$ from the instantaneous center of zero velocity, which is located at point A , Fig. 18–20*c*. Hence, $(v_G)_2 = r_{G/IC}(\omega_r)_2$ or $(v_G)_2 = 0.3(\omega_r)_2$. Thus,

$$\begin{aligned} T_2 &= \frac{1}{2} m_r (v_G)_2^2 + \frac{1}{2} I_G (\omega_r)_2^2 + \frac{1}{2} m_d (v_A)_2^2 + \frac{1}{2} I_A (\omega_d)_2^2 \\ &= \frac{1}{2} (5 \text{ kg}) [(0.3 \text{ m})(\omega_r)_2]^2 + \frac{1}{2} \left[\frac{1}{12} (5 \text{ kg}) (0.6 \text{ m})^2 \right] (\omega_r)_2^2 + 0 + 0 \\ &= 0.3 (\omega_r)_2^2 \end{aligned}$$

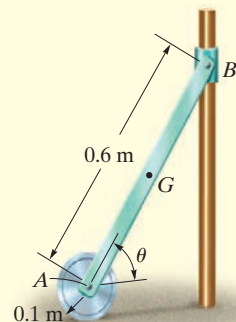
Conservation of Energy.

$$\begin{aligned} \{T_1\} + \{V_1\} &= \{T_2\} + \{V_2\} \\ \{0\} + \{12.74 \text{ J}\} &= \{0.3(\omega_r)_2^2\} + \{0\} \\ (\omega_r)_2 &= 6.52 \text{ rad/s} \end{aligned}$$

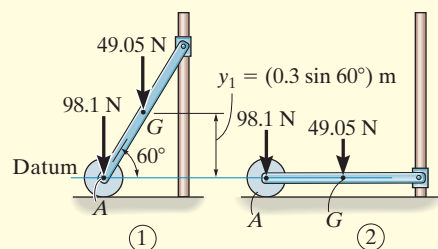
Ans.

NOTE: We can also determine the final kinetic energy of the rod using

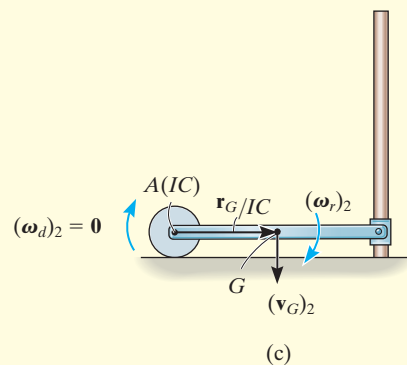
$$T_2 = \frac{1}{2} I_{IC} \omega_2^2.$$



(a)

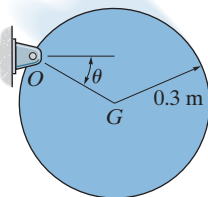


(b)

**Fig. 18–20**

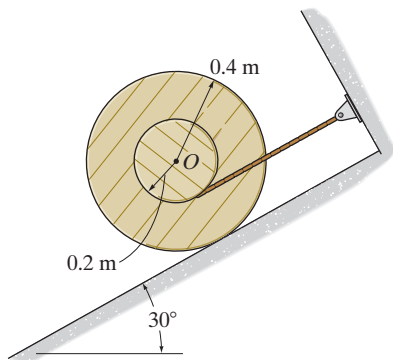
FUNDAMENTAL PROBLEMS

F18-7. If the 30-kg disk is released from rest when $\theta = 0^\circ$, determine its angular velocity when $\theta = 90^\circ$.



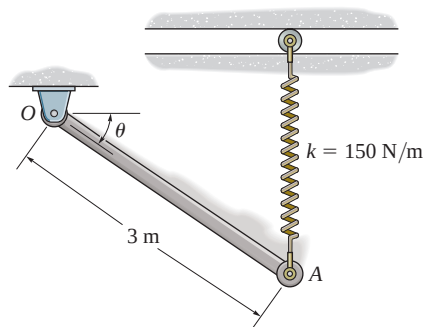
Prob. F18-7

F18-8. The 50-kg reel has a radius of gyration about its center O of $k_O = 300$ mm. If it is released from rest, determine its angular velocity when its center O has traveled 6 m down the smooth inclined plane.



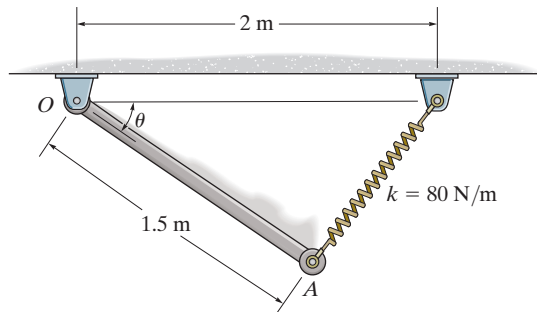
Prob. F18-8

F18-9. The 60-kg rod OA is released from rest when $\theta = 0^\circ$. Determine its angular velocity when $\theta = 45^\circ$. The spring remains vertical during the motion and is unstretched when $\theta = 0^\circ$.



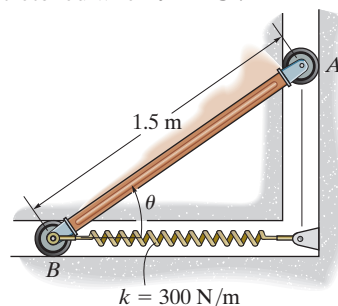
Prob. F18-9

F18-10. The 30-kg rod is released from rest when $\theta = 0^\circ$. Determine the angular velocity of the rod when $\theta = 90^\circ$. The spring is unstretched when $\theta = 0^\circ$.



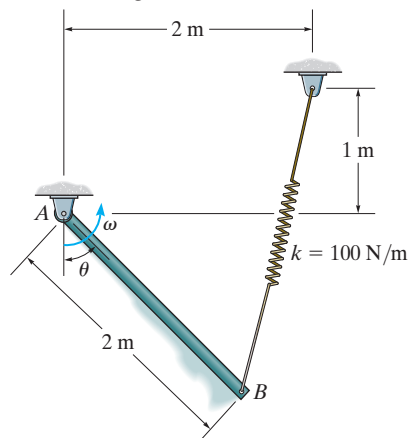
Prob. F18-10

F18-11. The 30-kg rod is released from rest when $\theta = 45^\circ$. Determine the angular velocity of the rod when $\theta = 0^\circ$. The spring is unstretched when $\theta = 45^\circ$.



Prob. F18-11

F18-12. The 20-kg rod is released from rest when $\theta = 0^\circ$. Determine its angular velocity when $\theta = 90^\circ$. The spring has an unstretched length of 0.5 m.

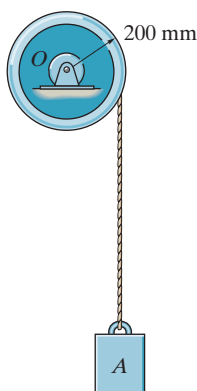


Prob. F18-12

PROBLEMS

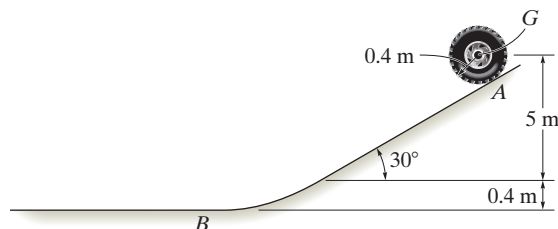
***18–36.** The spool has a mass of 20 kg and a radius of gyration of $k_O = 160$ mm. If the 15-kg block A is released from rest, determine the distance the block must fall in order for the spool to have an angular velocity $\omega = 8$ rad/s. Also, what is the tension in the cord while the block is in motion? Neglect the mass of the cord.

18–37. The spool has a mass of 20 kg and a radius of gyration of $k_O = 160$ mm. If the 15-kg block A is released from rest, determine the velocity of the block when it descends 600 mm.



Probs. 18–36/37

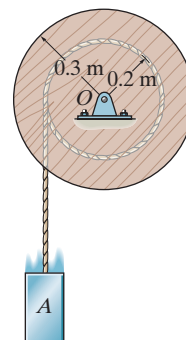
18–38. An automobile tire has a mass of 7 kg and radius of gyration of $k_G = 0.3$ m. If it is released from rest at A on the incline, determine its angular velocity when it reaches the horizontal plane. The tire rolls without slipping.



Prob. 18–38

18–39. The spool has a mass of 50 kg and a radius of gyration of $k_O = 0.280$ m. If the 20-kg block A is released from rest, determine the distance the block must fall in order for the spool to have an angular velocity $\omega = 5$ rad/s. Also, what is the tension in the cord while the block is in motion? Neglect the mass of the cord.

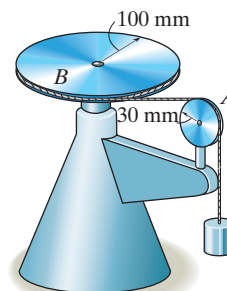
***18–40.** The spool has a mass of 50 kg and a radius of gyration of $k_O = 0.280$ m. If the 20-kg block A is released from rest, determine the velocity of the block when it descends 0.5 m.



Probs. 18–39/40

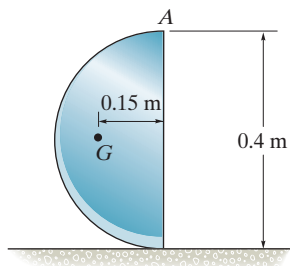
18–41. The assembly consists of a 3-kg pulley A and 10-kg pulley B . If a 2-kg block is suspended from the cord, determine the block's speed after it descends 0.5 m starting from rest. Neglect the mass of the cord and treat the pulleys as thin disks. No slipping occurs.

18–42. The assembly consists of a 3-kg pulley A and 10-kg pulley B . If a 2-kg block is suspended from the cord, determine the distance the block must descend, starting from rest, in order to cause B to have an angular velocity of 6 rad/s. Neglect the mass of the cord and treat the pulleys as thin disks. No slipping occurs.



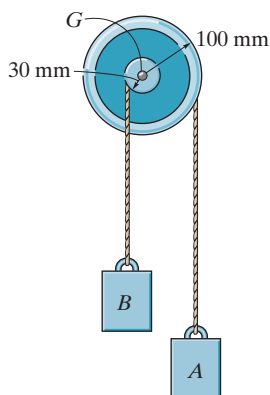
Probs. 18–41/42

18–43. The 15-kg semicircular segment is released from rest in the position shown. Determine the velocity of point A when it has rotated counterclockwise 90° . Assume that the segment rolls without slipping on the surface. The moment of inertia about its mass center is $I_G = 0.25 \text{ kg} \cdot \text{m}^2$.



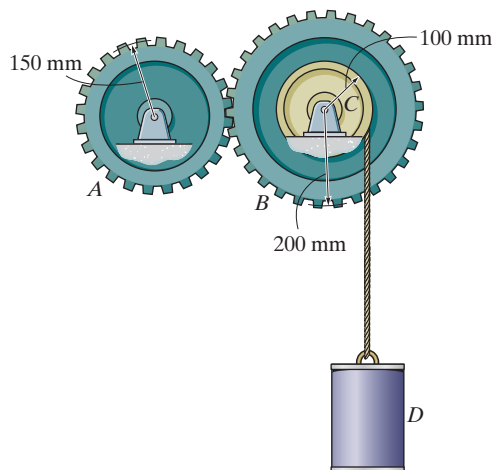
Prob. 18–43

***18–44.** The compound disk pulley consists of a hub and attached outer rim. If it has a mass of 3 kg and a radius of gyration of $k_G = 45 \text{ mm}$, determine the speed of block A after A descends 0.2 m from rest. Blocks A and B each have a mass of 2 kg. Neglect the mass of the cords.



Prob. 18–44

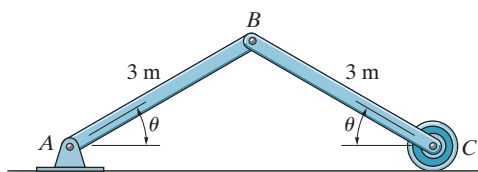
18–45. Determine the speed of the 50-kg cylinder after it has descended a distance of 2 m, starting from rest. Gear A has a mass of 10 kg and a radius of gyration of 125 mm about its center of mass. Gear B and drum C have a combined mass of 30 kg and a radius of gyration about their center of mass of 150 mm.



Prob. 18–45

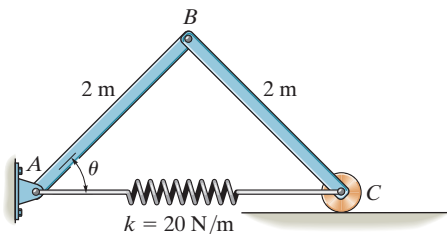
18–46. The assembly consists of two 10-kg bars which are pin connected. If the bars are released from rest when $\theta = 60^\circ$, determine their angular velocities at the instant $\theta = 0^\circ$. The 5-kg disk at C has a radius of 0.5 m and rolls without slipping.

18–47. The assembly consists of two 10-kg bars which are pin connected. If the bars are released from rest when $\theta = 60^\circ$, determine their angular velocities at the instant $\theta = 30^\circ$. The 5-kg disk at C has a radius of 0.5 m and rolls without slipping.



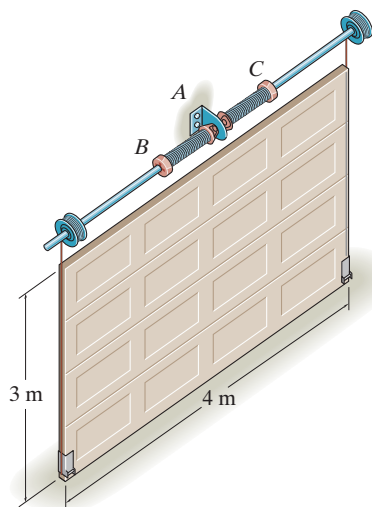
Probs. 18–46/47

***18–48.** The two 12-kg slender rods are pin connected and released from rest at the position $\theta = 60^\circ$. If the spring has an unstretched length of 1.5 m, determine the angular velocity of rod BC , when the system is at the position $\theta = 0^\circ$. Neglect the mass of the roller at C .



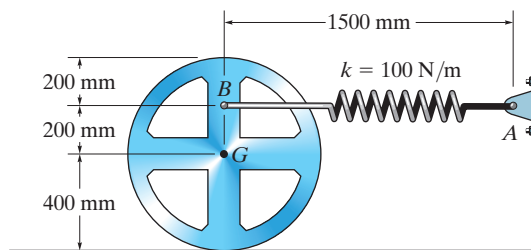
Prob. 18–48

18–49. The uniform garage door has a mass of 150 kg and is guided along smooth tracks at its ends. Lifting is done using the two springs, each of which is attached to the anchor bracket at A and to the counterbalance shaft at B and C . As the door is raised, the springs begin to unwind from the shaft, thereby assisting the lift. If each spring provides a torsional moment of $M = (0.7\theta) \text{ N} \cdot \text{m}$, where θ is in radians, determine the angle θ_0 at which both the left-wound and right-wound spring should be attached so that the door is completely balanced by the springs, i.e., when the door is in the vertical position and is given a slight force upward, the springs will lift the door along the side tracks to the horizontal plane with no final angular velocity. *Note:* The elastic potential energy of a torsional spring is $V_e = \frac{1}{2}k\theta^2$, where $M = k\theta$ and in this case $k = 0.7 \text{ N} \cdot \text{m/rad}$.



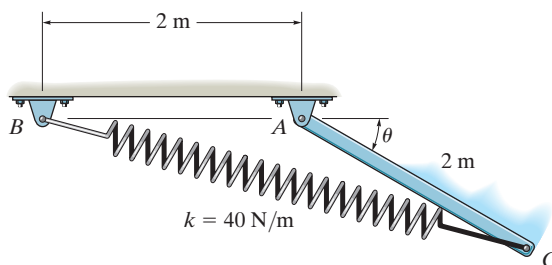
Prob. 18–49

18–50. The 40-kg wheel has a radius of gyration about its center of gravity G of $k_G = 250 \text{ mm}$. If it rolls without slipping, determine its angular velocity when it has rotated clockwise 90° from the position shown. The spring AB has a stiffness $k = 100 \text{ N/m}$ and an unstretched length of 500 mm. The wheel is released from rest.



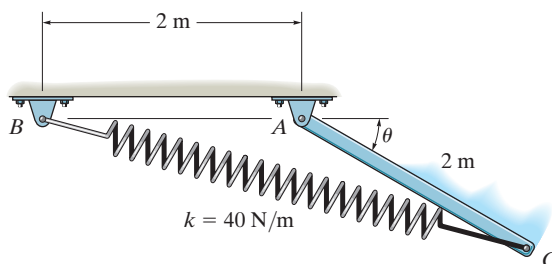
Prob. 18–50

18–51. The 12-kg slender rod is attached to a spring, which has an unstretched length of 2 m. If the rod is released from rest when $\theta = 30^\circ$, determine its angular velocity at the instant $\theta = 90^\circ$.



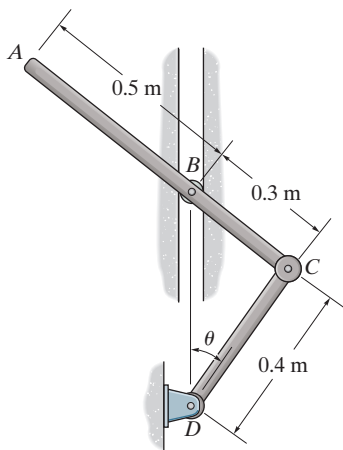
Prob. 18–51

***18–52.** The 12-kg slender rod is attached to a spring, which has an unstretched length of 2 m. If the rod is released from rest when $\theta = 30^\circ$, determine the angular velocity of the rod the instant the spring becomes unstretched.



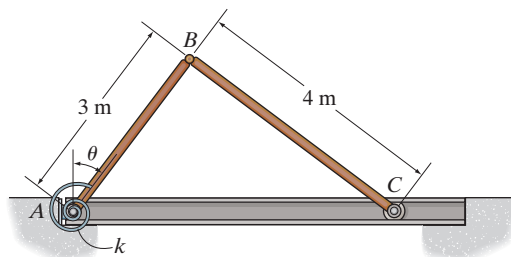
Prob. 18–52

18-53. The 6-kg rod ABC is connected to the 3-kg rod CD . If the system is released from rest when $\theta = 0^\circ$, determine the angular velocity of rod ABC at the instant it becomes horizontal.



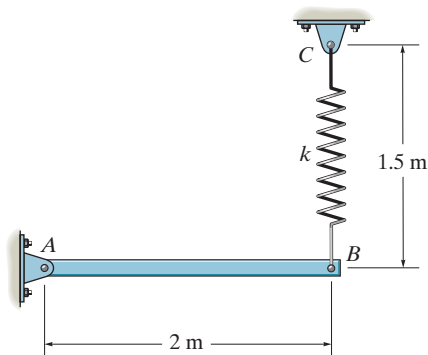
Prob. 18-53

18-55. The torsional spring at A has a stiffness of $k = 2000 \text{ N} \cdot \text{m}/\text{rad}$ and is uncoiled when $\theta = 0^\circ$. Determine the angular velocity of the bars, AB and BC , when $\theta = 0^\circ$, if they are released from rest at the closed position, $\theta = 90^\circ$. The bars have a mass per unit length of $20 \text{ kg}/\text{m}$.



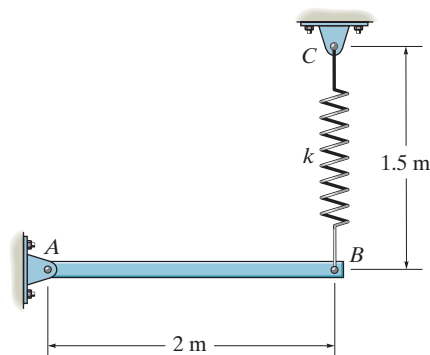
Prob. 18-55

18 **18-54.** The slender 6-kg bar AB is horizontal and at rest and the spring is unstretched. Determine the stiffness k of the spring so that the motion of the bar is momentarily stopped when it has rotated clockwise 90° after being released.



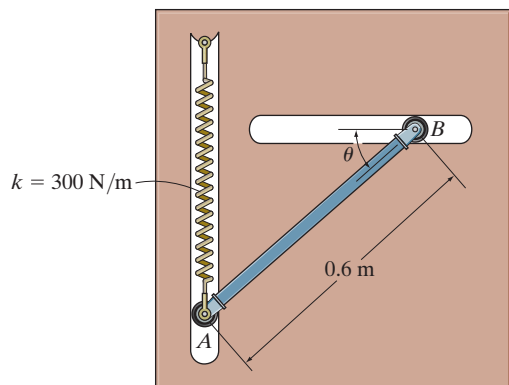
Prob. 18-54

***18-56.** The slender 6-kg bar AB is horizontal and at rest and the spring is unstretched. Determine the angular velocity of the bar when it has rotated clockwise 45° after being released. The spring has a stiffness of $k = 12 \text{ N}/\text{m}$.



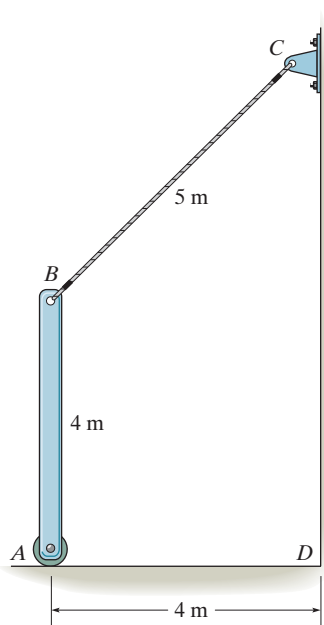
Prob. 18-56

18-57. A spring having a stiffness of $k = 300 \text{ N/m}$ is attached to the end of the 15-kg rod, and it is unstretched when $\theta = 0^\circ$. If the rod is released from rest when $\theta = 0^\circ$, determine its angular velocity at the instant $\theta = 30^\circ$. The motion is in the vertical plane.



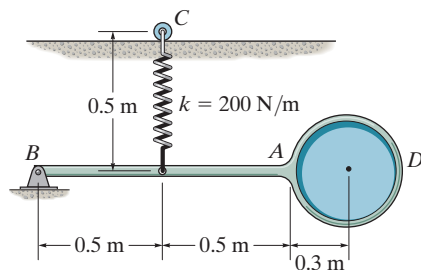
Prob. 18-57

18-58. The slender 15-kg bar is initially at rest and standing in the vertical position when the bottom end A is displaced slightly to the right. If the track in which it moves is smooth, determine the speed at which end A strikes the corner D . The bar is constrained to move in the vertical plane. Neglect the mass of the cord BC .



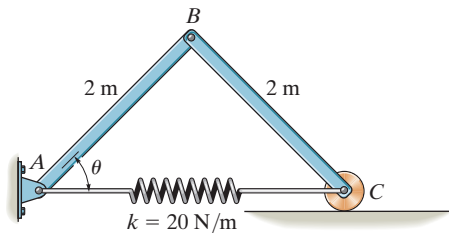
Prob. 18-58

18-59. The pendulum consists of a 6-kg slender rod fixed to a 15-kg disk. If the spring has an unstretched length of 0.2 m , determine the angular velocity of the pendulum when it is released from rest and rotates clockwise 90° from the position shown. The roller at C allows the spring to always remain vertical.



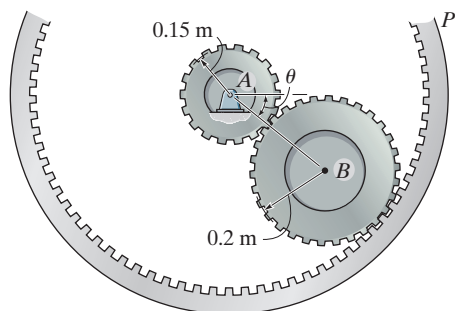
Prob. 18-59

***18-60.** The two 12-kg slender rods are pin connected and released from rest at the position $\theta = 60^\circ$. If the spring has an unstretched length of 1.5 m , determine the angular velocity of rod BC , when the system is at the position $\theta = 30^\circ$.



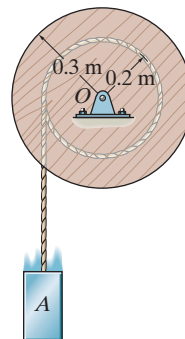
Prob. 18-60

18–61. If the 40-kg gear B is released from rest at $\theta = 0^\circ$, determine the angular velocity of the 20-kg gear A at the instant $\theta = 90^\circ$. The radii of gyration of gears A and B about their respective centers of mass are $k_A = 125$ mm and $k_B = 175$ mm. The outer gear ring P is fixed.



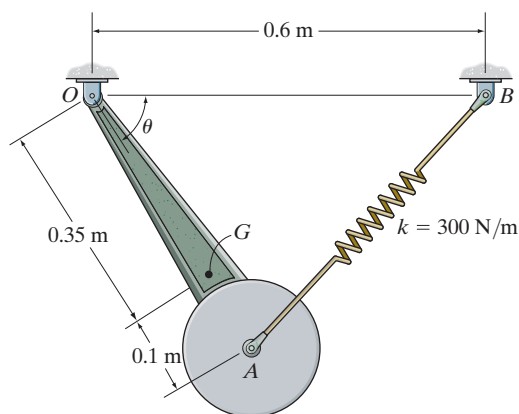
Prob. 18–61

18–63. The spool has a mass of 50 kg and a radius of gyration $k_O = 0.280$ m. If the 20-kg block A is released from rest, determine the distance the block must fall in order for the spool to have an angular velocity $\omega = 5$ rad/s. Also, what is the tension in the cord while the block is in motion? Neglect the mass of the cord.



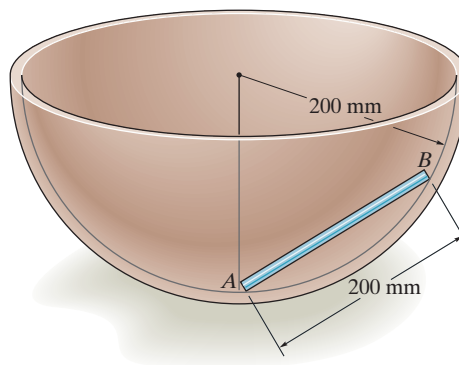
Prob. 18–63

18–62. The 30 kg pendulum has its mass center at G and a radius of gyration about point G of $k_G = 300$ mm. If it is released from rest when $\theta = 0^\circ$, determine its angular velocity at the instant $\theta = 90^\circ$. Spring AB has a stiffness of $k = 300$ N/m and is unstretched when $\theta = 0^\circ$.



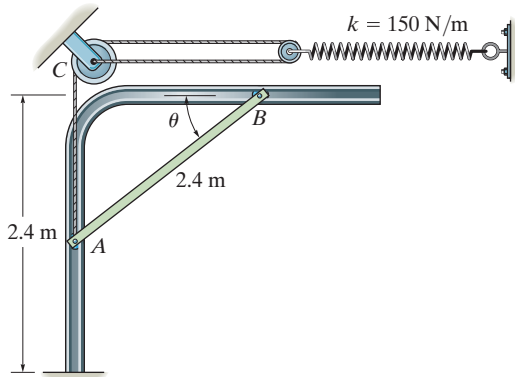
Prob. 18–62

***18–64.** The 500-g rod AB rests along the smooth inner surface of a hemispherical bowl. If the rod is released from rest from the position shown, determine its angular velocity at the instant it swings downward and becomes horizontal.



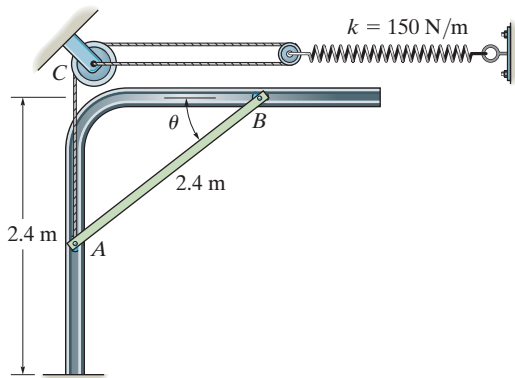
Prob. 18–64

18–65. The motion of the uniform 40-kg garage door is guided at its ends by the track. Determine the required initial stretch in the spring when the door is open, $\theta = 0^\circ$, so that when it falls freely it comes to rest when it just reaches the fully closed position, $\theta = 90^\circ$. Assume the door can be treated as a thin plate, and there is a spring and pulley system on each of the two sides of the door.



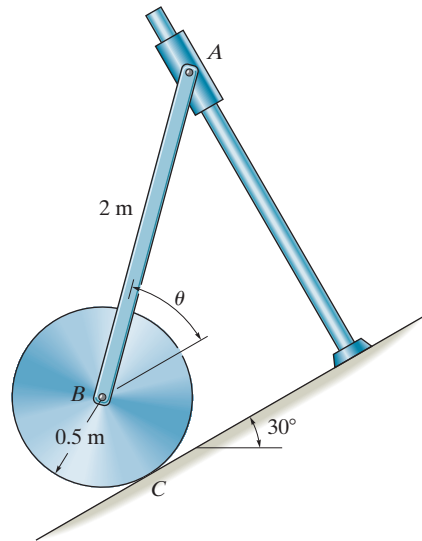
Prob. 18–65

18–66. The motion of the uniform 40-kg garage door is guided at its ends by the track. If it is released from rest at $\theta = 0^\circ$, determine the door's angular velocity at the instant $\theta = 30^\circ$. The spring is originally stretched 0.3 m when the door is held open, $\theta = 0^\circ$. Assume the door can be treated as a thin plate, and there is a spring and pulley system on each of the two sides of the door.



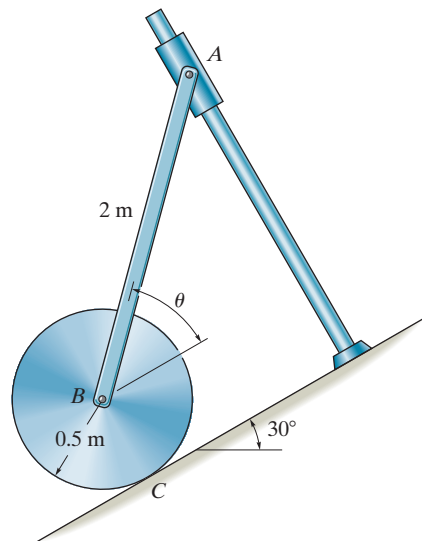
Prob. 18–66

18–67. The system consists of a 30-kg disk, a 12-kg slender rod BA , and a 5-kg smooth collar A . If the disk rolls without slipping, determine the velocity of the collar at the instant $\theta = 0^\circ$. The system is released from rest when $\theta = 45^\circ$.



Prob. 18–67

***18–68.** The system consists of a 30-kg disk, a 12-kg slender rod BA , and a 5-kg smooth collar A . If the disk rolls without slipping, determine the velocity of the collar at the instant $\theta = 30^\circ$. The system is released from rest when $\theta = 45^\circ$.



Prob. 18–68

CONCEPTUAL PROBLEMS

C18-1. The bicycle and rider start from rest at the top of the hill. Show how to determine the speed of the rider when he freely coasts down the hill. Use appropriate dimensions of the wheels, and the mass of the rider, frame and wheels of the bicycle to explain your results.



Prob. C18-1

C18-3. The operation of this garage door is assisted using two springs AB and side members BCD , which are pinned at C . Assuming the springs are unstretched when the door is in the horizontal (open) position and $ABCD$ is vertical, determine each spring stiffness k so that when the door falls to the vertical (closed) position, it will slowly come to a stop. Use appropriate numerical values to explain your result.



Prob. C18-3

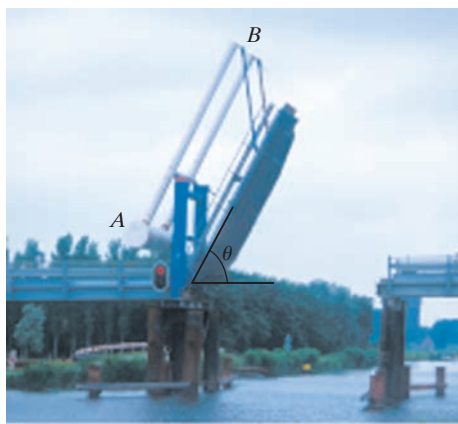
18

C18-2. Two torsional springs, $M = k\theta$, are used to assist in opening and closing the hood of this truck. Assuming the springs are uncoiled ($\theta = 0^\circ$) when the hood is opened, determine the stiffness k ($\text{N} \cdot \text{m}/\text{rad}$) of each spring so that the hood can easily be lifted, i.e., practically no force applied to it, when it is closed in the unlocked position. Use appropriate numerical values to explain your result.



Prob. C18-2

C18-4. Determine the counterweight of A needed to balance the weight of the bridge deck when $\theta = 0^\circ$. Show that this weight will maintain equilibrium of the deck by considering the potential energy of the system when the deck is in the arbitrary position θ . Both the deck and AB are horizontal when $\theta = 0^\circ$. Neglect the weights of the other members. Use appropriate numerical values to explain this result.

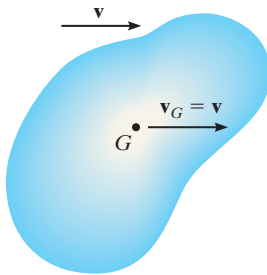


Prob. C18-4

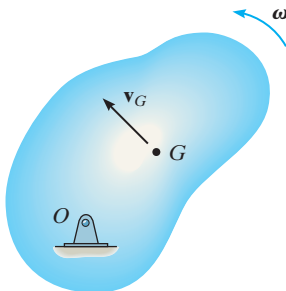
CHAPTER REVIEW

Kinetic Energy

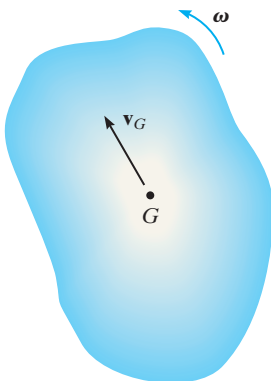
The kinetic energy of a rigid body that undergoes planar motion can be referenced to its mass center. It includes a scalar sum of its translational and rotational kinetic energies.



Translation



Rotation About a Fixed Axis



General Plane Motion

Translation

$$T = \frac{1}{2}mv_G^2$$

Rotation About a Fixed Axis

$$T = \frac{1}{2}mv_G^2 + \frac{1}{2}I_G\omega^2$$

or

$$T = \frac{1}{2}I_O\omega^2$$

General Plane Motion

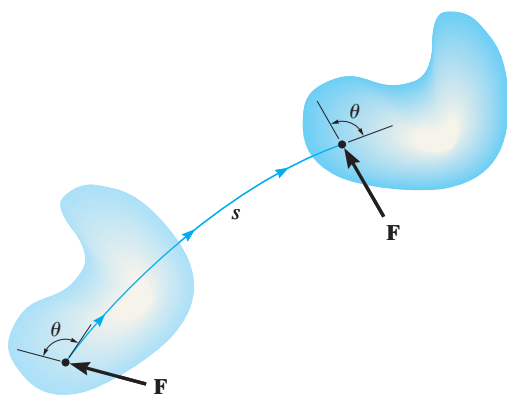
$$T = \frac{1}{2}mv_G^2 + \frac{1}{2}I_G\omega^2$$

or

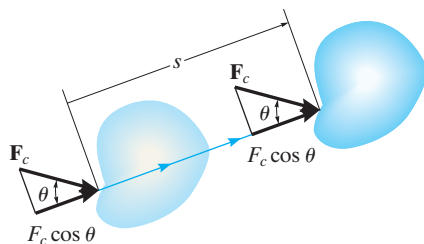
$$T = \frac{1}{2}I_C\omega^2$$

Work of a Force and a Couple Moment

A force does work when it undergoes a displacement ds in the direction of the force. In particular, the frictional and normal forces that act on a cylinder or any circular body that rolls *without slipping* will do no work, since the normal force does not undergo a displacement and the frictional force acts on successive points on the surface of the body.

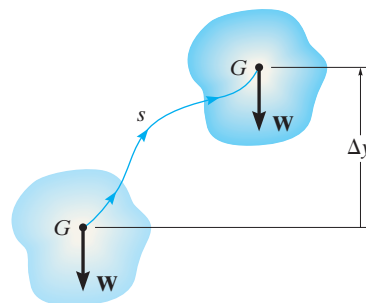


$$U_F = \int F \cos \theta \, ds$$



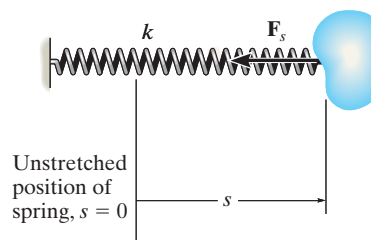
$$U_{F_c} = (F_c \cos \theta)s$$

Constant Force



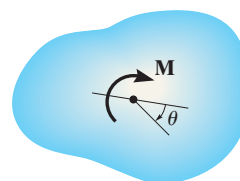
$$U_W = -W \Delta y$$

Weight



$$U = -\frac{1}{2} k s^2$$

Spring



$$U_M = \int_{\theta_1}^{\theta_2} M \, d\theta$$

$$U_M = M(\theta_2 - \theta_1)$$

Constant Magnitude

Principle of Work and Energy

Problems that involve velocity, force, and displacement can be solved using the principle of work and energy. The kinetic energy is the sum of both its rotational and translational parts. For application, a free-body diagram should be drawn in order to account for the work of all of the forces and couple moments that act on the body as it moves along the path.

Conservation of Energy

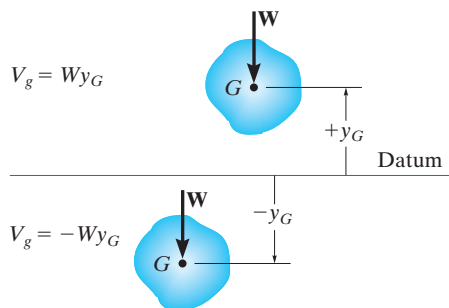
If a rigid body is subjected only to conservative forces, then the conservation-of-energy equation can be used to solve the problem. This equation requires that the sum of the potential and kinetic energies of the body remain the same at any two points along the path.

The potential energy is the sum of the body's gravitational and elastic potential energies. The gravitational potential energy will be positive if the body's center of gravity is located above a datum. If it is below the datum, then it will be negative. The elastic potential energy is always positive, regardless if the spring is stretched or compressed.

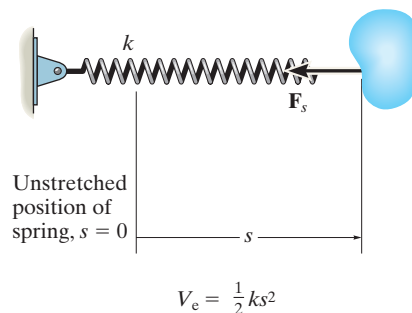
$$T_1 + \Sigma U_{1-2} = T_2$$

$$T_1 + V_1 = T_2 + V_2$$

$$\text{where } V = V_g + V_e$$



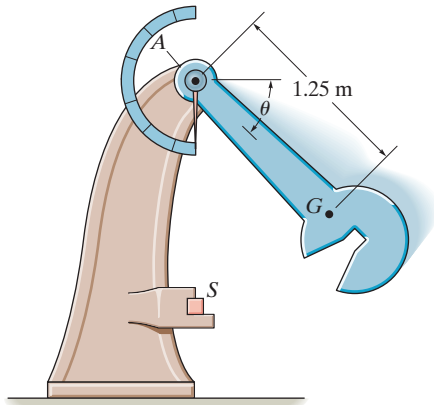
Gravitational potential energy



Elastic potential energy

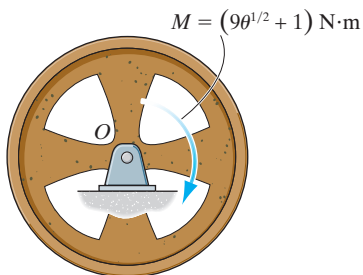
REVIEW PROBLEMS

R18-1. The pendulum of the Charpy impact machine has a mass of 50 kg and a radius of gyration of $k_A = 1.75$ m. If it is released from rest when $\theta = 0^\circ$, determine its angular velocity just before it strikes the specimen S , $\theta = 90^\circ$



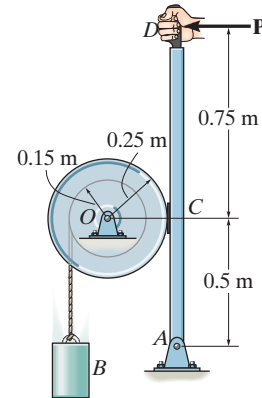
Prob. R18-1

R18-2. The 50-kg flywheel has a radius of gyration of $k_O = 200$ mm about its center of mass. If it is subjected to a torque of $M = (9\theta^{1/2} + 1)$ N·m, where θ is in radians, determine its angular velocity when it has rotated 5 revolutions, starting from rest.



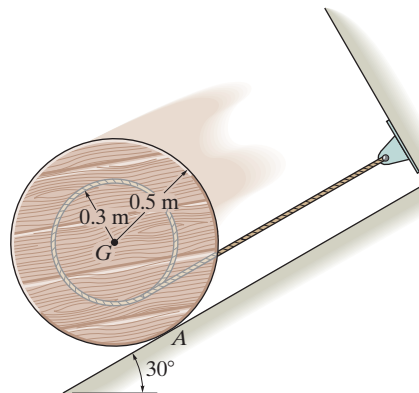
Prob. R18-2

R18-3. The drum has a mass of 50 kg and a radius of gyration about the pin at O of $k_O = 0.23$ m. Starting from rest, the suspended 15-kg block B is allowed to fall 3 m without applying the brake ACD . Determine the speed of the block at this instant. If the coefficient of kinetic friction at the brake pad C is $\mu_k = 0.5$, determine the force $\mathbf{P}$ that must be applied at the brake handle which will then stop the block after it descends *another* 3 m. Neglect the thickness of the handle.



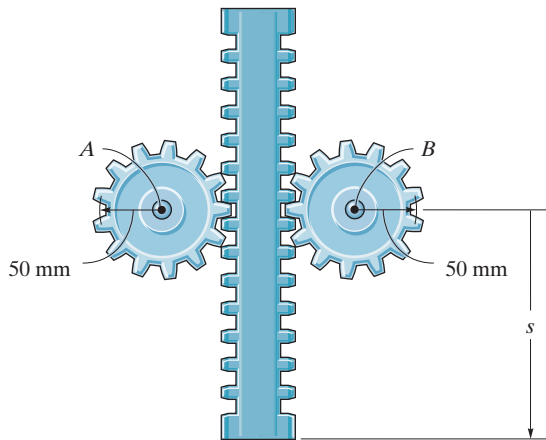
Prob. R18-3

R18-4. The spool has a mass of 60 kg and a radius of gyration of $k_G = 0.3$ m. If it is released from rest, determine how far its center descends down the smooth plane before it attains an angular velocity of $\omega = 6$ rad/s. Neglect the mass of the cord which is wound around the central core. The coefficient of kinetic friction between the spool and plane at A is $\mu_k = 0.2$.



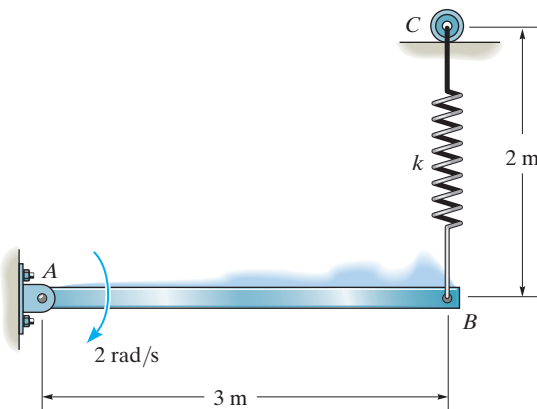
Prob. R18-4

R18-5. The gear rack has a mass of 6 kg, and the gears each have a mass of 4 kg and a radius of gyration of $k = 30$ mm at their centers. If the rack is originally moving downward at 2 m/s, when $s = 0$, determine the speed of the rack when $s = 600$ mm. The gears are free to turn about their centers A and B .



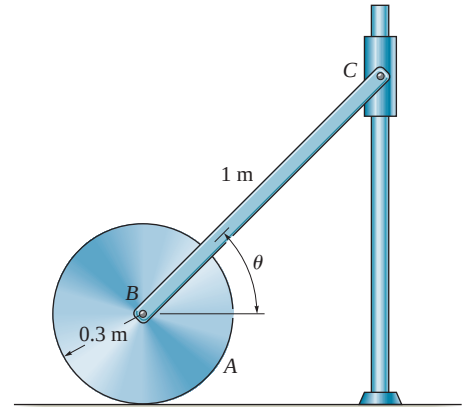
Prob. R18-5

R18-6. At the instant shown, the 25-kg bar rotates clockwise at 2 rad/s. The spring attached to its end always remains vertical due to the roller guide at C . If the spring has an unstretched length of 1 m and a stiffness of $k = 90$ N/m, determine the angular velocity of the bar the instant it has rotated 30° clockwise.



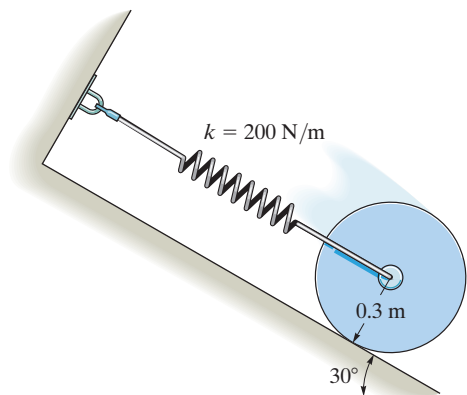
Prob. R18-6

R18-7. The system consists of a 10-kg disk A , 2-kg slender rod BC , and a 0.5-kg smooth collar C . If the disk rolls without slipping, determine the velocity of the collar at the instant the rod becomes horizontal, i.e., $\theta = 0^\circ$. The system is released from rest when $\theta = 45^\circ$.



Prob. R18-7

R18-8. At the instant the spring becomes undeformed, the center of the 40-kg disk has a speed of 4 m/s. From this point determine the distance d the disk moves down the plane before momentarily stopping. The disk rolls without slipping.



Prob. R18-8

Chapter 19



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The impulse that this tugboat imparts to this ship will cause it to turn in a manner that can be predicted by applying the principles of impulse and momentum.